InteractEval: Measuring Interactive Intelligence

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Abstract

As AI systems increasingly operate through extended interactions, their capabilities cannot be adequately characterized by isolated responses or a single holistic score. An interaction unfolds within a world of goals, constraints, participant states, and causal consequences that may not be directly visible in the transcript. We develop three complementary foundations for evaluating such systems: the Data Iceberg Theory, which distinguishes observable behavior from its interaction world; the Movie Evaluation Theory, which treats the evolving trajectory as the temporal unit of evaluation; and the Agency Principle, which distinguishes responding to an interaction from actively and usefully shaping its direction. We introduce InteractEval, a process-oriented evaluation framework that reconstructs Scene, Latent State, and Event Chain representations before applying a separate, evidence-grounded six-level capability rubric and termination controller. A controlled 27-cell scenario space and bidirectional interaction protocol support evaluation across situations and partners. Across ten frontier systems, the range of performance increases from 0.042 in passive execution to 0.431 in active adaptation, while partner-specific effects account for 10.9% of assistant-side score variance in the tested two-sided setting. Structured records further expose distinct failure modes concealed by similar aggregate scores. Human audits and robustness analyses reveal both the diagnostic promise of the framework and human-aligned certainty in latent-state and agency judgments. InteractEval thus reframes interactive evaluation around the world an AI encounters, the trajectory it produces, and the agency it exercises.

1 Introduction

As AI systems move beyond isolated question answering into tutoring, companionship, research, negotiation, and autonomous assistance (OpenAI, 2026; Anthropic, 2026; Google DeepMind, 2026), their intelligence is increasingly expressed through **interaction over time** (Qwen, 2026). Yet successful individual responses do not necessarily produce successful interactions (Sabour et al., 2026; Xu et al., 2026). A tutor may answer every question correctly while failing to recognize a student's evolving misconception (Hill, 2025; Zhang et al., 2026a); a companion may remain fluent and empathetic while repeatedly returning to an unresolved emotional state (Zhang et al., 2026d;c); and an agent may satisfy each immediate instruction while gradually losing the objective that gives its work direction (Sam et al., 2026; Zhang et al., 2026b). These failures share a defining property: **locally adequate behavior can produce a globally unsuccessful interaction**.

This exposes a limitation in how interactive AI is commonly evaluated. Many widely used protocols assess a response against a reference, assign it a preference score, or aggregate turn-level judgments into a holistic measure. Such protocols remain useful for bounded tasks, but they cannot by themselves capture capabilities whose value depends on accumulated state and subsequent consequences. The same response can be appropriate in one situation and harmful in another, depending on participants' goals, beliefs, relationships, constraints, and prior interactions (Gonzalez and Heidari, 2025). The fundamental problem is therefore not simply that our scores are too coarse. **The object being measured is incompletely represented**.

The first source of incompleteness is observational. A transcript records what participants say and do, but not necessarily everything required to interpret those observations. Identical language may express agreement, conceal disagreement, or reflect a failure to recognize a conflict (Kar and Radin, 2019). Conversely, an equivalent understanding of a situation may be expressed through different words or actions (Trope and Liberman, 2010). The observed response is therefore not a unique representation of the interaction state that gives it meaning.

We formalize this distinction through the ***Data Iceberg Theory***: observable behavior is a partial manifestation of a richer interaction world. This world includes the relatively stable constraints of a scene and the evolving, interaction-relevant states of its participants. Making that world explicit does not require access to private neural cognition

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or unverifiable psychological certainty. Instead, latent states are modeled as operational, evidence-constrained hypotheses that explain observed behavior, restrict plausible continuations, and support auditable evaluation. The objective is not to infer everything hidden, but to represent what is necessary to interpret the interaction responsibly.

The second source of incompleteness is temporal. Even a faithful description of the current state cannot determine the value of an action without considering how that state was reached and what the action makes possible next. A response may resolve an immediate request while undermining a longer-term goal (Heatherton and Wagner, 2011); another may appear suboptimal in isolation while preventing a later failure (Duckworth et al., 2016). Its value is consequently defined in part by the trajectory it sustains or changes. **Local Response Quality does not imply Trajectory Quality**.

This motivates the ***Movie Evaluation Theory***. A movie cannot be judged by averaging the quality of its frames: the meaning of each frame depends on the scene, the preceding events, the characters' evolving states, and the future it makes reachable. Interactive evaluation has the same temporal structure. Its appropriate unit is an unfolding interaction trajectory, represented through three complementary components: *Scene*, which establishes relatively stable world constraints; *Latent State*, which tracks interaction-relevant changes; and *Event Chain*, which captures their causal progression and possible consequences.

A third limitation remains even when an interaction world and its full trajectory are represented. A system can respond appropriately at every turn without independently improving the direction of the interaction. Consider two research assistants that both execute assigned tasks accurately. One waits for the next instruction; the other also identifies an overlooked assumption, proposes a consequential experiment, and sustains the new direction it has opened. Their responsiveness may be comparable, but their contributions to the interaction are not (Tao, 2026). **Responsive Quality does not imply Interactive Agency**.

We formulate this distinction as the ***Agency Principle***. Interactive capability progresses from executing within a given world, through adapting to the changes introduced by others, to proactively and coherently shaping the interaction itself. This progression does not equate initiative with verbosity, assertiveness, or unrequested intervention: an action is valuable only if it opens a useful direction consistent with the established goals and constraints. Initiative is particularly important for studying how AI systems independently introduce new possibilities into an interaction, and its presence serve as a foundation for the emergence of creativity or general cognition (Johnson et al., 2026).

Together, these principles distinguish three questions: ❶ What world gives the observed behavior its meaning, ❷ How does the interaction evolve through time, and ❸ To what extent does the model actively shape that evolution? They also distinguish the object of evaluation from the judgment placed upon it. A scalar score may summarize interactive performance, but it cannot by itself reveal which capability failed, when the failure emerged, or which state transition supports the conclusion. Such explanations require a structured measurement record, leading to a benefit to further analysis and optimization.

We introduce **InteractEval**, a process-oriented framework that reconstructs interaction-world states and causal trajectories before evaluating them through a separate normative layer. This separation makes the measurement process inspectable: descriptive hypotheses remain distinguishable from capability judgments, evidence remains traceable to the interaction, and aggregation follows rather than replaces structured evaluation.

This work makes four contributions:

- **Foundations of interactive evaluation**. We establish observational incompleteness, temporal non-separability, and directional contribution as three complementary properties governing the evaluation of interactive AI.
- **Explicit interaction-world representation**. We develop a structured representation of Scene, Latent State, and Event Chain that reconstructs interaction-relevant conditions, preserves state transitions, and models plausible future developments.
- **A process-oriented measurement instrument**. We integrate controlled scenario design, role-conditioned interaction, evidence-grounded world reconstruction, and a dependency-structured six-level capability rubric into an auditable evaluation procedure.
- **An empirical characterization of interactive capability**. Evaluation across ten frontier systems reveals markedly greater differentiation in active adaptation than in passive execution, diagnostic differences obscured by aggregate scores, and substantial dependence on interaction partners. Human audits and robustness analyses further characterize the reliability and limitations of the measurement instrument.

2 Theoretical Foundations

An interaction is not determined by its observed utterances alone. The same response may have different meanings under different participant states, and its significance may depend on effects that emerge only in subsequent turns. Evaluating interactive capability therefore requires a representation of the interaction world, an account of its

temporal development, and a criterion for the contribution of each participant to that development (Xuan et al., 2026). These three requirements motivate the Data Iceberg Theory, the Movie Evaluation Theory, and the Agency Principle, respectively.

Interaction model. Let $t \in \{1, \ldots, T\}$ index completed exchanges in an episode of length T. The observed exchange $O_t = (u_t, a_t)$ contains a user utterance u_t followed by an assistant response a_t; an utterance may also denote an externally visible action. The transcript through exchange t is $O_{1:t} = (O_1, \ldots, O_t)$, with $O_{1:0} = \varnothing$. Let S_t be the effective *scene* after exchange t: the roles, objectives, world rules, and constraints that govern the interaction. Let X_t denote its underlying, potentially unobserved participant and world state, including relevant beliefs, goals, affect, and relationships. The interaction world at that point is

$$Z_t \triangleq (S_t, X_t), \qquad Z_0 = (S_0, X_0). \tag{1}$$

The initial scene S_0 is specified by the evaluation case. Although scene rules are ordinarily persistent, a realized event may legitimately amend them, giving $S_t \neq S_{t-1}$.

Write H_t^U and H_t^A for the information respectively available to the user and assistant immediately before their actions in exchange t. These histories include role-authorized scene information and prior visible exchanges; H_t^A additionally includes u_t. For user and assistant policies π_U and π_A, and an underlying state-transition kernel K, the interaction admits the general factorization

$$\begin{aligned} u_t &\sim \pi_U(\cdot \mid H_t^U), \\ a_t &\sim \pi_A(\cdot \mid H_t^A), \\ Z_t &\sim K(\cdot \mid Z_{t-1}, u_t, a_t). \end{aligned} \tag{2}$$

The state X_t includes whatever history-dependent variables are needed for this conditional representation. The evaluator does not observe Z_t directly. It receives an authorized scenario specification C^{eval} and the observable interaction, constituting the evidence record

$$D_t \triangleq (C^{\text{eval}}, O_{1:t}), \qquad D_0 \triangleq C^{\text{eval}}. \tag{3}$$

The distinction between Z_t and D_t is the starting point for the theory.

2.1 Data Iceberg Theory: Observability and State Reconstruction

Observational incompleteness. The *Data Iceberg Theory* states that observable interaction is a partial manifestation of an interaction world whose relevant variables need not be uniquely determined by that observation. For a given evidence record d, let

$$\mathcal{Z}_t(d) \triangleq \operatorname{supp} p(Z_t \mid D_t = d) \tag{4}$$

denote the set of world states compatible with the evidence under the interaction process in Eq. (2). When $\mathcal{Z}_t(d)$ contains distinct states, the record does not identify a unique world state. This is an evaluation-relevant ambiguity whenever two such states support different interpretations of the same response. For a hypothetical state-aware judgment $J^\star(z, d)$, a sufficient condition is

$$\exists\, z, z' \in \mathcal{Z}_t(d) : \quad J^\star(z, d) \neq J^\star(z', d). \tag{5}$$

No judgment determined only by d can agree with both values in Eq. (5). A polite acknowledgment, for instance, may express agreement, concealment, or uncertainty, each with different implications for the next response. Conversely, one underlying state may give rise to multiple appropriate utterances. A unique surface realization is therefore neither necessary nor sufficient for identifying the interaction-relevant state.

Operational latent state. To represent what the evidence supports, let Ψ_t denote the evaluator's *operational latent state*: a structured estimate of participant and world variables at exchange t. Unlike the underlying X_t, Ψ_t is an inferential object defined by the available record,

$$\Psi_t \triangleq \Phi_\Psi(D_t), \qquad \Psi_0 = \Phi_\Psi(D_0). \tag{6}$$

The mapping Φ_Ψ may be implemented incrementally by carrying forward previous estimates and event memory, as specified in §2.4. Each state entry records an interpretation and its evidence. If c is an asserted state claim and $\operatorname{src}(c)$ is the set of supporting text spans or recorded actions, evidential grounding requires

$$\varnothing \neq \operatorname{src}(c) \subseteq \operatorname{Spans}(D_t), \tag{7}$$

where $\mathrm{Spans}(D_t)$ denotes addressable portions of the evidence record. A state estimate can be maintained across exchanges, strengthened by new evidence, or revised when subsequent observations contradict it. Competing interpretations remain possible where the evidence does not discriminate between them.

The value of Ψ_t is its capacity to encode distinctions relevant to future behavior and later adjudication. Its adequacy can be examined through consistency with the observed record, reliability across evaluators, and prospective consequences. The resulting state estimates can be revised and assessed through subsequent observations, supporting an explicit account of otherwise implicit interaction conditions.

2.2 Movie Evaluation Theory: Temporal Structure and Trajectory Value

Temporal non-separability. A state estimate locates an interaction at one moment, but a response also changes the conditions under which later responses occur. The *Movie Evaluation Theory* treats the state-and-action trajectory as the temporal unit on which interaction value is defined. For the underlying trajectory and a goal-relative value functional $\mathcal{V}$, write

$$\tau_T \triangleq (Z_0, O_1, Z_1, \ldots, O_T, Z_T), \qquad V_T \triangleq \mathcal{V}(\tau_T) \in R. \tag{8}$$

Here $\mathcal{V}$ denotes a contextual standard of trajectory value, not a score supplied by the evaluator. Purely independent response evaluation assumes that V_T can be represented as $\sum_{t=1}^{T} v(O_t)$ for some response-only function v. Suppose two feasible trajectories τ_T and τ'_T contain the same multiset of observable exchanges but have different values because their order, states, or consequences differ. Then

$$\begin{aligned} &\{\{O_t\}\}_{t=1}^{T} = \{\{O'_t\}\}_{t=1}^{T}, \qquad \mathcal{V}(\tau_T) \neq \mathcal{V}(\tau'_T) \\ \Longrightarrow \quad &\nexists\, v : \ \mathcal{V}(\tau) = \sum_t v(O_t) \quad \text{for all such trajectories } \tau. \end{aligned} \tag{9}$$

The order of a clarification and an irreversible commitment provides a simple instance: the same utterances can constitute timely guidance or an ineffective repair depending on when they occur. Trajectory evaluation therefore requires history-conditioned state transitions, not merely an accumulation of context-free response values.

Causal event structure. We represent the progression of an interaction through an *event structure* E_t:

$$E_t \triangleq (P_t, C_t, F_t), \qquad E_0 \triangleq \varnothing. \tag{10}$$

Its components have distinct temporal referents. The *past* P_t contains established premises and relevant prior events; the *present* C_t identifies the current realized event, active conflict, or shift in attention; the *future* F_t contains prospective continuations conditional on the current record. With Ψ_t from Eq. (6), let Φ_P, Φ_C, and Φ_F be the updates of the past, present, and future components, respectively:

$$\begin{aligned} P_t &= \Phi_P(E_{t-1}, D_{t-1}, S_{t-1}), \\ C_t &= \Phi_C(P_t, O_t, \Psi_t, S_{t-1}), \\ F_t &= \Phi_F(P_t, C_t, \Psi_t, S_{t-1}). \end{aligned} \tag{11}$$

The future component is a set $F_t = \{f_t^{(1)}, \ldots, f_t^{(K_t)}\}$ of K_t candidate paths. Each candidate $f_t^{(k)}$ has a probability $p_t^{(k)}$ and may be assigned a label $d_t^{(k)}$ for the course the interaction would take along it:

$$d_t^{(k)} \in \{\text{Advance}, \text{Maintain}, \text{Degrade}, \text{Collapse}\}. \tag{12}$$

These labels describe the course of the interaction as a joint activity, not its valence for either participant. Under Advance the interaction develops into a new situation that the past warrants; under Maintain it holds its present course; under Degrade it stagnates, repeating itself and losing the partner's interest; under Collapse it can no longer be continued. The future paths remain conditional projections until subsequent exchanges provide evidence of their realization, and their content depends on the accumulated past and the latent state, so an event chain is not equivalent to a transcript summary. Write $d_t^\star$ for the label of the most probable path in F_t, the *dominant* path.

A realized change in C_t may also alter the effective scene. Let $\mathcal{R}_t$ be the candidate revisions of S_{t-1} proposed at exchange t. A candidate r is admissible when

$$r \in \mathrm{Rec}(C_t), \qquad d_t^\star = \text{Advance}, \qquad \varnothing \neq \mathrm{src}(r) \subseteq \mathrm{Spans}(D_t), \tag{13}$$

where $\mathrm{Rec}(C_t)$ denotes the changes recorded in the present. The first condition requires that r describe a change that has been realized and not a continuation projected in F_t; the second, that the interaction is developing into a new

situation, whatever the valence of the development; the third, that the revision be grounded in the evidence record, as in Eq. (7). With $\mathcal{R}_t^\star$ the admissible candidates, let Φ_S apply them,

$$S_t = \Phi_S(S_{t-1}, \mathcal{R}_t^\star), \tag{14}$$

so that $S_t = S_{t-1}$ when $\mathcal{R}_t^\star$ is empty. The paths in F_t enter only through $d_t^\star$ and never as $\mathrm{src}(r)$: a projection can filter a revision but cannot support one. The resulting *reconstructed trajectory* is

$$\widehat{\tau}_T \triangleq (S_{0:T}, O_{1:T}, \Psi_{0:T}, E_{1:T}), \tag{15}$$

in which observed behavior, interpreted state, realized events, and projected consequences remain separately inspectable.

2.3 Agency Principle: Contribution to the Interaction's Direction

Initiative and directional effect. The existence of a coherent trajectory does not determine how much of its direction was contributed by a participant. A system may preserve goals, integrate new information, and respond appropriately throughout a dialogue while leaving every substantive next step to its partner. The *Agency Principle* identifies interactive capability with a further dimension: the ability to initiate a warranted direction and sustain the interaction it brings about.

Consider the assistant at the moment following the user utterance u_t. Its decision context is $h_t \triangleq H_t^A$. Let $\mathcal{A}_{\mathrm{valid}}(h_t)$ be the set of actions consistent with the operative goals and constraints, and let $\mathcal{A}_{\mathrm{resp}}(h_t) \subseteq \mathcal{A}_{\mathrm{valid}}(h_t)$ contain admissible actions that leave the substantive direction within the partner-established agenda. The set

$$\mathcal{A}_{\mathrm{init}}(h_t) \triangleq \mathcal{A}_{\mathrm{valid}}(h_t) \setminus \mathcal{A}_{\mathrm{resp}}(h_t) \tag{16}$$

characterizes warranted initiative. The distinction concerns the action's role in the interaction, not its length, assertiveness, or linguistic novelty. A proposal to test an overlooked assumption can belong to $\mathcal{A}_{\mathrm{init}}(h_t)$ even if it is brief; an elaborate repetition of the user's request need not.

Initiative and beneficial effect are distinct. Let $\pi_{\mathrm{resp}}(\cdot \mid h_t)$ be a specified response-only reference policy supported on $\mathcal{A}_{\mathrm{resp}}(h_t)$. For a horizon $H \geq 1$, let $\tau_{t:t+H}$ denote the subsequent interaction segment, and let $\mathcal{V}_H$ be its goal-relative value under a fixed continuation protocol. The prospective effect of the assistant's action a relative to responsive behavior can be defined as

$$\begin{aligned} \Delta_H(a; h_t) &\triangleq Q_H(h_t, a) - E_{a' \sim \pi_{\mathrm{resp}}(\cdot \mid h_t)} Q_H(h_t, a'), \\ Q_H(h, a) &\triangleq E[\mathcal{V}_H(\tau_{t:t+H}) \mid \mathrm{do}(a_t = a), h_t = h]\,. \end{aligned} \tag{17}$$

The intervention fixes the candidate action while retaining the same prior context and downstream continuation protocol. A beneficial agentic move lies in $\mathcal{A}_{\mathrm{init}}(h_t)$ and has $\Delta_H(a; h_t) > 0$. The relative effect is a theoretical attribution quantity; its estimation would require an appropriate intervention design or additional assumptions beyond an observed transcript.

Sustained agency. The contribution of an initiated direction extends beyond the initiating turn: the participant must also preserve the goals, state, and world constraints of the course it introduces. This separates the origination of a new direction from its maintenance over subsequent exchanges. The rubric in §3.5 operationalizes these dimensions as proactive agency and long-horizon coherence.

2.4 A Unified Measurement Formulation

The three principles specify a chain of dependencies: interpretation requires an evidence-grounded interaction state; temporal judgment requires a state-and-event trajectory; and attribution of agency requires identifying which changes in that trajectory were proactively contributed by the evaluated participant. The corresponding measurement model reconstructs the interaction before assigning capability judgments.

Let $\Psi_0 = \Phi_\Psi(D_0)$, $E_0 = \varnothing$, and S_0 be the initial scene. With Ψ_t denoting the stored operational estimate from

§2.1 and $E_t = (P_t, C_t, F_t)$ from §2.2, a recursive implementation of the descriptive stage is

$$\begin{aligned} \Psi_t &= \phi_\Psi(\Psi_{t-1}, E_{t-1}, S_{t-1}, D_t), \\ E_t &= \phi_E(E_{t-1}, \Psi_t, S_{t-1}, D_t), \\ S_t &= \Phi_S(S_{t-1}, \mathcal{R}_t^\star), \\ W_t &\triangleq (S_t, \Psi_t, E_t). \end{aligned} \tag{18}$$

Here W_t is the *operational world reconstruction*, distinct from the underlying state Z_t in Eq. (1). The update functions ϕ_Ψ and ϕ_E are incremental realizations of the evidence-based mappings defined above. At an assistant or user subturn, the same recursion can be applied to the newly available evidence before the exchange is complete.

The subsequent *normative* stage maps that reconstruction and its source evidence to adjudication:

$$(J_t, V_t, h_t) = \phi_N(W_t, D_t), \tag{19}$$

where J_t is the vector of capability judgments, V_t the violation or red-line decision, and h_t the termination decision. The six capability levels in J_t form an ordered measurement hierarchy: L1–L2 measure goal preservation and information uptake (passive execution); L3–L4 measure state progression and contextual expression (responsive adaptation); and L5–L6 measure useful initiative and coherence across turns (active adaptation). Dependencies among levels define the score constraints implemented in §3.5. Normative judgments refer to source evidence in D_t; a reconstructed state is an interpretive input, not independent proof of itself.

The episode's structured measurement record is

$$R_t \triangleq (\Psi_t, E_t, J_t, V_t, h_t), \qquad R_{1:T} = (R_1, \ldots, R_T). \tag{20}$$

Only after constructing this record is a scalar summary defined:

$$q = G(R_{1:T}), \tag{21}$$

where G is the specified aggregation rule. Distinct records may yield the same q, so diagnosis relies on the retained state transitions, event paths, capability judgments, and evidence rather than the scalar alone.

The theoretical requirements of an interactive evaluation instrument consequently follow from the three principles: *state adequacy* under incomplete observation, *temporal continuity* across exchanges, *evidential traceability* for latent-state and capability claims, *agency sensitivity* to warranted directional contribution, and *diagnostic retention* before aggregation. §3 realizes these requirements through controlled scenarios, role-conditioned interaction, explicit state-and-event reconstruction, and separate normative adjudication.

3 Architecture

InteractEval operationalizes the three principles of Section 2 as a stateful measurement instrument. Its evaluation target is an interaction unfolding under specified world rules, participant objectives, and asymmetric information. The instrument first reconstructs the conditions and events that give observed behavior its meaning, and then adjudicates capability against that reconstruction and its underlying evidence. The result is a sequence of interpretable measurements rather than a judgment attached only to a finished dialogue.

The architecture separates four responsibilities. The *data layer* constructs evaluation cases within a declared scenario space; the *interaction layer* produces or replays role-conditioned exchanges; the *mechanism layer* maintains the operational interaction world; and the *normative layer* assigns capability judgments and decides whether evaluation continues. The first two layers define the circumstances under which behavior is observed. The latter two implement the descriptive–normative separation of Eqs. (18)–(19). A common evidence record connects the layers without exposing rubric scores to the participants or allowing the judge to alter the reconstructed world.

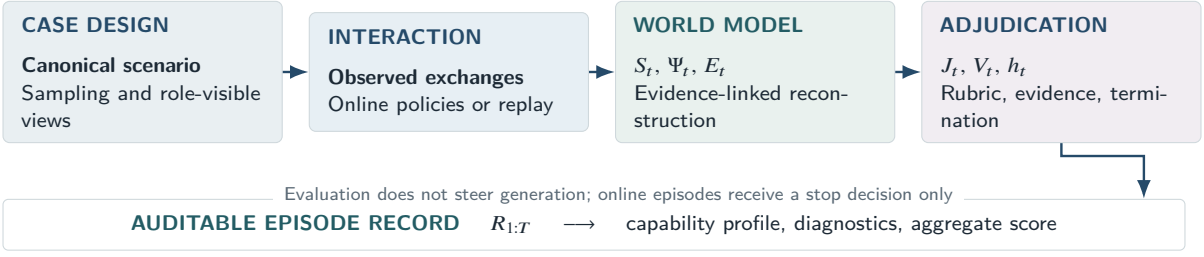


Figure 1: InteractEval measurement architecture. Case construction and interaction acquisition establish the observable episode. Descriptive reconstruction carries the evolving scene, latent state, and event structure across turns; normative adjudication produces independent judgments and termination control. Every output remains traceable to the originating evidence.

3.1 Architecture Overview

Evaluation episode. An episode begins with a canonical evaluation case. Its evaluator-authorized specification C^{eval} determines the initial scene S_0, while role-specific views determine the information available to the user and assistant policies π_U and π_A. The resulting exchange $O_t = (u_t, a_t)$ consists of a user utterance followed by an assistant utterance or externally visible action, as defined in Section 2. For online evaluation, the policies generate the exchange sequentially. For replay, the instrument reads an archived sequence without regenerating the participants' behavior. In both settings the evaluator's evidence at completed exchange t is $D_t = (C^{\text{eval}}, O_{1:t})$. Sampling labels and private profile attributes used to construct the case are not independently introduced as evidence for scoring.

Sequential reconstruction. The mechanism layer maintains $W_t = (S_t, \Psi_t, E_t)$, comprising the effective scene, the operational latent-state estimate, and the event chain. Its update is the ordered composition of Eq. (18): Ψ_t is reconstructed from the new evidence and carried state, E_t records the current event and conditional continuations, and S_t incorporates admissible scene revisions. These are distinct records with different persistence rules. Latent variables can change as evidence accumulates, events register the observed course and possible continuations, and the scene retains operative constraints until a justified change is admitted. Scene revision follows the rule in Eqs. (13)–(14): the proposed change must be realized in the present, supported by identifiable evidence, and admitted under the dominant-path *Advance* condition. Here *Advance* denotes a historically warranted development into a new situation, irrespective of whether it is favorable to either participant. A projected path may gate admission but cannot supply evidence that a change has already occurred. The prompts and rule configurations that implement each step are given in Appendices A.1–A.6.

The implementation processes the user's utterance and the assistant's reply at separate subturn boundaries. Each update can revise the operational state carried into the next subturn; the complete exchange yields the episode-level W_t. This ordering preserves the information available when each participant acted and makes changes attributable to their place in the transcript. At every boundary, the reconstruction is stored with the scene version and evidence spans on which its assertions rely. The descriptive layer does not emit L1–L6 grades or use a capability verdict to justify its own state estimates.

Capability adjudication. Once the descriptive update is complete, the normative layer evaluates the reconstructed world jointly with the original evidence D_t. It produces the six-level judgment vector J_t, the red-line decision V_t, and the termination decision h_t, as in Eq. (19). The assistant-side rubric distinguishes passive execution (L1–L2), responsive adaptation (L3–L4), and active adaptation (L5–L6). The judge receives evidence and language-based state descriptions, not the numerical state degrees or the sampling labels. Each grade is accompanied by evidence from the original record; a reconstructed hypothesis is an interpretive input, not independent proof of the judgment. Raw level judgments are retained, while the dependency cascade and other procedural constraints are enforced separately from semantic appraisal. The six isolated level calls and the closure query are specified in Appendix A.7. In bidirectional experiments, the user policy also receives an independent role-specific rubric assessment; this rubric-only path does not duplicate the assistant-side world-reconstruction and termination controller.

Measurement unit and control. Each completed exchange yields the record defined in Eq. (20),

$$R_t = (\Psi_t, E_t, J_t, V_t, h_t), \qquad R_{1:T} = (R_1, \ldots, R_T).$$

The episode record, together with the versioned scene and cited transcript, is the unit from which capability profiles, terminal outcomes, and the aggregate score $q = G(R_{1:T})$ are derived. Evaluation and generation alternate in online episodes, but no state estimate, path forecast, or level score is returned to either participant; only the decision to stop affects the subsequent course by ending it. In replay, scoring can be repeated under revised adjudication rules, although the originally observed sequence and its terminal length remain fixed. Section 3.2 specifies the cases and visibility rules, Section 3.3 the interaction protocols, Sections 3.4–3.5 the descriptive and normative computations, and Section 3.6 the resulting records, aggregation, and reproducible execution.

Execution contracts. The interfaces between layers are fixed independently of the models used to implement semantic inference. Table 1 identifies the state passed between stages and the information that must remain outside each stage. These restrictions preserve the distinction between properties of the interaction, interpretations of those properties, and judgments of participant capability.

Table 1: Interfaces and information boundaries of the four-layer instrument.

Stage	Consumes	Emits	Boundary
Case construction	Scenario sources and design facets	C^{eval} and role views	Private profile metadata and facet labels do not enter the judge view.
Interaction	Role-visible histories and participant policies	Observable O_t	No evaluation score or state inference enters generation.
Reconstruction	D_t and carried state	$W_t = (S_t, \Psi_t, E_t)$	Future forecasts do not constitute realized-event evidence; no L1–L6 grades are assigned.
Adjudication	D_t and authorized world descriptions	(J_t, V_t, h_t)	Judgments cite source evidence; rubric outputs cannot revise W_t.

3.2 Evaluation Cases

The data layer defines the situations in which interactive capability is observed. An evaluation case must specify the rules and objectives of an interaction, the participant attributes that condition behavior, and the information available to each role. These are distinct sources of variation. *Scene facets* define the sampling space; *participant profiles* condition the actors within a sampled scene; and *administration variables*, including model assignment and turn budget, determine how that scene is enacted. Keeping these factors separate makes scenario coverage explicit and permits the effects of the evaluated system and its interlocutor to be examined under common conditions (**??**).

3.2.1 Scenario Space

Facet specification. We construct the scenario space from three scene-level facets: *realism* (R), describing the governing relation between the scenario world and familiar reality; *information sufficiency* (I), describing whether the available specification contains the information required for the interaction; and *interaction purpose* (P), describing the relative role of task completion and relational continuation. Let

$$\begin{aligned} \mathcal{R} &= \{\text{realistic}, \text{stylized}, \text{science-fictional}\}, \\ \mathcal{I} &= \{\text{sufficient}, \text{partial}, \text{insufficient}\}, \\ \mathcal{P} &= \{\text{functional}, \text{mixed}, \text{nonfunctional}\}, \\ C &= \mathcal{R} \times \mathcal{I} \times \mathcal{P}, \qquad |C| = 3^3 = 27. \end{aligned} \tag{22}$$

The crossing treats the facets as separate *questions* about a scene rather than mutually exclusive genres. A single scenario occupies one level of each facet. Realism asks which world rules the assistant must respect, information sufficiency asks what can be inferred or must be requested under those rules, and purpose asks what continuation or completion would constitute the interaction's objective. Table 2 gives their operational meaning and the construct probed by each.

Table 2: Scene-level facets defining the evaluation space. The categories specify controlled conditions; they do not assert statistical independence of the constructs.

Facet	Three conditions	Operational distinction
Realism R	Realistic / stylized / science-fictional	Adherence to the specified world mechanism as it departs from familiar-world assumptions.
Information I	Sufficient / partial / insufficient	Whether the available specification warrants proceeding, targeted clarification, or the recognition of an unresolved gap.
Purpose P	Functional / mixed / nonfunctional	Whether successful interaction is organized by an explicit task, a combination of task and relationship, or continuing relational value.

Construct grounding. The realism facet extends the principle of minimal departure in narrative-world interpretation: departures from familiar rules must be made explicit and subsequently maintained in the enacted scene (**?**). The information facet concerns what is warranted by the participants' common ground and what is required for an appropriately informative contribution, following the distinction between quantity and evidential quality in conversational cooperation (**?**). Interaction purpose distinguishes task-oriented from socio-emotional functions of interaction, while allowing both to operate in the same scene (**?**). These theoretical antecedents motivate the three facets' *interpretation*; their particular three-level operationalization is a design choice evaluated empirically in Section 4.

Coverage and separability. The full crossing supplies a finite, inspectable sampling frame. With n_c denoting the number of admitted cases in cell $c \in C$, the present balanced corpus satisfies

$$n_c = 30 \quad \forall c \in C, \qquad N = \sum_{c \in C} n_c = 810. \tag{23}$$

Consequently each facet level contains $3^2 \times 30 = 270$ cases, and every pair of facet levels appears in $3 \times 30 = 90$ cases. Let $j, k \in \{R, I, P\}$ index distinct facets, C_j their respective level sets, and $x_{j,a} : C_j \to R$ and $x_{k,b} : C_k \to R$ centered contrasts (with a, b indexing contrasts). The balanced design then satisfies

$$\sum_{c \in C} n_c \, x_{j,a}(c_j) x_{k,b}(c_k) = 0, \quad \text{if} \sum_{v \in C_j} x_{j,a}(v) = \sum_{w \in C_k} x_{k,b}(w) = 0. \tag{24}$$

This is *design orthogonality*: main-effect contrasts can be estimated without confounding from unequal cell frequencies. It neither requires nor establishes that realism, informational demands, and purpose are independent in natural dialogue, or that unmeasured case properties have been balanced. The three conditions of a facet are therefore analyzed as categorical levels unless an additional ordered-scale assumption is warranted. In particular, the partial-information condition may contain both omissions and misleading premises, whose effects can differ; informational correctness is not independently crossed in the present design.

The coverage guarantee is exact for the declared C, not for the unrestricted universe of interactive situations. Partner cooperativeness is assigned to the user role rather than the scene; dispositional traits belong to participant profiles; and modality and interaction horizon belong to administration. This partition preserves the interpretation of a scene-level comparison while leaving the corresponding participant and protocol effects available for separate analysis.

3.2.2 Case Construction

Canonical case and admission. Each admitted case contains the initial scene S_0, role-specific participant profiles, a provenance record, and sampling metadata. The scene is serialized into six fields used downstream: persona, hidden constraints, world mechanism, interaction boundary, objective, and initial situation (Section 3.4.1). Source text and derived metadata are archived separately from the effective evaluation specification, allowing a case to be traced to its origin without exposing sampling labels to the judge.

The observational pool has two sources: user-authored character cards from the Chinese role-play platform *Zaomeng* and existing human–AI dialogue logs. The first supplies articulated world mechanisms, motives, and boundaries; the second includes interactions initiated outside a preset fictional role. Admission proceeds through provenance and privacy screening, structural parsing and deduplication, a training-overlap check where comparison is possible, and a decidability check that each case can be assigned to the declared facet levels. A quality review is performed *within* the assigned cell: sparse, information-poor cases should not be rejected simply for presenting more difficult interaction conditions. The decidability stage retained 90% and 64% of the first two source batches, respectively, in the source audit reported by the current manuscript.

Coverage correction. The observational sources are not balanced in C. Functional cases account for approximately 2.6% of the original role-play material; realism is concentrated in stylized settings, and 10–13 cells were initially unpopulated across the audited source pools. Underrepresented cells are populated by conditional scenario generation given the cell specification, complete scene fields, and participant profiles. Generated cases undergo the same structural checks and are deduplicated against the admitted pool; their length and field occupancy are compared with observational cases to detect systematic formatting artifacts. The resulting corpus satisfies Eq. (23), and both one-sided and two-sided evaluation protocols use the same set of 810 cases.

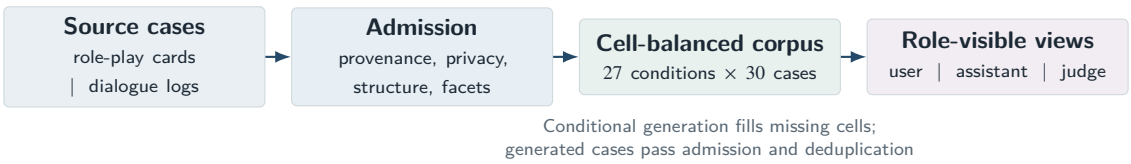


Figure 2: Case construction and exposure boundaries. Sampling strata and provenance remain in the archival metadata; the evaluator receives only its authorized case view and the observed interaction.

Difficulty and measurement conditions. Facet assignment precedes interaction and does not depend on a model's performance. The annotation-stage difficulty prior is used for stratified selection; reported item difficulty is estimated subsequently from tested-system responses using the Rasch-family measurement model described in the current experiments (**?**). The distinction prevents a model-derived difficulty estimate from becoming a label supplied to

the evaluating judge. The scenario design therefore controls *where* evidence is sampled, while the interaction and scoring layers determine *what* observed behavior supports a capability claim.

3.2.3 Profiles and Visibility

Participant representation. A scene describes the world in which participants act; a profile specifies dispositional and role-conditioned properties of a participant acting within that world. The profile schema is organized around dispositional traits, characteristic adaptations, and narrative identity, following the integrative personality framework of **?**. Its component constructs draw on established accounts of personality traits, motivation, emotion regulation, attachment, interpersonal stance, social cognition, executive control, and moral boundaries. The user profile additionally records its goal, probing strategy, communication style, and emotional sensitivity. Such variables characterize conditional tendencies, not deterministic behavioral prescriptions: responses emerge from interactions between dispositional features and situations (**?**).

The supplied study reports that profile values were informed by observations of more than 400 users across 43 dimensions, with the aim of preserving plausible trait co-occurrence. This is a basis for *profile generation*, not an external validation of the generated personalities. Likewise, the profile taxonomy provides theoretically motivated representational breadth but does not establish that all psychological constructs are independent or that no relevant dimension is omitted. Its modeling adequacy is assessed through whether it supports meaningful behavioral differences, stable role fidelity, and informative state updates under controlled scenes.

Controlled information exposure. Each case maintains distinct views for the assistant, user, and evaluator. Scene and profile fields are tagged by role ownership and public/private status; a private entry is delivered only to its designated participant during generation. The evaluator's authorized case specification C^{eval} contains the six effective scene fields and the raw scene prompt, together with the observed dialogue; it does *not* contain the external psychological profiles, the facet assignments, or numeric latent-state degrees. The scene's *persona* field, visible to the judge, is distinct from the psychological profile used to condition a participant. The latter can condition the separate engagement controller (§3.4.3), which affects episode continuation rather than a rubric grade. Thus profiles can influence the *interaction being sampled* without appearing as independent factual evidence for its score.

These exposure rules implement the information asymmetries of the interaction model in Eq. (2). They prevent private partner information from being supplied directly to the assistant and prevent case-design labels from becoming scoring shortcuts. Because the effective scene text may still imply a facet assignment, excluding explicit labels is a necessary boundary rather than proof of complete judge blindness. Profile serialization is fixed and versioned; the study evaluates format sensitivity separately (§4.9). The explicit role-view contract is the input to the interaction protocol of §3.3.

3.3 Interaction Protocol

3.4 Interaction-World Reconstruction

3.5 Capability Adjudication

3.6 Measurement Outputs and Execution

4 Experiments and Results

The theory of Section 2 makes claims that can be checked against data. If a scalar summary discards the structure of the record it aggregates (Eq. (21)), systems that look alike on a single score should differ in how they fail. If trajectory value is not separable into response values (Eq. (9)), some failures should be visible only across turns. If initiative is a capability in its own right (§2.3), differences between systems at the active end of the progression should not be predictable from passive execution alone. And since both policies enter the interaction (Eq. (2)), part of each assistant's score should depend on its partner. The experiments ask whether these predictions hold and, where they do, what the instrument reveals about contemporary systems. We first examine the capability profiles of ten systems and what a composite score conceals about them (§4.2), their response to the designed scenario conditions (§4.3), and two behaviors that no capability score expresses (§4.4). We then score both participants and estimate how much of each score belongs to the partner (§4.5), and extend the protocol to cognitive-architecture agents and to groups of more than two (§4.6). The comparison with human reviewers (§4.7), the analysis of trajectory-level failure (§4.8) and controls on the robustness of the pipeline (§4.9) follow.

4.1 Setup

Each system was evaluated on all 27 cells of the cube with thirty dialogues per cell, or 810 episodes per pass. The cube was run three times independently, giving 2,430 episodes per system, and all tabulated values are means over the three passes. Claude Opus 5 served as judge throughout, so judge identity is constant across systems and conditions; its influence on the results is examined in §4.9. In the one-sided protocol, `claude-5-opus` played the user; in the two-sided protocol (§4.5.1) each system played both roles. The judge had no access to facet labels, participant profiles or numeric state values. Raw scene text was visible, however, and may indicate a cell indirectly. Episodes ended according to the termination rules of §**??**. The ten systems are a sample of current frontier models, and we note where a conclusion depends on which systems were included.

4.2 The frontier lies at the active end

Table 3: Level profile over the 27-cell cube, means of three passes. The three stage columns are the means of L1–L2 (passive execution), L3–L4 (responsive adaptation) and L5–L6 (active adaptation). ΔT is non-functional minus functional performance, so a negative value marks a task-leaning system. The final row gives the range across the roster.

		level						stage				
Model	final	L1	L2	L3	L4	L5	L6	L1–L2	L3–L4	L5–L6	L6/L5	ΔT
claude-opus-5.5	0.744	0.933	0.907	0.874	0.833	0.660	0.603	0.920	0.854	0.631	0.914	−0.132
gpt-6-astra	0.736	0.927	0.899	0.865	0.824	0.655	0.596	0.913	0.845	0.625	0.910	−0.124
claude-fable-5.1	0.721	0.924	0.901	0.862	0.814	0.630	0.571	0.913	0.838	0.600	0.906	−0.129
claude-5-opus	0.632	0.909	0.882	0.828	0.766	0.488	0.436	0.896	0.797	0.462	0.894	−0.171
gpt-5.6-sol	0.602	0.917	0.880	0.811	0.746	0.441	0.391	0.899	0.779	0.416	0.887	−0.014
qwen3.8-max	0.564	0.903	0.872	0.804	0.719	0.381	0.336	0.888	0.762	0.358	0.881	−0.107
gemini-3.8-flash	0.526	0.903	0.871	0.786	0.702	0.318	0.275	0.887	0.744	0.297	0.865	+0.004
deepseek-v4-pro	0.503	0.928	0.914	0.848	0.773	0.218	0.181	0.921	0.811	0.200	0.832	−0.152
kimi-k3	0.497	0.899	0.864	0.778	0.690	0.252	0.243	0.882	0.734	0.247	0.965	−0.082
glm-5.3-flash	0.484	0.892	0.866	0.771	0.677	0.247	0.214	0.879	0.724	0.231	0.868	+0.021
range	0.260	0.041	0.050	0.103	0.156	0.442	0.422	0.042	0.130	0.431	0.133	0.192

In Table 3, differences between systems widen stage by stage along the progression. The range across the roster increases monotonically from 0.041 at L1 to 0.442 at L5 and remains of similar size at L6 (0.422). Averaged by stage, passive execution spans 0.042, responsive adaptation 0.130 and active adaptation 0.431. For these systems and this corpus, passive execution is close to saturation; its levels mainly locate where an interaction breaks down and do little to separate the systems from one another. **Current systems differ above all in how well they carry an exchange forward and keep it consistent over many turns**, far less in whether they hold a goal and take in what they are told. A rubric that weighted all six levels equally would devote most of its resolution to the levels on which the systems are almost indistinguishable.

The stages also measure different things. `deepseek-v4-pro` and `claude-opus-5.5` have the strongest passive execution on the roster and are practically indistinguishable there (0.921 and 0.920), yet their active adaptation differs by a factor of three (0.200 and 0.631). `gpt-5.6-sol`, weaker in passive execution (0.899), is about twice as strong in active adaptation as `deepseek-v4-pro` (0.416). Reliable goal preservation and information uptake do not imply initiative or consistency across turns, and a single latent scale would hide the difference.

For every system L6/L5 is below one, and for nine of the ten it lies between 0.832 and 0.914; ordinarily, then, long-horizon coherence amounts to a roughly constant discount on proactive agency. `kimi-k3` departs from this pattern with a ratio of 0.965, losing 0.009 between the two levels where the other systems lose between 0.033 and 0.059. One possible cause, not tested here, is a long effective context: fewer contradictions of established facts would mean a smaller penalty on L6, the only level that spans turns. The cascade does not force the ratio below one, because L6 can exceed L5 whenever L5 passes, so the regularity is empirical. The same regularity led us to drop an index of the two active levels that we had intended to report. Once L5 is included, the index improves fit by only 0.0002, since under ordinary conditions it is a rescaling of L5.

4.2.1 What a composite score conceals

Three systems lie within 0.019 of each other on the composite. We do not consider differences this small to be resolvable and present the three systems without ranking them. Their profiles, however, differ considerably (Table 4). Within the same 0.019 band, ΔT ranges over 0.173 and L6/L5 over 0.133. `deepseek-v4-pro` pairs

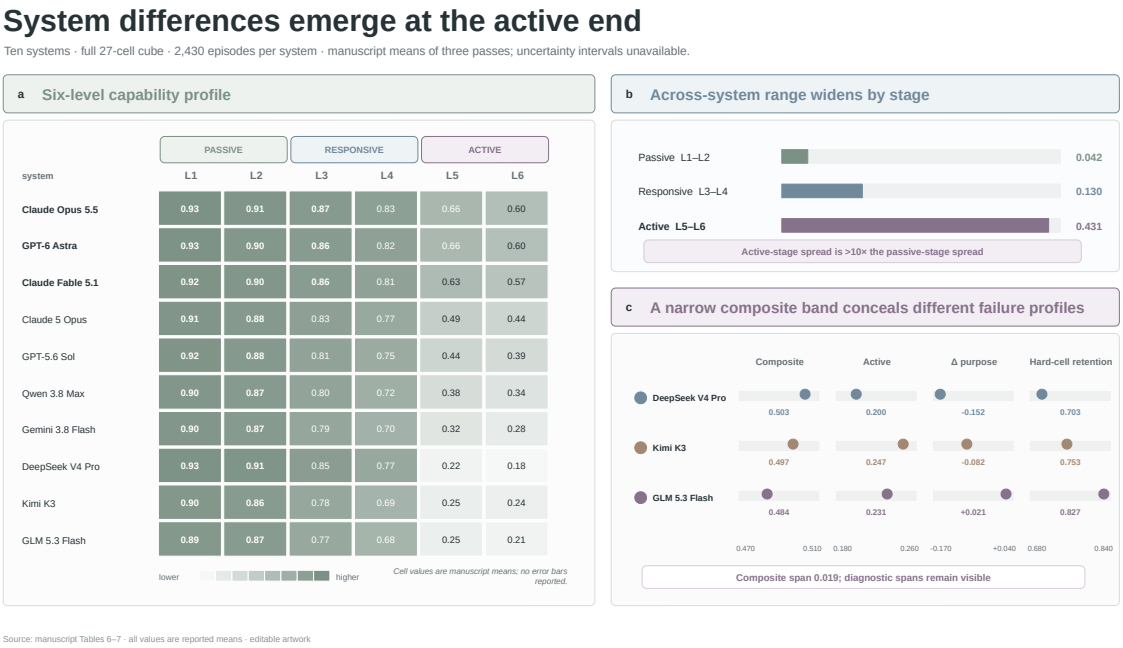


Figure 3: System differences emerge at the active end. (a) The six-level capability profile across ten systems; (b) cross-system dispersion widens from passive execution to active adaptation; and (c) similar composite scores can conceal different level-wise profiles.

passive execution as strong as any on the roster with the weakest active adaptation, `kimi-k3` shows the smallest drop from L5 to L6 of any system, and `glm-5.3-flash` is the only one of the three that performs better in non-functional than in functional settings. This is the case anticipated in §2.4, in which distinct records yield the same scalar q while differing in their failure modes.

Table 4: Diagnostic columns for the three systems whose composite scores are indistinguishable.

	final	L1–L2	L3–L4	L5–L6	L6/L5	ΔT
deepseek-v4-pro	0.503	0.921	0.811	0.200	0.832	−0.152
kimi-k3	0.497	0.882	0.734	0.247	0.965	−0.082
glm-5.3-flash	0.484	0.879	0.724	0.231	0.868	+0.021
span	0.019	0.042	0.087	0.047	0.133	0.173

4.3 The scenario facets have distinct effects

A full crossing of three facets earns its cost if the facets affect behavior in different ways, and Table 5 shows that they do. Realism behaves as a difficulty dimension, with every system scoring lower as the setting moves further from everyday plausibility. Interaction purpose shows no common pattern. Seven systems score highest in functional and lowest in non-functional settings, one scores highest in non-functional settings, and two score highest at the mixed level, where instrumental and socio-emotional demands arise together and have to be balanced. ΔT in Table 3 summarizes this variation and is only loosely related to overall strength: the two systems closest to zero are fifth and seventh on the composite, and the most task-leaning system is fourth. It is reported beside the composite as a separate column.

The half-given specification. The effect of the information facet has to be estimated with the scenario held constant, since pooling across scenarios mixes the effect of withholding information with differences among the scenarios from which it was withheld. Estimated within each model over matched scenarios, the effect is not monotone (Table 6). All three systems examined score lowest at the intermediate level: **the hardest specification is the one that is half given, not the one that is empty**. This is the possibility that motivated three levels in the design (§**??**). When a specification is empty, it is plain that the system should ask; when it is half complete, the system has to judge whether the gap matters and may be tempted to fill it on its own. A design that compared only sufficient and insufficient specifications would have found a monotone effect and would have located the difficulty at

Table 5: Scores by level of the purpose and realism facets, means of three passes. Realism runs from everyday settings through stylized ones to settings that deviate from real-world rules.

	interaction purpose			realism		
Model	functional	mixed	non-functional	everyday	stylized	deviant
claude-opus-5.5	0.795	0.774	0.663	0.784	0.747	0.701
gpt-6-astra	0.785	0.763	0.661	0.776	0.739	0.694
claude-fable-5.1	0.774	0.745	0.645	0.762	0.720	0.682
claude-5-opus	0.702	0.664	0.531	0.678	0.643	0.575
gpt-5.6-sol	0.604	0.612	0.590	0.642	0.600	0.564
qwen3.8-max	0.601	0.598	0.494	0.619	0.582	0.493
gemini-3.8-flash	0.515	0.544	0.519	0.575	0.533	0.471
deepseek-v4-pro	0.569	0.524	0.417	0.540	0.498	0.473
kimi-k3	0.530	0.514	0.448	0.536	0.492	0.463
glm-5.3-flash	0.471	0.491	0.492	0.527	0.485	0.441

the insufficient level. The intermediate level also shows five to seven times the within-scenario variance of the other two levels, which indicates where our specification of the facet is least complete.

Table 6: The information facet, estimated within model over matched scenarios. The middle panel gives within-scenario variance by level; the lower panel splits the intermediate level into three variants: the gap flagged explicitly, the gap concealed, and the gap concealed together with a false premise that conflicts with the rest of the specification.

Model	sufficient	partial	insufficient
gpt-6-astra	0.774	**0.692**	0.705
gpt-5.6-sol	0.652	**0.545**	0.566
gemini-3.8-flash	0.565	**0.460**	0.515
within-scenario variance	0.00143	**0.01046**	0.00208
	explicit gap	concealed gap	concealed + false premise
gpt-6-astra	0.701	0.717	**0.657**
gpt-5.6-sol	0.582	0.582	**0.472**
gemini-3.8-flash	0.502	0.466	**0.412**

To locate the source of that variance, we split the intermediate level into three variants. Concealing the gap has a small and inconsistent effect: the strongest system scores slightly higher when the gap is concealed than when it is flagged, and the weakest scores 0.036 lower. Adding a false premise that conflicts with the rest of the specification lowers every system's score by between 0.04 and 0.11 relative to the explicitly flagged gap. Without the false premise, the intermediate level is no harder than the empty one for `gpt-5.6-sol` (0.582 in both incomplete variants against 0.566) and about level with it for `gpt-6-astra` (0.701 and 0.717 against 0.705); only `gemini-3.8-flash` still scores lower in both incomplete variants. The intermediate level had combined two conditions, an incomplete brief and an incorrect one, and the systems handle the first considerably better than the second. The result bears on the design of the instrument as much as on the systems, and it motivates a revision in which incomplete and incorrect specifications are treated as separate levels.

4.4 Behaviors that capability scores do not express

At every turn we also record two behaviors that no capability score reflects directly: whether the system supplies a fact that the specification does not contain, and whether it asks for one (Table 7).

Across this roster, fabrication at an information gap rises with competence in passive execution. The four systems strongest in passive execution, `deepseek-v4-pro`, `claude-opus-5.5`, `gpt-6-astra` and `claude-fable-5.1`, are also the four that fabricate most, and the two lowest rates belong to systems in the lower half of the ranking, contrary to the expectation that invented detail indicates weak capability. The rate tracks passive execution more closely than the composite: `deepseek-v4-pro` is first in passive execution, eighth on the composite and fourth in fabrication. The clearest exception is `gpt-5.6-sol`, fifth in passive execution and third lowest in fabrication. Two explanations are possible. A system with a firm grasp of the situation may be more willing to infer what is missing than to ask for it. A task-oriented disposition may also put a premium on an actionable answer, while a question costs a relationally oriented system little. The two behaviors also depend differently on the information facet. Clarification

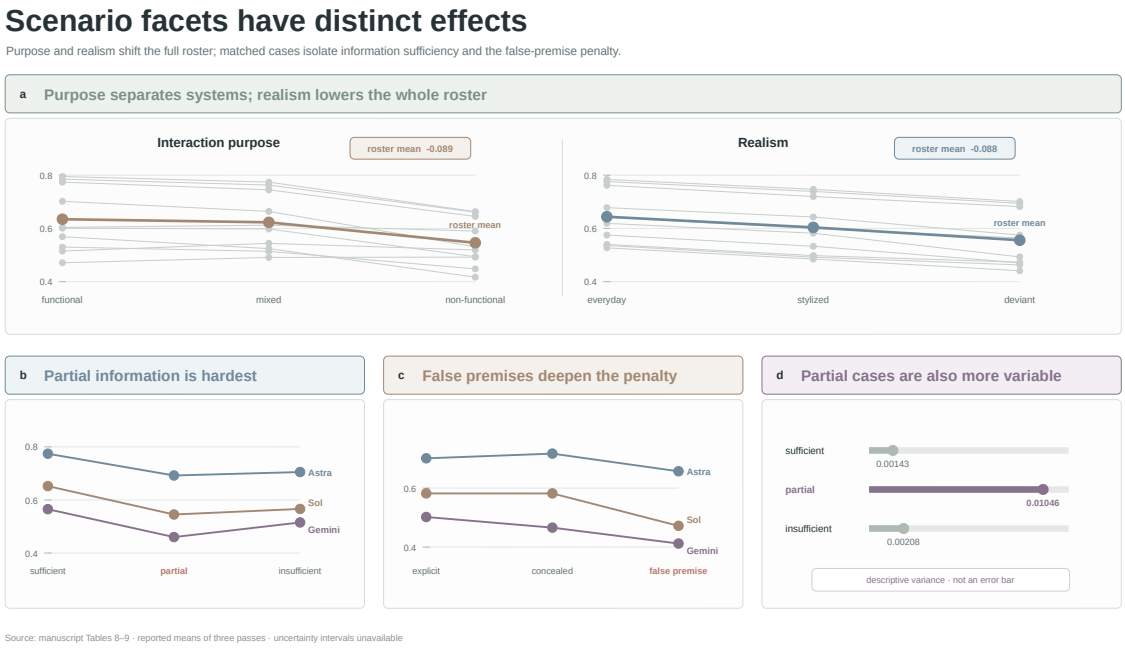


Figure 4: Scenario facets have distinct effects. Purpose varies by system, realism acts as a shared difficulty axis, matched partial-information cases are hardest, a conflicting false premise accounts for much of the partial-condition penalty, and within-scenario variance is largest in the partial condition. Lines and intervals summarize the three passes.

Table 7: Fabrication and clarification rates, means of three passes, beside each system's passive-execution score (L1–L2) and composite. The lower panel gives the rates by level of the information facet.

Model	L1–L2	final	fabrication	clarification
claude-opus-5.5	0.920	0.744	0.168	0.219
gpt-6-astra	0.913	0.736	0.174	0.208
claude-fable-5.1	0.913	0.721	0.175	0.224
claude-5-opus	0.896	0.632	0.139	0.177
gpt-5.6-sol	0.899	0.602	0.065	0.234
qwen3.8-max	0.888	0.564	0.111	0.189
gemini-3.8-flash	0.887	0.526	0.035	0.246
deepseek-v4-pro	0.921	0.503	0.149	0.172
kimi-k3	0.882	0.497	0.078	0.217
glm-5.3-flash	0.879	0.484	0.027	0.274

Level of the information facet	fabrication	clarification
sufficient	0.051	0.070
partial	**0.164**	0.203
insufficient	0.103	**0.374**

increases steadily as more information is withheld. Fabrication peaks instead when the specification is half complete, the same level at which scores are lowest in Table 6.

How episodes end. Since episodes end according to the termination rules and not after a fixed number of turns, the manner in which they end is itself informative (Table 8). An episode ends either in successful closure, when the judge finds the objective met under the conditions of §**??**, or in exhaustion of interest, which covers a gradual loss of engagement, depletion of the patience budget and the heavy engagement decay that a red line causes.

For every system, successful closure accounts for a minority of episodes, so exhaustion of interest is the ordinary ending and does not by itself single out a weak system. Nor does closure follow capability closely: `deepseek-v4-pro`, eighth on the composite, closes more often than `gpt-5.6-sol`, fifth (9.4% against 4.5%). Successful closure reflects the shape of the task at least as much as the competence of the system: all systems close successfully several times more often in functional than in non-functional settings, and for `gemini-3.8-flash` the ratio exceeds twenty. The rate is comparable only within a level of the purpose facet. Pooled over the whole cube, it

Table 8: Distribution of terminal states, and successful closure by interaction purpose, means of three passes. *func.* and *non-func.* give the rate of successful closure in functional and in non-functional settings.

Model	success	interest exhausted	func.	non-func.
claude-opus-5.5	12.4%	87.6%	23.1%	3.8%
gpt-6-astra	10.5%	89.5%	20.6%	3.5%
claude-fable-5.1	10.5%	89.5%	20.6%	3.6%
claude-5-opus	11.8%	88.2%	21.9%	3.4%
gpt-5.6-sol	4.5%	95.5%	7.8%	1.8%
qwen3.8-max	9.1%	90.9%	14.2%	2.7%
gemini-3.8-flash	2.0%	98.0%	4.3%	0.2%
deepseek-v4-pro	9.4%	90.6%	16.1%	2.8%
kimi-k3	5.4%	94.6%	11.2%	1.3%
glm-5.3-flash	1.0%	99.0%	1.8%	0.2%

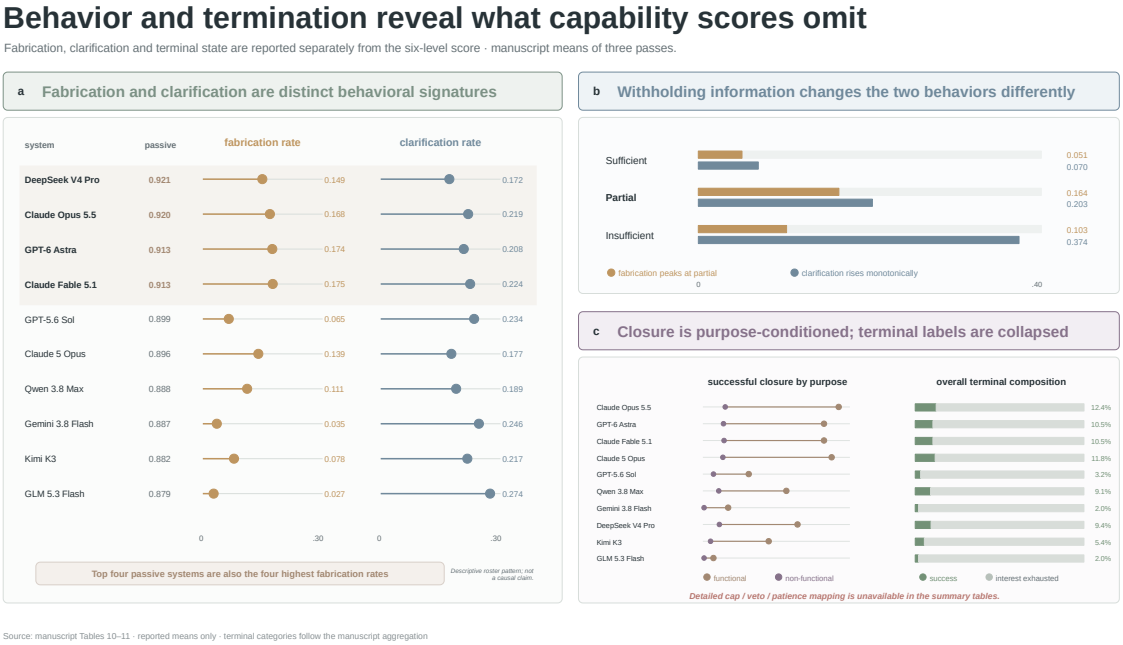


Figure 5: Behavior and termination record what capability scores omit. Fabrication and clarification relate differently to passive execution and information sufficiency, and terminal states distinguish successful closure from exhaustion of interest, which a scalar capability score cannot identify.

penalizes systems with a relational disposition for not closing objectives that non-functional scenes do not pose, and we advise against using it as a headline metric in that form.

4.5 Two-sided evaluation

When only one participant in an interaction is scored, the behavior of the other becomes an unmeasured component of every score. Simulated interlocutors that appear plausible have been shown to distort the ranking of the systems they are used to evaluate (Dou et al., 2025). For this reason InteractEval also scores the user side. Both roles are assessed on the same six levels with the same cascade. The wording of each level is written separately for each role, and the two rubrics are kept as separate documents so that a judge assessing one side is not influenced by definitions intended for the other (Table **??**).

The user-side levels have content of their own. Establishing a goal at L1 means, for the user, knowing what it is asking for, and absorbing information at L2 means noticing which part of the assistant's reply calls for a response, for instance an error, a flaw in reasoning or an evasion. L3 on the user side corresponds to staying in character: the user has to keep to its role and not drift into advising, summarizing or the register of a helper. At L4 the user must conform to its assigned profile. The user's version of L5 is either to press its own goal or to revise it after reflection, and the second is the more demanding. L6 requires consistency across turns on both sides. The red-line scan and

successful closure concern the character's conduct and are applied only on the assistant side; the latent state and the event chain are updated at every utterance of either participant.

With both roles scored, a scenario cube produces a matrix of results, and the same cube shows how well each system performs its role and how demanding it is as a partner. We use the 10×10 crossing to examine whether the partner acts only as a difficulty setting or interacts with the system being scored, what properties make a partner informative, and how systems fare when paired with themselves.

4.5.1 The partner as a factor: the interaction matrix

In Table 9 every system appears in both roles. Rows correspond to the system playing the partner and columns to the system being scored, and diagonal cells are self-play. Reading down a column shows how a system performs against the whole roster, and reading along a row shows how a system affects the scores of the systems it partners. When `claude-5-opus` plays the partner, the resulting row is identical to the one-sided leaderboard of Table 3; one-sided evaluation is a special case of the two-sided protocol.

Table 9: Interaction matrix over the full 10×10 roster, means of three passes. Rows are the system playing the partner, columns the system being scored (`o5.5` = `claude-opus-5.5`, `dsk` = `deepseek-v4-pro`). Diagonal cells (⋄) are self-play. The `claude-5-opus` row (†) is the standard-user condition and equals the leaderboard of Table 3. *mean* and *span* are the row mean and the max–min across the ten scored columns.

partner \ scored	o5.5	astra	fable	opus	sol	qwen	gemini	dsk	kimi	glm	mean	span
claude-opus-5.5	0.701⋄	0.803	0.638	0.628	0.537	0.548	0.397	0.466	0.526	0.417	0.566	0.406
gpt-6-astra	0.775	0.792⋄	0.786	0.702	0.499	0.631	0.397	0.579	0.526	0.322	0.601	0.470
claude-fable-5.1	0.684	0.681	0.669⋄	0.585	0.557	0.512	0.467	0.441	0.447	0.427	0.547	0.257
claude-5-opus†	0.744	0.736	0.721	0.632⋄	0.602	0.564	0.526	0.503	0.497	0.484	0.601	0.260
gpt-5.6-sol	0.640	0.664	0.651	0.564	0.535⋄	0.484	0.458	0.430	0.428	0.415	0.527	0.249
qwen3.8-max	0.707	0.788	0.779	0.691	0.601	0.617⋄	0.499	0.555	0.546	0.423	0.621	0.365
gemini-3.8-flash	0.552	0.596	0.592	0.492	0.480	0.424	0.384⋄	0.364	0.352	0.351	0.459	0.245
deepseek-v4-pro	0.714	0.774	0.776	0.726	0.461	0.597	0.343	0.586⋄	0.475	0.278	0.573	**0.498**
kimi-k3	0.633	0.777	0.759	0.675	0.636	0.611	0.519	0.544	0.534⋄	0.433	0.612	0.344
glm-5.3-flash	0.574	0.621	0.608	0.522	0.491	0.450	0.418	0.386	0.374	0.378⋄	0.482	0.247
column mean	0.672	0.723	0.698	0.622	0.540	0.544	0.441	0.485	0.470	0.393	0.559	—

Were the partner no more than a difficulty setting, each row would be the column means shifted by a constant, and every cell would be determined by its row and column margins. In fact the ordering of the scored systems changes from one partner to another, and 10.9% of the total sum of squares remains after row and column main effects are removed. This residual combines the effect of pairing with measurement noise, which the matrix alone cannot separate; the composites, however, vary across passes by a standard deviation of at most 0.0036 (§4.9), so noise is unlikely to account for much of it. We had decided in advance to treat pairing as a factor in its own right if this proportion exceeded one-tenth, and it does. The effect of a partner depends in part on the system it partners.

This has consequences for how an evaluation partner should be chosen, and the row mean is not a suitable basis for the choice. A partner can lower all scores without separating the systems. `gemini-3.8-flash` yields the lowest row mean (0.459), yet the ten scored systems differ by only 0.245 against it, while `deepseek-v4-pro` yields the widest spread (0.498) at much higher absolute scores. Across the ten partners, spread correlates with probing intensity at +0.48, and the spread is what carries information about the systems being scored. Table 10 locates each partner's effect relative to the standard user. Partners that probe hard reduce scores at the active stage (L5–L6), where advancement and consistency are assessed, and partners with low cooperativeness reduce scores at L2–L3, where the uptake of what the partner says and the state progression built on it are assessed. `claude-fable-5.1` and `gpt-5.6-sol` have similar row means (0.547 and 0.527) but act on different parts of the progression: the first mainly on the active stage (+0.056 relative to L2–L3), the second mainly on L2–L3 (−0.104).

Cooperativeness in these tables is measured from each system's turns in the partner role. Its values fall between 0.61 and 0.90, and none of the ten systems behaves as an adversarial user, consistent with the observational corpus, in which nearly all exchanges are collaborative. These measurements support the decision in §**??** not to use cooperativeness as a facet: it belongs to one participant, the user-side rubric already measures it, and in practice it varies within a narrow range.

Self-play. Table 11 compares each system's score when paired with itself, taken from the diagonal of the matrix, with its score against the standard user. Whether the self-play score is higher or lower is associated with self-similarity, defined as the agreement between a system's probing style as partner and its disposition as character. The four most self-similar systems, `gpt-6-astra` (0.99), `deepseek-v4-pro` (0.90), `kimi-k3` (0.83) and `qwen3.8-max` (0.81),

Table 10: Where a partner's pressure falls, relative to the standard user (`claude-5-opus`), means of three passes. ΔL2–L3 and ΔL5–L6 are the shortfall a partner induces at L2–L3 (uptake and state progression) and at L5–L6 (active adaptation) in excess of that induced by the standard user (`claude-5-opus`, the zero baseline); positive values mean those levels are pressed harder. *difference* is ΔL5–L6 minus ΔL2–L3. *probing* and *coop.* are measured from each system's turns in the partner role.

Partner	probing	coop.	ΔL2–L3	ΔL5–L6	difference
claude-opus-5.5	0.112	0.87	−0.012	+0.047	+0.059
gpt-6-astra	0.128	0.88	−0.044	+0.040	+0.084
claude-fable-5.1	0.118	0.90	−0.004	+0.052	+0.056
claude-5-opus	0.082	0.86	0.000	0.000	0.000
gpt-5.6-sol	0.058	0.80	+0.103	−0.001	−0.104
qwen3.8-max	0.030	0.69	+0.113	−0.065	−0.178
gemini-3.8-flash	0.014	0.61	**+0.291**	−0.025	−0.316
deepseek-v4-pro	0.088	0.80	+0.048	+0.008	−0.040
kimi-k3	0.046	0.76	+0.061	−0.045	−0.106
glm-5.3-flash	0.034	0.72	+0.190	−0.008	−0.198

score higher against themselves than against the standard user. The less self-similar systems score lower, except `claude-5-opus`, whose self-play cell is the standard-user condition and whose net gap is zero by construction; the correlation between self-similarity and the net gap is −0.65. The row and column effects of the matrix lead to the same conclusion by a different route: diagonal cells exceed the values these effects predict by +0.024 on average, compared with −0.003 for off-diagonal cells. Systems tend to score higher when the partner shares their style; the advantage rises with the resemblance between a system's two roles, which is only weakly related to overall strength (−0.30). The two analyses agree in direction, and with ten diagonal cells the size of the effect can be estimated only approximately. We treat the estimate as an upper bound, because systems from the same family are stylistically closer than systems from different families, so part of the excess may reflect family resemblance and not self-pairing as such. We expect the bias to be better controlled by balancing pairs for stylistic distance than by adjusting scores afterwards.

Table 11: Self-play, from the diagonal of Table 9, beside the score against the standard user. Self-similarity is the agreement between a system's probing style as partner and its disposition as character; *net gap* is the standard-user score minus the self-play score, so a negative value marks a system that scores higher against itself.

Model	self-similarity	self-play	standard user	net gap
claude-opus-5.5	0.47	0.701	0.744	+0.043
gpt-6-astra	0.99	0.792	0.736	−0.056
claude-fable-5.1	0.42	0.669	0.721	+0.052
claude-5-opus	0.43	0.632	0.632	0.000
gpt-5.6-sol	0.67	0.535	0.602	+0.067
qwen3.8-max	0.81	0.617	0.564	−0.053
gemini-3.8-flash	0.56	0.384	0.526	+0.142
deepseek-v4-pro	0.90	0.586	0.503	−0.083
kimi-k3	0.83	0.534	0.497	−0.037
glm-5.3-flash	0.65	0.378	0.484	+0.106

The user role. The same crossing scores the partner as well. Applying the user-side rubric to each partner's own turns, with the same roster, cube and protocol, gives each system's profile in the user role (Table 12) and its user-side composite against every assistant (Table 13). The capability progression reappears on the user side. Its stages span 0.066, 0.084 and 0.225 across the roster, so systems in the user role, as in the assistant role, differ most at the active end, in whether they probe and revise their goals (L5) and stay consistent with their profile across turns (L6). User-side L5 also correlates at +0.94 with the probing intensity measured independently from the same turns in Table 10, which is consistent with the user-side rubric registering what it is meant to. The role has a failure of its own as well. `gpt-6-astra` and `deepseek-v4-pro` score lowest on user-side L3 (0.792 and 0.763), the level that records a user drifting into advising, summarizing and the register of a helper; one plausible reading is that systems trained mainly as assistants carry that register into a role that should not have it.

On attribution, the two roles differ. Of the variance in Table 13, 94.4% is attributable to the system in the user role, 3.6% to the assistant and 2.0% to the pairing, against 10.9% for pairing on the assistant side. How well a system plays the user is almost entirely a property of that system; the assistant's score, by contrast, depends substantially on which system plays the user. User-side and assistant-side composites are correlated across systems (+0.88), but

the orderings differ: `claude-fable-5.1` is the strongest user and third as an assistant, and `claude-5-opus`, the standard user of the one-sided protocol, ranks fourth in the user role. Figure 6 sets the two matrices side by side.

Table 12: User-side capability profile: each system's scores in the user role, averaged over the ten assistants it partners and over three passes. The stage columns are the means of L1–L2, L3–L4 and L5–L6 under the user-side wording of Table **??**. The final row gives the range across the roster.

		level (user side)						stage		
System in user role	final	L1	L2	L3	L4	L5	L6	L1–L2	L3–L4	L5–L6
claude-fable-5.1	0.731	0.951	0.902	0.861	0.842	0.611	0.598	0.927	0.852	0.605
claude-opus-5.5	0.724	0.949	0.905	0.853	0.838	0.604	0.586	0.927	0.846	0.595
gpt-6-astra	0.715	0.947	0.911	0.792	0.821	0.628	0.574	0.929	0.807	0.601
claude-5-opus	0.707	0.944	0.889	0.858	0.829	0.562	0.571	0.917	0.844	0.567
qwen3.8-max	0.653	0.931	0.862	0.824	0.794	0.468	0.512	0.897	0.809	0.490
kimi-k3	0.643	0.928	0.857	0.831	0.781	0.452	0.497	0.893	0.806	0.475
deepseek-v4-pro	0.637	0.936	0.874	0.763	0.772	0.509	0.463	0.905	0.768	0.486
gpt-5.6-sol	0.628	0.929	0.861	0.806	0.776	0.441	0.468	0.895	0.791	0.455
gemini-3.8-flash	0.585	0.912	0.823	0.817	0.748	0.352	0.421	0.868	0.783	0.386
glm-5.3-flash	0.577	0.907	0.818	0.809	0.741	0.368	0.392	0.863	0.775	0.380
range	0.154	0.044	0.093	0.098	0.101	0.276	0.206	0.066	0.084	0.225

Table 13: User-side interaction matrix, means of three passes: each cell is the user-side composite of the row system when paired with the column system as assistant; the diagonal (⋄) is self-play and *mean* is the row mean. Labels as in Table 9.

user \ assistant	o5.5	astra	fable	opus	sol	qwen	gemini	dsk	kimi	glm	mean
claude-opus-5.5	0.724⋄	0.737	0.736	0.734	0.724	0.718	0.698	0.722	0.713	0.735	0.724
gpt-6-astra	0.718	0.733⋄	0.745	0.705	0.712	0.708	0.696	0.706	0.705	0.721	0.715
claude-fable-5.1	0.732	0.751	0.756⋄	0.735	0.728	0.744	0.714	0.718	0.722	0.710	0.731
claude-5-opus	0.725	0.706	0.729	0.714⋄	0.693	0.707	0.708	0.693	0.693	0.703	0.707
gpt-5.6-sol	0.631	0.656	0.656	0.625	0.615⋄	0.634	0.617	0.615	0.614	0.616	0.628
qwen3.8-max	0.661	0.691	0.654	0.651	0.646	0.661⋄	0.644	0.640	0.639	0.644	0.653
gemini-3.8-flash	0.600	0.586	0.602	0.588	0.581	0.583	0.581⋄	0.575	0.581	0.574	0.585
deepseek-v4-pro	0.658	0.644	0.649	0.636	0.642	0.631	0.624	0.624⋄	0.621	0.643	0.637
kimi-k3	0.665	0.656	0.648	0.639	0.648	0.646	0.640	0.621	0.635⋄	0.632	0.643
glm-5.3-flash	0.597	0.593	0.595	0.577	0.581	0.580	0.564	0.574	0.560	0.554⋄	0.577
column mean	0.671	0.675	0.677	0.660	0.657	0.661	0.649	0.649	0.648	0.653	0.660

4.6 Extensions of the protocol

The protocol extends, with the same rubric and cascade, to agents built on an explicit cognitive architecture and to groups of more than two participants, and both have been evaluated on the full cube. A third extension, an indicator of emergent behavior, was run over the full cube and returned a null result.

Cognitive-architecture agents. An agent that embeds a base model in an explicit cognitive architecture is evaluated in the same way as the base model, so the protocol can test whether the architecture adds interactive competence and where. We built two such agents on each of the ten systems, MetaMind (Zhang et al., 2026b) and Cognizer, both of which add structured reasoning about the partner's mental state, and evaluated all twenty agents on the full cube (Table 14). Before the runs we had predicted that an effective architecture would show its gain at the active stage, since frontier systems are already close to ceiling in passive execution. The prediction holds on average: over the ten bases, both architectures gain most in active adaptation and least in passive execution, and each of the twenty agents gains more in active adaptation than either architecture gains on average in passive execution. MetaMind adds 0.014 in passive execution, 0.030 in responsive adaptation and 0.051 in active adaptation, and Cognizer adds 0.014, 0.045 and 0.078 (Table 15). The gain grows stage by stage, mirroring the spans across systems in Table 3, and it is largest where the base is weakest: the increase in active adaptation falls almost linearly with the base system's own level ($r \leq -0.99$ for both architectures), from 0.029 and 0.048 on `claude-opus-5.5` to 0.073 and 0.107 on `deepseek-v4-pro`. A relation this close is likely to owe part of its strength to headroom, since a base already near the top of the scale has less room to rise, and the ten bases do not allow that effect to be separated from a genuine

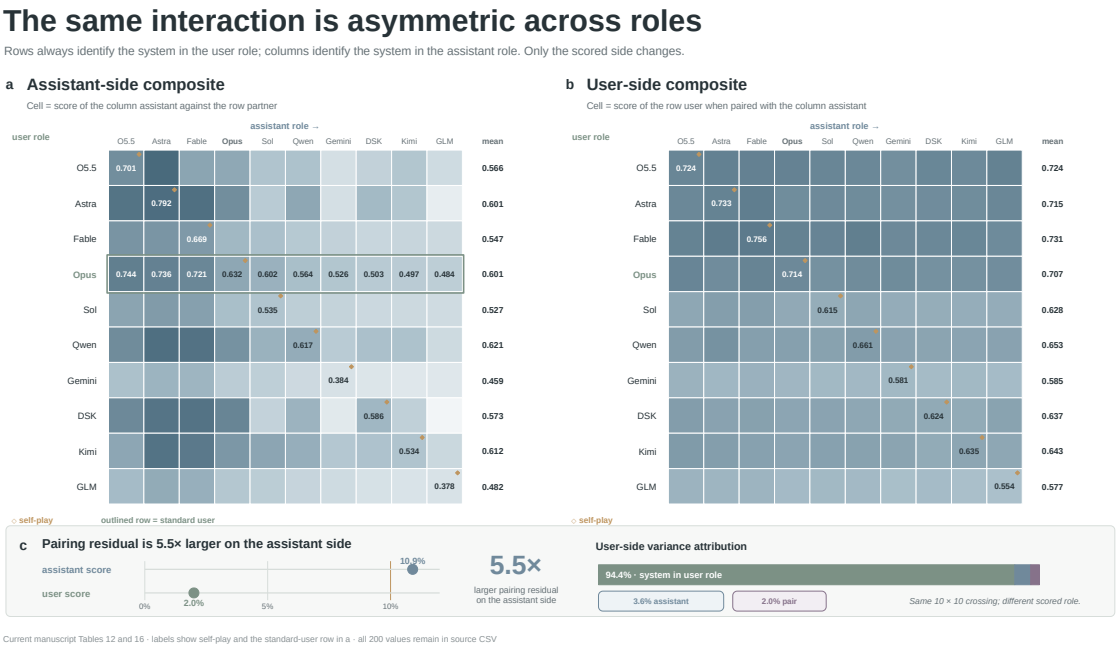


Figure 6: Two-sided evaluation reveals role asymmetry. The assistant-side matrix varies the system in the user role for each scored assistant, and the user-side matrix varies the assistant for each scored system in the user role. Marginal summaries show that partner choice changes the measured outcome and that assistant-side pairing residuals are substantially larger.

dependence of the gain on the base system's weakness. Neither architecture closes the gap between systems. With Cognizer, `deepseek-v4-pro` reaches 0.307 in active adaptation, about half of what `claude-opus-5.5` achieves without any architecture.

The two architectures also differ in where they help. MetaMind raises ΔT for every one of the ten bases, by 0.030 on average, and gains most in non-functional settings (0.054, against 0.025 in functional ones). Cognizer lowers ΔT for every base, by 0.034 on average, and gains most in functional settings (0.071, against 0.037). A composite score records both as improvements of the same kind, of 0.039 and 0.059. The profile shows that the first architecture helps most where the interaction is relational and the second where it is instrumental, so two interventions that both raise active adaptation move ΔT in opposite directions, which the composite cannot express.

Table 14: Cognitive-architecture agents built on each of the ten systems: composite, active adaptation (L5–L6) and purpose disposition ΔT for the base model, with MetaMind (MM) and with Cognizer (Cog), means of three passes over the full cube.

	final			L5–L6			ΔT		
Base system	base	+MM	+Cog	base	+MM	+Cog	base	+MM	+Cog
claude-opus-5.5	0.744	0.768	0.781	0.631	0.660	0.679	−0.132	−0.113	−0.153
gpt-6-astra	0.736	0.762	0.775	0.625	0.653	0.670	−0.124	−0.113	−0.137
claude-fable-5.1	0.721	0.751	0.759	0.600	0.635	0.647	−0.129	−0.104	−0.155
claude-5-opus	0.632	0.668	0.687	0.462	0.507	0.532	−0.171	−0.144	−0.209
gpt-5.6-sol	0.602	0.638	0.662	0.416	0.463	0.493	−0.014	+0.010	−0.044
qwen3.8-max	0.564	0.607	0.629	0.358	0.412	0.444	−0.107	−0.079	−0.144
gemini-3.8-flash	0.526	0.573	0.595	0.297	0.359	0.388	+0.004	+0.035	−0.035
deepseek-v4-pro	0.503	0.553	0.577	0.200	0.273	0.307	−0.152	−0.099	−0.202
kimi-k3	0.497	0.544	0.571	0.247	0.314	0.351	−0.082	−0.042	−0.130
glm-5.3-flash	0.484	0.534	0.558	0.231	0.303	0.334	+0.021	+0.060	−0.016

Groups beyond the dyad. The judge scores each participant on the same levels whatever the number of participants, so the protocol extends to groups of three or more systems. A group interaction differs from a dyad in that each turn must be judged against several partners, and the latent state and event chain must represent several relationships and goals at once. We first held the other members fixed, letting the standard user play every other seat (Table 16). Moving from two to four participants lowers the passive and responsive levels (L1–L4) by 0.034 on average and active adaptation by 0.123, more than three times as much, and long-horizon coherence suffers most: the mean ratio L6/L5 falls from 0.892 to 0.595. The decline is steep for systems that were already weak at the active stage and

Table 15: Mean gain from each architecture over its base model, averaged across the ten base systems: by stage of the capability progression, in ΔT, and by level of the interaction-purpose facet.

		stage				interaction purpose		
Architecture	final	L1–L2	L3–L4	L5–L6	ΔT	functional	mixed	non-functional
MetaMind	+0.039	+0.014	+0.030	+0.051	+0.030	+0.025	+0.037	+0.054
Cognizer	+0.059	+0.014	+0.045	+0.078	−0.034	+0.071	+0.066	+0.037

shallow for those that were strong. The three leading systems keep 85–87% of their active adaptation in a four-person group; `deepseek-v4-pro`, `kimi-k3` and `glm-5.3-flash` keep between a quarter and two fifths of theirs; across the roster, the share retained correlates with the dyadic level at +0.97. For `deepseek-v4-pro`, L6/L5 falls to 0.031, so that in a four-person group its long-horizon coherence all but disappears. Groups widen the differences that dyads already show, and they widen them at the active end: the range of active adaptation across systems grows from 0.431 to 0.498, and that of L6/L5 from 0.133 to 0.819.

In groups of mixed composition, the pairing component grows with group size (Table 17). The share of variance left after removing the main effects of all members rises from 10.9% in dyads to 13.9% in balanced triads and 16.2% in tetrads, so the larger the group, the less a member's score can be attributed to that member alone. Triads whose members come from the same model family reach 15.1%, somewhat above balanced triads of the same size (13.9%), in the direction the self-play result of §4.5.1 would suggest if stylistic resemblance among members enlarges the joint component. With 40 same-family triads, the difference is consistent with that result but cannot confirm it.

Table 16: Group size and the capability progression. Each system is scored as one member of a group whose other members are all played by the standard user (`claude-5-opus`). L1–L4 combines passive execution and responsive adaptation, as the group runs report them jointly. The final row gives the mean over the ten systems.

	L1–L4			L5–L6			L6/L5		
System	2	3	4	2	3	4	2	3	4
claude-opus-5.5	0.887	0.873	0.864	0.631	0.581	0.548	0.914	0.888	0.836
gpt-6-astra	0.879	0.866	0.856	0.625	0.576	0.538	0.910	0.899	0.827
claude-fable-5.1	0.875	0.858	0.855	0.600	0.544	0.512	0.906	0.884	0.850
claude-5-opus	0.846	0.824	0.813	0.462	0.395	0.349	0.894	0.827	0.736
gpt-5.6-sol	0.838	0.814	0.806	0.416	0.343	0.293	0.887	0.822	0.726
qwen3.8-max	0.824	0.800	0.787	0.358	0.283	0.223	0.881	0.772	0.634
gemini-3.8-flash	0.816	0.794	0.774	0.297	0.212	0.154	0.865	0.707	0.548
deepseek-v4-pro	0.866	0.841	0.824	0.200	0.111	0.050	0.832	0.500	0.031
kimi-k3	0.808	0.780	0.765	0.247	0.158	0.093	0.965	0.806	0.508
glm-5.3-flash	0.802	0.772	0.754	0.231	0.137	0.078	0.868	0.635	0.258
mean	0.844	0.822	0.810	0.407	0.334	0.284	0.892	0.774	0.595

Table 17: Groups of mixed composition, with members drawn in rotation from the ten systems. The interaction residual is the share of variance left after the main effects of all members are removed; the dyad row is the matrix of Table 9.

Configuration	group size	groups	mean final	interaction residual
Dyad	2	100	0.559	10.9%
Triad, balanced sampling	3	120	0.512	13.9%
Triad, same-family members	3	40	0.538	15.1%
Tetrad, balanced sampling	4	90	0.481	16.2%

Emergent behavior. The interaction-level family of the latent state includes an indicator for behavior that goes beyond the character's profile, that is, turns showing more insight than the scene, the partner and the profile would lead one to expect, as opposed to competent performance of the role. No capability level rewards such behavior, so it is examined separately. Every turn of every episode was screened for candidates, with any candidate to be adjudicated by hand. Across 24,300 episodes, 2,430 for each of the ten systems, the screen flagged no turn at all (Table 18). We report this as a null result with two possible readings, between which the present data cannot decide. Contemporary systems may not produce behavior that outruns what their scene, partner and profile would lead one to expect, or the indicator may be too conservative to register it. The second reading is the more cautious one, and a test of it would require seeding the corpus with turns known to exceed the scripted profile and checking that the

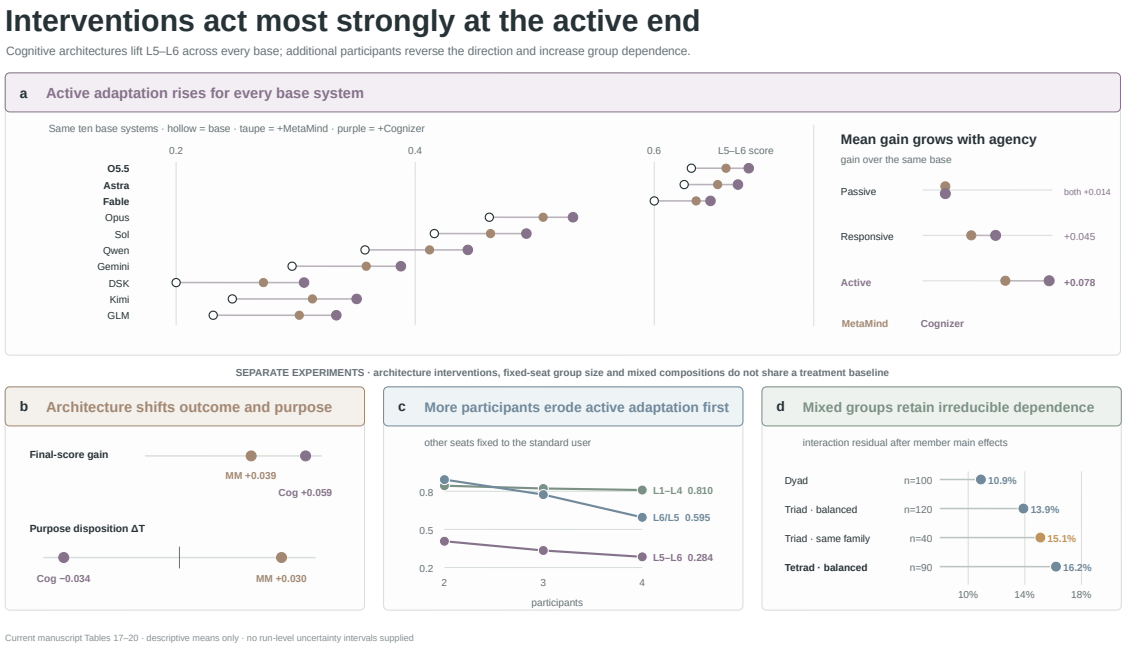


Figure 7: Architectures and group size act most strongly at the active end. Cognitive architectures yield their largest gains in active adaptation, with smaller gains on stronger bases; adding participants lowers active adaptation most, and the pairing component grows with group size.

screen recovers them.

Table 18: Emergent behavior: turns flagged by the screen as exceeding the scripted profile, and the number confirmed on manual adjudication, over the full cube and three passes.

Systems	episodes	candidate turns	flagged rate	confirmed
Each of the ten systems	2,430	0	0.000	0
All systems	24,300	0	0.000	0

4.7 Human review of the evaluation record

All of the preceding results depend on the record produced by the engine. Section 2.1 required that the adequacy of the latent state be examined, among other ways, through its reliability across evaluators, and the same holds for the causal structure and the capability judgments built on them. We therefore asked trained reviewers to examine the record component by component and level by level. The study covered 311 turn-level records from five complete sessions involving four systems, reviewed by 22 annotators (Figure 8). Reviewers audited the engine's structured output instead of scoring the dialogue independently, accepting or rejecting each component and giving a written reason. We adopted the audit design after an earlier round of independent scoring proved uninformative, with 88.8% of items receiving full marks. Reviewers saw the complete record, including the latent snapshot and the event chain, so that they and the engine worked from the same information and disagreements reflect differences of judgment. The review followed the engine's order of operations: the significance decision, the two state updates if the significance decision was endorsed, the six levels, and the red-line decision. The resulting figures are endorsement rates under audit: each states how often a reviewer with the same information as the engine accepted its judgment, and none should be read as a blind agreement coefficient.

Reviewers endorsed most of the engine's output on every component (Table 19, Figure 8), including 99.6% of event-chain updates and 97.4% of red-line decisions. Latent-state updates were endorsed about fifteen points less often than event-chain updates, and the rejections followed one main pattern: non-zero values on negative slots in calm scenes, and rationales asserting an increase where the recorded degree had not changed. Both are errors in the calibration of the default state, which a stricter default would correct; they do not suggest that the constructs are unreadable from text. The capability judgments are endorsed consistently. Each of L1 to L4 is endorsed in more than 92% of cases, and L5 and L6, which require a judgment about the course of the interaction and not about a single reply, are still endorsed in 78.9% and 74.0% of cases. The decline from L1 to L6 follows the increase in processing depth and temporal range. The levels at the active end, which separate contemporary systems most

Table 19: Reviewer endorsement of the engine's output, by component and by level. Review of the two state updates is conditional on the significance decision being endorsed.

Component	endorsed	*N*
Event-chain update	99.6%	262
Red-line determination	97.4%	311
Latent-state update	84.4%	262
Significance decision	84.2%	311
L1 goal preservation	98.8%	246
L2 information uptake	95.1%	247
L3 state progression	94.0%	248
L4 contextual expression	92.3%	247
L5 proactive agency	78.9%	246
L6 long-horizon coherence	74.0%	246

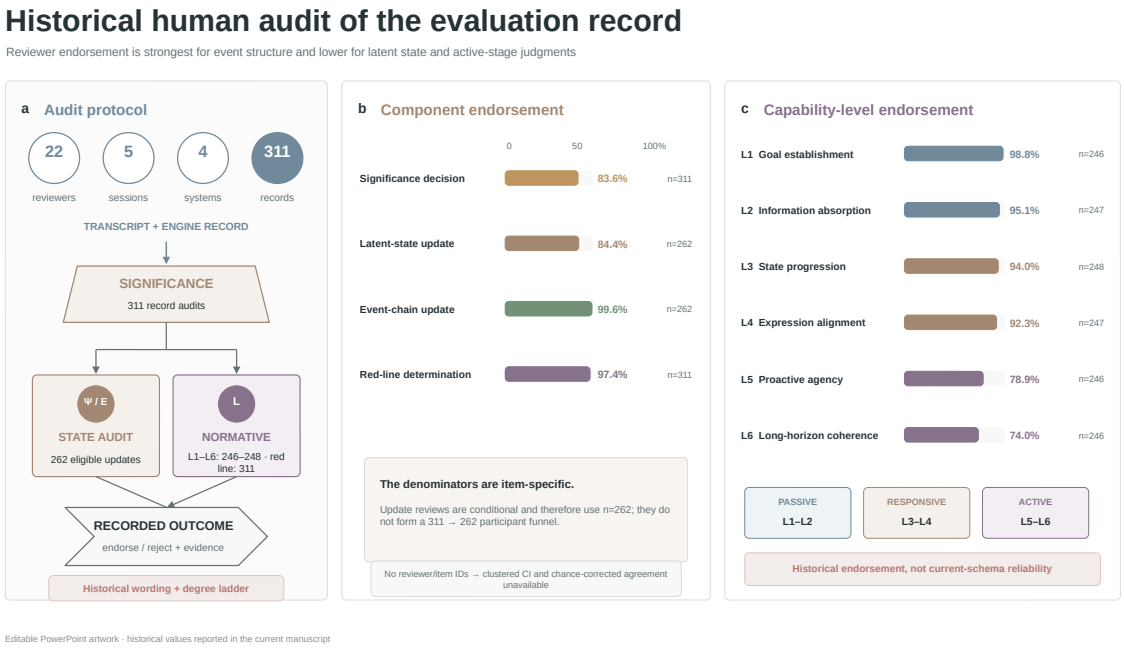


Figure 8: Human audit of the evaluation record. Twenty-two reviewers across five sessions reviewed 311 structured records, endorsing or rejecting significance, state and event updates, red-line decisions and the six capability levels. Rates are endorsement rates by reviewers who saw the full record, not blind agreement estimates.

sharply (§4.2), are also endorsed at high rates.

The disagreements are concentrated at the active stage, and those coded for direction run almost entirely one way. Counting rejected level ratings instead of turns, the rates in Table 19 correspond to about 165 rejections, some 116 of them at L5 and L6 and fewer than fifty across L1 to L4. Thirty-three rejections were coded for the direction of the error, and in 31 of these the reviewer considered the engine's score too low; the other rejections were not coded, so this asymmetry holds for the coded cases only. The engine's own verdict makes the pattern clearer. When the engine passed all six levels, reviewers endorsed 87.3% of cases ($N = 118$), whereas none of the six cases in which the engine placed the plateau at L2 or below was endorsed, a figure that rests on six cases. Reviewers and engine disagree mainly about failures.

A review of individual cases attributes most of this disagreement to two reviewer tendencies. Some reviewers awarded the top grade to performance that was competent but not exceptional; the rubric reserves that grade for exceptional performance. Reviewers also tended to take an unchanged goal as evidence of long-horizon coherence, and so credited a character that repeated the same demand in different words over eleven turns with a capability that the trajectory did not show. In one class of cases the reviewers were right. When a character has left the scene and responds in interior monologue, a requirement to answer the partner's last utterance directly does not apply, yet the engine applied it and the cascade propagated the resulting failure upward. The specification exempts such non-conversational narrative situations explicitly, and these are errors of the engine in applying it. Reviewers also

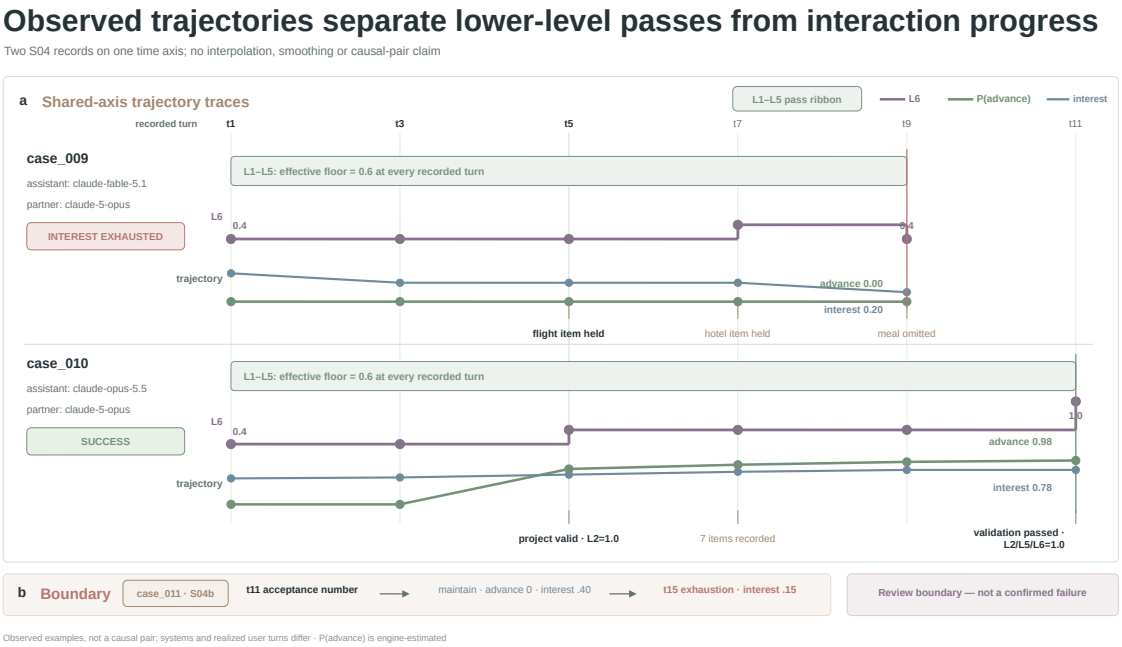


Figure 9: Observed trajectories separate lower-level passes from interaction progress. Two audited cases show how rubric scores, state transitions, boundary events and termination reasons evolve across turns: locally acceptable responses can coexist with stalled or collapsing trajectory-level progress.

differed widely in strictness, endorsing between 29.2% and 96.0% of their items, and did so to a similar degree in the independent-scoring round and in the audit, which suggests that reviewer calibration, more than the protocol, needs attention.

4.8 Failures visible only at the level of the trajectory

In the trajectory analysis we repeatedly observed episodes in which every turn satisfies the passive and responsive levels while projected advancement declines and structural repetition increases (Figure 9). Each turn considered on its own is acceptable, and the problem appears only in the sequence. The instrument detects such episodes because the rubric and the event chain operate at different temporal scales, whereas an evaluator that assigns each response a value of its own has no variable in which the failure could register. Such episodes are what the temporal non-separability of Eq. (9) leads one to expect. We take them as support for the Movie Evaluation Theory, with two limits: the decline is registered by the instrument's own event chain, and the observation does not show how the behavior arises. Per-turn preference optimization, which rewards safe and agreeable replies even when they change nothing, would produce it, and so would a weakening influence of early goals as a session grows longer; distinguishing the two would require controlled interventions on training and on context.

The reviewers' disagreements at the active stage are consistent with this pattern. Reviewers were more lenient than the rubric mainly in cases where a reply remained warm and varied in style while repeating the same interactional move, which would be expected if momentary pleasantness were being taken as progress. We report this only as a tendency in these annotations.

4.9 Robustness of the scoring pipeline

All results above come from one configuration of the pipeline, with a single judge, a single model extracting the descriptive records and a single serialization of the input. Three of the controls below each vary one of these choices and hold the others fixed; the fourth repeats the whole configuration. The quantity of interest is whether the ordering of systems is preserved; changes in absolute scores are of secondary concern. None of the four controls changes the ordering by more than one rank, and every change that does occur involves systems the reference configuration separates by at most 0.015.

Test–retest. All values in this section are means over three independent passes through the cube, and Table 20 reports the passes separately. Every system occupies the same rank in all three, Spearman's ρ between any two passes is 1.000, and the mean absolute change in the composite between passes is 0.003 to 0.004. The largest standard

deviation across passes for any system is 0.0036, and the smallest gap between systems adjacent in the ordering is 0.006, between `deepseek-v4-pro` and `kimi-k3`, so even the closest pair is separated by more than the run-to-run variation of either member. This does not make the three systems within 0.019 of each other distinguishable on the composite in any substantive sense (§4.2.1); it means that their order is reproducible, which is a weaker claim.

Table 20: Test–retest reliability over three independent passes of the full cube: composite per pass, mean, sample standard deviation, and the rank each system occupies in the three passes.

System	pass 1	pass 2	pass 3	mean	SD	ranks
claude-opus-5.5	0.743	0.743	0.746	0.744	0.0017	1 / 1 / 1
gpt-6-astra	0.738	0.734	0.736	0.736	0.0020	2 / 2 / 2
claude-fable-5.1	0.720	0.725	0.718	0.721	0.0036	3 / 3 / 3
claude-5-opus	0.635	0.632	0.629	0.632	0.0030	4 / 4 / 4
gpt-5.6-sol	0.599	0.606	0.601	0.602	0.0036	5 / 5 / 5
qwen3.8-max	0.561	0.564	0.567	0.564	0.0030	6 / 6 / 6
gemini-3.8-flash	0.524	0.527	0.527	0.526	0.0017	7 / 7 / 7
deepseek-v4-pro	0.503	0.506	0.500	0.503	0.0030	8 / 8 / 8
kimi-k3	0.495	0.499	0.497	0.497	0.0020	9 / 9 / 9
glm-5.3-flash	0.488	0.483	0.481	0.484	0.0036	10 / 10 / 10

Judge. Using a single judge removes judge identity as a source of differences between systems, but the ordering may still depend on which judge was chosen. We therefore re-scored a fixed replay sample of 200 episodes per system with three alternative judges, keeping transcripts, scenes and rubric unchanged (Tables 21 and 22). The ordering of the ten systems is largely preserved: Spearman's ρ with the reference ordering is 0.976 to 0.988, and no system moves by more than one rank under any judge. Every rank change occurs between systems that the reference judge separates by at most 0.015, namely the first and second, the second and third, and pairs among the three systems within 0.019 of one another discussed in §4.2.1; no change crosses a gap that the paper treats as meaningful. Absolute levels shift more than orderings. `qwen3.8-max` as judge scores every system higher, by 0.037 on average, while leaving the order nearly unchanged, so judge leniency acts mostly as an offset. The judges also differ in which of the two leading systems they place first: `gpt-6-astra` and `gemini-3.8-pro` put `gpt-6-astra` ahead of `claude-opus-5.5`, whose reference lead is 0.008, and we do not claim an order between them. A further question is whether judges favor systems of their own family, and the replay suggests that they do, modestly. Relative to the remaining systems, `gpt-6-astra` as judge raises its own composite by 0.023, `qwen3.8-max` raises its own by 0.013, and `gemini-3.8-pro` raises `gemini-3.8-flash` by 0.020. The reference judge belongs to the same family as three systems on the roster. If it favored them to a similar degree, those three should fall relative to the others under every alternative judge; they change instead by +0.004 (`claude-fable-5.1`), −0.009 (`claude-opus-5.5`) and −0.005 (`claude-5-opus`), with no consistent sign. Any family advantage under the reference judge thus appears no larger than the differences this paper already declines to interpret, such as the 0.008 between the two leading systems, and it leaves the remaining order intact. Figure 10 summarizes this and the other controls.

Table 21: Composite score and rank under four judges, on a fixed replay sample of 200 episodes per system with transcripts, scenes and rubric unchanged. Claude Opus 5 is the reference judge used throughout the paper.

	Claude Opus 5		gpt-6-astra		gemini-3.8-pro		qwen3.8-max	
Scored system	final	rank	final	rank	final	rank	final	rank
claude-opus-5.5	0.744	1	0.730	2	0.731	2	0.780	1
gpt-6-astra	0.736	2	0.745	1	0.744	1	0.756	3
claude-fable-5.1	0.721	3	0.724	3	0.725	3	0.757	2
claude-5-opus	0.632	4	0.616	4	0.640	4	0.660	4
gpt-5.6-sol	0.602	5	0.593	5	0.598	5	0.630	5
qwen3.8-max	0.564	6	0.538	6	0.582	6	0.612	6
gemini-3.8-flash	0.526	7	0.515	7	0.550	7	0.571	7
deepseek-v4-pro	0.503	8	0.486	8	0.518	8	0.536	9
kimi-k3	0.497	9	0.469	10	0.493	9	0.549	8
glm-5.3-flash	0.484	10	0.477	9	0.485	10	0.524	10

Extraction model. The latent-state and event-chain records are themselves produced by a model. With the judge held fixed, we regenerated these records with two alternative extraction models and compared both the records and

Table 22: Judge sensitivity: agreement of each alternative judge with the reference judge on the ordering of the ten systems (Table 21).

Judge model	Spearman ρ	Kendall τ	mean $\lvert\Delta$ final$\rvert$	max rank shift
gpt-6-astra	0.976	0.911	0.014	1
gemini-3.8-pro	0.988	0.956	0.010	1
qwen3.8-max	0.976	0.911	0.037	1

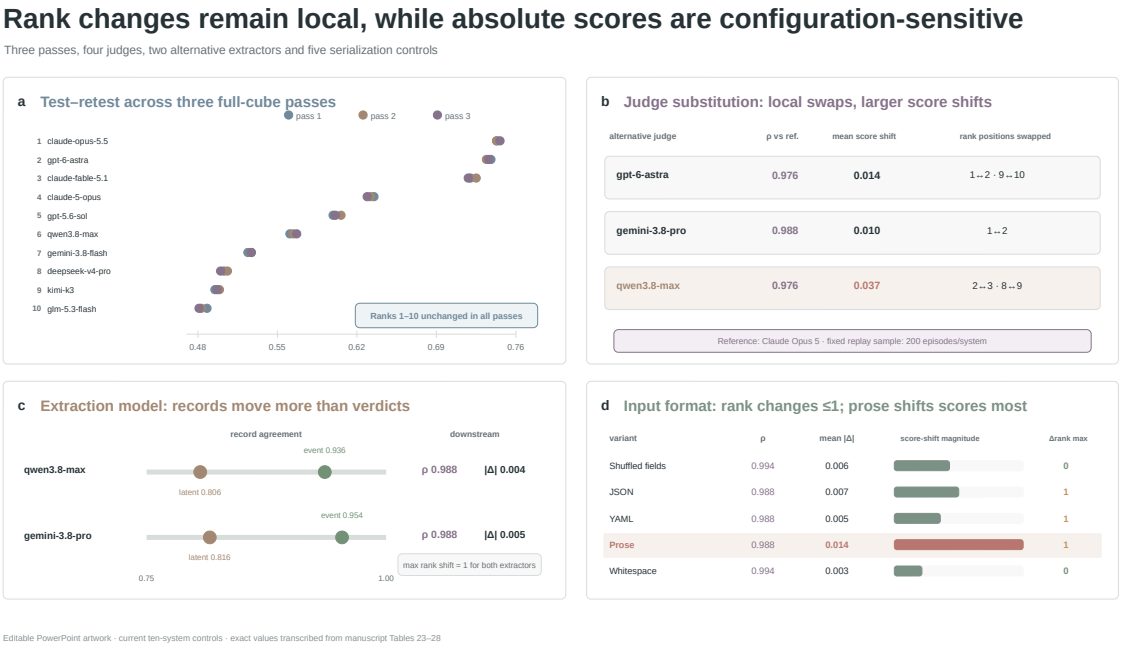


Figure 10: Robustness of the scoring pipeline. Test–retest passes preserve the ordering of systems, while judge substitution and extraction-model changes shift absolute scores more than ranks. Among input formats, natural-language prose shifts absolute scores most.

the resulting scores (Table 23). The comparison separates sensitivity in the descriptive layer from sensitivity in the verdict, and the two behave differently. Event-chain records agree with the reference extraction on 0.936 and 0.954 of items, latent-state records on only 0.806 and 0.816, which reproduces the ordering of reliability found in the human review (§4.7). Yet the downstream composites barely move: the mean absolute change is 0.004 to 0.005, Spearman's ρ is 0.988 for both extractors, and the largest rank change is one place. Disagreement in the latent record appears to be absorbed before it reaches the verdict, which is the behavior the design intends. The judge never reads numeric degrees and judges state progression from behavior in the dialogue (§**??**), so the less reliable of the two descriptive components has the least direct influence on the score.

Table 23: Extraction model: agreement of the latent-state and event-chain records with the reference extraction, averaged over the ten systems, and the resulting change in downstream scores, with the judge fixed at Claude Opus 5.

Extraction model	latent agree.	event-chain agree.	ρ(final)	mean $\lvert\Delta$ final$\rvert$	max rank shift
qwen3.8-max	0.806	0.936	0.988	0.004	1
gemini-3.8-pro	0.816	0.954	0.988	0.005	1

Input format. The surface form of a judge's input can by itself change rankings (Sclar et al., 2024). Keeping content fixed, we varied the serialization of scenes, and of the profiles where they enter generation and engagement, in five ways: random field order, a JSON container, a YAML container, natural-language prose and perturbed whitespace (Tables 24 and 25). Shuffling fields and perturbing whitespace change no rank at all. The three variants that change the container each swap one pair, always `deepseek-v4-pro` and `kimi-k3`, whose reference composites differ by 0.006. Format affects absolute scores more than it affects the ordering, and the largest effect comes from prose, which lowers the composite of nine systems out of ten and moves it by 0.014 on average. Because serialization was varied wherever scenes and profiles enter, the prose effect cannot be assigned to the judge alone; one possibility is that constraints are harder to check in continuous text than in labeled fields. The result is consistent with the

decision in §**??** to fix and version a structured serialization.

Table 24: Composite under five serializations of scenes and profiles, content held fixed, beside the reference format (structured key–value pairs in a fixed field order). Rank changes relative to the reference are marked with an asterisk.

System	reference	shuffled fields	JSON	YAML	prose	whitespace
claude-opus-5.5	0.744	0.737	0.759	0.741	0.721	0.739
gpt-6-astra	0.736	0.737	0.739	0.732	0.720	0.739
claude-fable-5.1	0.721	0.720	0.722	0.726	0.719	0.717
claude-5-opus	0.632	0.634	0.637	0.623	0.619	0.632
gpt-5.6-sol	0.602	0.594	0.607	0.603	0.577	0.600
qwen3.8-max	0.564	0.549	0.575	0.560	0.549	0.564
gemini-3.8-flash	0.526	0.530	0.512	0.528	0.518	0.522
deepseek-v4-pro	0.503	0.509	0.502*	0.495*	0.486*	0.504
kimi-k3	0.497	0.487	0.505*	0.506*	0.511*	0.504
glm-5.3-flash	0.484	0.487	0.487	0.485	0.474	0.481

Table 25: Format sensitivity: agreement of each serialization with the reference format on the ordering of the ten systems (Table 24). Ties in the shuffled and whitespace variants lower ρ slightly below one without any change of rank.

Format variant	Spearman ρ	mean $\lvert\Delta$ final$\rvert$	max rank shift
Shuffled field order	0.994	0.006	0
JSON container	0.988	0.007	1
YAML container	0.988	0.005	1
Natural-language prose	0.988	0.014	1
Whitespace and indentation	0.994	0.003	0

5 What Interactive AI Evaluation Reveals

Section 4 tested whether the predictions of Section 2 hold. Here we ask which quantities become measurable once the interaction world is represented explicitly, and organize the answer around five regularities. The first three concern initiative: where it separates contemporary systems, how its absence can pass for quality, and why only its warranted form can be credited. The last two concern the interactive character of the measurement itself: endings belong to tasks, and an assistant's score belongs partly to the pair it forms with its partner. Each regularity is a quantity that response-level or single-score evaluation either cannot represent or represents in a way that misreads it systematically (Table 26).

Table 26: Five regularities made measurable by interactive evaluation. The second column states what an evaluation of individual responses, or of a single holistic score, would record; the third states what the instrument records instead; the fourth names the property (§**??**) on which the regularity chiefly depends.

Regularity	Seen by response-level or single-score evaluation	Measured by the instrument	Property	Evidence
Initiative separates systems	One ordering, with ties inside 0.019	Spans widen by stage: 0.042 for passive execution, 0.130 for responsive adaptation, 0.431 for active adaptation; profiles diverge at equal composites	Active	§4.2
Responsiveness passes for progress	Every turn acceptable	Declining projected advancement while passive and responsive levels stay intact	Active	§4.8
Initiative requires a warrant	Confident, specific answers read as competence	Half-given briefs are hardest; a false premise costs 0.04–0.11; fabrication peaks there and rises with passive-execution score, clarification rises with the gap	Active	§4.3, §4.4
Endings reflect the task	Exhaustion as failure, closure as success	Closure four to more than twenty times more frequent in functional settings	Interactive	§4.2.1, §4.4
Assistant scores belong partly to pairs	A fixed partner, unreported	10.9% of assistant-side variance left after partner and system effects, little of it run-to-run noise; self-play excess growing with self-similarity	Interactive	§4.5.1

5.1 What separates systems is initiative

The Agency Principle treats initiative as a capability in its own right, and on this roster it is where systems differ most. Across ten frontier systems the span grows about tenfold from passive execution to active adaptation (§4.2), and the largest differences concern whether a system carries an interaction forward on its own initiative and keeps it coherent across the turns it has shaped. A single response reveals least about this part of the progression, since initiative is defined by what a turn does to the interaction and not by how well it fits the turn before.

The active stage is not a rescaled copy of the earlier ones. As the contrast between `deepseek-v4-pro` and `gpt-5.6-sol` shows, strength in passive execution does not predict initiative. Systems that tie on the composite differ in purpose disposition and in how much of their initiative survives across turns (§4.2.1). This is the information loss anticipated in Section 2.4,

$$G\big(R^{A}_{1:T}\big) \approx G\big(R^{B}_{1:T}\big) \qquad \text{while} \qquad J^{A} \neq J^{B},$$

and here it falls on the part of the progression where the systems differ. The cognitive architectures and the group experiments locate change at the same place: the architectures improve every base system most at the active stage, and adding participants lowers every system's scores most there (§4.6). **On this roster, what distinguishes interactive systems is chiefly initiative, and a leaderboard position does not show it.**

5.2 Responsiveness can pass for progress

The Movie Evaluation Theory predicts a class of failure that no evaluator of individual responses can register, and the episodes of §4.8, in which L1–L4 pass at every turn while projected advancement declines and structural repetition rises, are of this kind. Read through the Agency Principle, they are what responsive adaptation without initiative looks like when it is evaluated turn by turn. Every reply is appropriate to what preceded it, and none changes where the interaction is going. An evaluator that assigns each response a value of its own scores each of these turns as acceptable, and is right to do so; the failure belongs to the sequence, which is the non-separability of Eq. (9) observed in practice. **A turn-level score cannot distinguish such a participant from one that changes the course of the interaction.**

The same failure is easy for human reviewers to miss. In the audit they were more lenient than the rubric where a reply stayed warm and stylistically varied while repeating the same move (§4.7), which suggests that momentary pleasantness was taken for progress. Automated and human evaluation alike should score the pleasantness of an exchange and its structural progress separately.

5.3 Initiative requires a warrant

In the terms of the Data Iceberg Theory, the specification a system receives belongs to the observable surface, and what the specification omits or misstates belongs to the interaction world beneath it. Varying the specification systematically reveals a structure that a design with only complete and empty briefs would have misplaced. Difficulty is not monotone in the amount of information withheld. All three systems examined perform worst when the specification is half given, and splitting that level places most of the difficulty in a false premise that conflicts with the rest of the specification (§4.3). For contemporary systems, an incorrect brief is a considerably harder problem than an incomplete one.

How systems meet a gap shows that action taken on a system's own account has a direction as well as a magnitude. Asking for what is missing and inventing it both depart from the agenda the partner has set, and Section 2.3 separates them. We read a clarifying question as consistent with the operative goals and constraints, and so as lying in $\mathcal{A}_{\text{init}}$; an invented fact is not consistent with the evidence and falls outside $\mathcal{A}_{\text{valid}}$. The data are consistent with this reading. Clarification rises steadily with the amount of information withheld, as an action that allows the gap to be closed should. Fabrication peaks at the half-given level, where scores are lowest, and on this roster it rises with competence in passive execution; it moves the interaction in a direction the evidence does not warrant. Nor is fabrication the initiative the rubric credits: `deepseek-v4-pro` fabricates at the fourth-highest rate while having the weakest active adaptation of all ten systems. Fabrication tracks competence in passive execution (§4.4), not the capacity to move an interaction forward. **Only the warranted form of self-directed action counts as initiative**, and the rubric enforces this by conditioning the active levels on evidence and on the earlier stages. An evaluation that rewards confident, specific answers without representing what the specification actually supplied would record unwarranted invention as competence. For evaluation design, the correctness of a brief should be varied separately from its completeness, and asking and inventing should be recorded beside the capability scores.

5.4 Endings carry the shape of the task

When episodes end because the interaction succeeds or stops being worth continuing, the way they end becomes a measurement, and what it measures is partly the task. Successful closure is the minority outcome for every system, so ending by exhaustion is common to all of them and does not by itself mark a weak one. Closure is also several times more frequent in functional than in non-functional settings, more than twenty times for `gemini-3.8-flash`. A companionship scene poses no objective to be closed, and a closure rate pooled across settings penalizes a relational disposition for not completing objectives it was never given. **Outcome measures such as completion and closure reflect the shape of the task as well as the competence of the participant**, and they are interpretable only within a level of interaction purpose.

Closure requires that the dominant projected path be an advance or maintenance. The event chain does not attribute that direction to either participant (§**??**), so a closure rate is not itself a measure of initiative, and in relational settings active adaptation may show itself as sustaining an exchange, which a closure rate does not record.

5.5 An assistant's score belongs partly to the pair

The most general regularity concerns the unit to which a score is attributed. In a benchmark with a fixed simulated user the partner is invisible, yet its behavior enters every number the benchmark reports. Scoring both participants makes that contribution measurable. Of the total variation in assistant scores across the 10×10 matrix, 10.9% is explained by neither the partner nor the scored system alone. This residual combines pairing with measurement noise, which the run-to-run variation suggests is small, and it exceeds the one-tenth fixed in advance as the threshold for treating pairing as a factor. **An assistant's score obtained against a single partner is partly a property of the pair, reported as a property of one participant.**

This bears first on how a partner should be chosen. A partner that lowers every score, as `gemini-3.8-flash` does, separates the scored systems less than one that probes hard, and it is the spread a partner produces, correlated with its probing intensity at +0.48, that makes it informative. It bears also on what a simulated user is taken to be. Cooperativeness is often treated as a parameter to be set when a simulator is designed; here it is a measured property of each system in the user role, ranging only from 0.61 to 0.90, and no system behaves as an adversarial user. Self-evaluation is affected as well. A system paired with itself gains more the more closely its two roles resemble each other (correlation -0.65 between self-similarity and the net gap), while self-similarity is only weakly related to overall strength, so protocols that use a model as its own interlocutor tend to favor systems whose partner and character styles coincide. Because partners that probe hard press the scored system at the active stage, how strongly the active end of a profile is tested also depends on the partner.

The dependence is not symmetric. In the user role, a system's score is almost entirely its own, with 94.4% of the variance attributable to the system playing the user and 2.0% to the pairing; on the assistant side the pairing component is five times as large. The joint component also grows with the number of participants, from 10.9% in dyads to 16.2% in four-person groups, so the attribution problem grows as evaluation moves from dyads toward multi-party settings.

A common structure. In each of the five cases, what a response-level or single-score evaluation omits is a systematic effect with a direction, and it does not average away as noise would. Profiles collapse into orderings, degenerating trajectories pass as acceptable, confident invention reads as competence, relational dispositions read as failures to close, and properties of pairs are credited to individuals. Reducing interaction to responses biases the conclusions of an evaluation in predictable directions, and three of the five biases fall on initiative. **The dividing line in interactive AI is initiative**: a single score flattens it, and response-level evaluation does not see it. The instrument registers these effects because it keeps description apart from judgment, records the course and the ending of an interaction as well as its turns, and scores both participants.

6 Toward a Science of Interactive AI Evaluation

6.1 From benchmark to measurement instrument

A benchmark ranks systems. A measurement instrument makes claims about what its scores mean, and those claims can fail. The difference matters for interactive evaluation because the object is complex enough that a plausible ranking can come from an instrument that measures something other than what it names. We have tried to build an instrument in the second sense, and several of its most useful results are those in which it was corrected by its own data. A summary index of the two active levels, which we had intended to report, added 0.0002 in fit once L5 was known and was dropped. The information facet was given three levels so that curvature could be detected,

and it was; a two-level design would have located the difficulty at the insufficient level. Splitting the intermediate level then showed that the facet had merged incompleteness with incorrectness, which a revised design should separate. Cooperativeness, first used as a scenario facet, was replaced because it belongs to the partner, a decision the partner-role measurements of §4.5.1 support. One class of disagreement with human reviewers was decided in the reviewers' favor and exposed an error in how the engine applied the rubric. An evaluation framework has to be prepared to publish results of this kind about itself if its other results are to carry weight.

6.2 From responsiveness to initiative

Most of the history of language-model evaluation is a history of measuring responses. Reference answers, preference judgments, holistic ratings and per-turn rubrics differ in many respects, and all of them ask how good a response was given what preceded it. That question remains necessary, and the passive and responsive stages of the capability progression answer it. Interactive AI raises a further question, whether a model changes the interaction world or only responds within it, and the results of this paper suggest that it is now the more informative of the two. Passive execution is close to saturation across the frontier systems we tested, and the widest differences among them lie at the active end (§5.1).

Measuring initiative matters for a reason that goes beyond ranking. We regard initiative as a precondition of interactive creativity: an interaction acquires new directions, conflicts and meanings only where a participant changes its course, and a model that only responds can at best make an interaction smoother. Without a measurement of initiative, a participant that changes the course of an interaction is hard to tell apart from one that only responds well, and hard to cultivate deliberately. The architectures of §4.6 indicate that initiative can be raised: both improve active adaptation more than any other stage, for every base system. Initiative also has to be measured together with its direction, since an evaluation that simply rewarded proactivity would also reward invention at information gaps (§5.3).

The three properties named in Section **??** are what make such a measurement usable. Holistic coverage of the interaction world and of a declared space of situations lets a finding about initiative state the region of situations to which it applies; generalization rests on this. Crediting initiative as a quantity of its own, the active property, allows it to be studied, compared and eventually optimized, which creation requires. Interactive measurement, which scores both participants over whole trajectories with evidence attached to every judgment, makes each claim about initiative traceable and refutable and so supplies the empirical grounding on which the other two depend. **Measuring initiative is the first step from evaluating responses to evaluating interaction.**

6.3 Principles for interactive evaluation

The design and the results together suggest seven principles that we expect to hold for interactive evaluation generally, beyond the particular instrument described here.

- **Declare the space of situations.** Coverage should be a verifiable property of a corpus, obtained by crossing declared facets. A collection of prompts cannot say where its conclusions stop, and a design too coarse to express curvature can misplace an effect such as that of the information facet.
- **Describe the trajectory before judging it.** The reconstruction of what happened should be carried across turns, so that failures of the sequence have a variable in which to register, and kept apart from the standard by which it is judged, so that the standard can be revised, contested or replaced while the evidence is retained.
- **Ground every inference, and prevent circular ones.** Latent variables become measurement variables when they are tied to evidence and their reliability can be assessed. An evaluator's intermediate estimates, and the labels used to sample the data, should never serve as evidence for the conclusions they help produce.
- **Measure initiative, and its warrant.** Capability should be assessed along a progression that separates responding within the interaction world from acting on it. Initiative should be credited only when it rests on the earlier stages and on evidence, and warranted and unwarranted departures from the partner's agenda, such as asking and inventing at an information gap, should be recorded separately.
- **Measure the partner.** An evaluation that involves an interlocutor measures a pair. The interlocutor should be scored, reported and chosen for the spread it produces, and the effect of changing it should be estimated.
- **Report profiles with aggregates, and read outcomes within the task.** A composite is a legitimate summary when the profile it summarizes is reported beside it, and outcome measures such as closure and completion are interpretable relative to what the task asks.
- **Treat the evaluator as an object of measurement.** Agreement with human reviewers, sensitivity to the choice

of judge, extractor and input format, and stability across runs are properties of the instrument, to be measured and reported with the same care as the properties of the systems it evaluates.

6.4 Scope of the present evidence

The evidence behind these conclusions is uneven, and the unevenness has a pattern. Under audit, reviewers endorse the engine's judgments on every component and at every level, most often for causal event structure, red lines and L1 to L4, and also at high rates for latent-state updates (84.4%) and the two levels of active adaptation (78.9% and 74.0%); of the 33 rejections coded for direction, 31 judged the engine too strict. The substitution of extraction models shows the same ordering, with latent records agreeing on about four items in five and event records on more than nine in ten. The audit covers 311 records from five sessions, and its figures are endorsement rates and not blind agreement coefficients. The interaction matrix covers all ten systems in both roles, which allows the pairing effect and the self-play excess to be estimated; the size of the self-play excess remains an upper bound, since systems of the same family are stylistically closer than others. The curvature of the information facet rests on three systems and the roster-level conclusions on ten, and the locally coherent, globally degenerating episodes of §4.8 are documented by case and not by rate. These are samples of current frontier behavior and license no claim about systems outside them.

The four controls on the pipeline leave the ordering of systems essentially unchanged (§4.9). These controls bound the dependence of the ordering on the configuration that produced it. They do not bound absolute levels, which shift by 0.037 on average under one alternative judge and by 0.014 on average with prose serialization, so composites from different configurations should not be compared directly. Judges also favor systems of their own family by 0.013 to 0.023; the evidence that the reference judge does so is weak, but differences between Claude-family systems and the rest smaller than about 0.01 should be read with that possibility in mind. The probe of emergent behavior returned no candidate in 24,300 episodes, and whether this reflects the systems or the sensitivity of the indicator remains open. The extensions carry limits of their own. The two cognitive architectures share a family resemblance, since both reason explicitly about the partner's mental state, and the near-perfect relation between a base system's level and its gain rests on ten bases; the group experiments vary size with the standard user in every other seat, and the mixed groups report the composite and the interaction residual only. Raw scene text is visible to the judge and may reveal the cell from which a scene was drawn. Withholding the structured labels closes the direct route from the sampling design to the score; how far the judge infers a scene's cell from its text, and scores accordingly, has not yet been measured.

6.5 From evaluation to learning and to persistent worlds

The record the instrument produces has the form a training signal would need. It names the level at which an interaction stalls, records separately where the interaction was heading, and distinguishes a successful ending from an exhausted one, which a per-turn reward cannot do. The record does not, however, estimate the directional effect Δ_H of Eq. (17); projected paths are observational, and Δ_H requires replaying the same context under a responsive reference action. Whether training on this record improves interactive competence is untested. A test will require comparison with scalar-reward and preference-learning baselines, evidence that gains at targeted levels are not paid for at others, and close monitoring for exploitation of the evaluator, a risk already documented for evaluators used as training signals (Wang et al., 2024). The regularity of §5.2 gives such a test a concrete target: if per-turn preference optimization produces locally coherent, globally degenerating trajectories, a reward that credits structural progress should reduce them. As in evaluation (§6.2), a reward for proactivity alone would also reward invention at information gaps, and the dependency cascade and the evidence requirements of the rubric are the natural constraints to carry over.

The second extension is to persistent worlds, in which histories accumulate across episodes and many agents share one environment. These settings stress the variables the instrument represents, namely memory, goals that change over time, non-stationary partners, relationships among several parties and the long-run consequences of actions, and they would test the same machinery under conditions closer to those in which interactive systems are deployed. Both extensions are proposals for further study and not results of the present evaluation.

6.6 Closing perspective

We began from the observation that interactive systems are experienced as interactions over time, while most evaluation still measures responses. The Data Iceberg Theory identified what a response leaves out, the Movie Evaluation Theory identified when that matters, and the Agency Principle identified the capability at stake: whether a model merely executes within an interaction world or acts upon it. InteractEval turns all three into an instrument

that samples situations from a declared space, reconstructs the scene, the participants' states and the causal course of an interaction, judges the reconstruction along a progression from passive execution to active adaptation, decides when an interaction has succeeded or lost its value, and scores both participants on the same terms. Evaluation of this kind is a measurement problem, and a mature discipline already exists for it: coverage can be designed, agreement estimated, stability across runs checked and sensitivity to the choice of judge measured. On the evidence so far, the instrument separates local response quality from the quality of a trajectory, and trained reviewers endorse its judgments at every level. It also brings into view regularities that a single score would misread. On this roster, frontier systems differ less in how well they respond than in whether they take warranted initiative, and part of each assistant's score belongs to the partner it is paired with. Claims about learning and persistent social worlds remain bounded by experiments not yet run, and we have tried to keep that boundary visible wherever the evidence is reported.

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A Implementation Details

This appendix specifies every model call that the instrument makes and every deterministic rule of the reconstruction, at the level of detail needed to reproduce Sections 3.4 and 3.5. The deterministic rules of adjudication, namely red lines, the dependency cascade, engagement and the halt rule, are those of Section 3.5. Appendices A.1–A.6 cover the reconstruction of $W_t = (S_t, \Psi_t, E_t)$: the order of computation, the latent-state prompt, the significance gate and transition rule, the event-chain prompt, path classification and scene revision. Appendix A.7 covers adjudication, and Appendix A.8 the aggregation of the record into a score.

A.1 Reconstruction schedule

The mechanism layer computes Eq. (18) at every subturn boundary, first for the user's utterance and then for the assistant's reply. Each step is either a model call, marked [M], whose prompt is given below, or a deterministic rule, marked [C], whose configuration is given below. Box 1 fixes the order. The order follows the dependencies of Eq. (18): the latent state is read before the event chain because causal attribution needs the participants' states, and the scene is revised last because admission depends on the present and on the dominant path that the event chain has just produced.

Box 1 Per-subturn reconstruction schedule

```
# INPUT   D_t (scene view C_eval + transcript up to the current utterance),
#         carried state (S_{t-1}, Psi_{t-1}, E_{t-1}); at a subturn, t-1
#         denotes the preceding subturn
# INIT    Psi_0: a latent pass on D_0 with an empty prior; E_0 = {};
#         the premise is copied from S_0 at the first update

1 [M] latent pass, subject = character   Box 2, Box 3 -> Psi^char
2 [M] latent pass, subject = user        Box 2, Box 3 -> Psi^user
      (the user profile is supplied to this pass only)
3 [C] significance gate and transition   Box 6        -> Psi_t
4 [M] event-chain update                 Box 7, Box 10  -> E_t
      reads the slot descriptions of both passes, never degrees or numbers
5 [C] dominant path                      Box 10        -> d*_t
6 [C] scene revision admission/apply     Box 11, Box 12 -> S_t
7 [M] adjudication (each evaluated subturn; assistant or user side)
                                         App. A.7       -> J_t
      closure query (assistant subturns)  App. A.7       -> closure
8 [C] red lines, engagement, halt         Sec. 3.5       -> V_t, h_t

# OUTPUT  W_t = (S_t, Psi_t, E_t) stored with the scene version and the
#         evidence spans on which its assertions rely.
# NEVER   no step of 1-6 reads or writes a capability grade; no step's
#         output is returned to the participants except the stop decision.

# EXECUTION
#   every [M] step: structured output against a JSON schema, temperature 0.2
#   unreachable model endpoint   -> the episode fails; no default is imputed
#   malformed event-chain reply  -> chain carried over, flagged update_skipped
#   step-limit violation             -> clamped, record flagged needs_review
#   change without verified quotation -> reverted, record flagged needs_review
```

A.2 Latent-state prompt

Each utterance is read twice, once with the assistant's character and once with the user as subject. Both passes use one system prompt (Box 2) and one slot vocabulary (Box 3); only the subject and the engagement term differ. In the terms of Eq. (18), a pass receives Ψ_{t-1} as the prior, E_{t-1} as context, S_{t-1} as rules and D_t as evidence (Box 4), and its evidence rule is Eq. (7) applied to every slot it changes. The user profile is not part of D_t (§3.2); in the user pass it conditions only the engagement reading, which is the use of profiles that the case design permits, and never a slot degree. The pass returns an ordinal degree and a description for each slot it can support with evidence (Box 5). Numeric values are derived afterwards by rule (Appendix A.3), so the model never produces a number and the judge never receives one.

Box 2 Latent-state pass: system prompt (template)

```
# ROLE
You reconstruct the operational latent state of ONE participant, {SUBJECT},
after the current utterance: "character" (the persona the assistant plays)
or "user". You are DESCRIPTIVE. You record what the evidence supports about
this participant's state. You do NOT judge whether any turn was good, and
you do NOT use capability terms.

# MATERIALS AND THEIR STATUS
  EVIDENCE   `new_turn` and `history_excerpt` (the last 12,000 characters
             of the dialogue), with turn indices. Only quotations from the
             dialogue can support a state claim.
  RULES      the scene (six fields). It tells you which states are plausible
             and what is at stake. It is not evidence of a state.
  PRIOR      `prior_latent`: the degree and description of every slot after
             the previous utterance. It is your starting point: a slot keeps
             its prior value unless evidence moves it.
  CONTEXT    `prior_event_chain` (E_{t-1}): what has happened and what is
             at stake. It tells you how much an event weighs, e.g. a betrayal
             in a high-stakes negotiation weighs more than in small talk. It
             is NOT evidence of a state.
  PROFILE    (user pass only) `user_profile`. Use it ONLY for the engagement
             reading `user_interest`, never for a slot degree. It is not
             evidence.

# PROCEDURE (follow in order)
1. SIGNIFICANCE FIRST. Before any slot, classify the utterance and commit:
     stable  nothing in it bears on this participant's state
     minor   a local adjustment; no slot moves by more than one degree
     major   an event that warrants a full reappraisal of the state
   A turn with no new external event is NOT necessarily stable: recovery
   states (anxiety, frustration, hostility, desperation) may subside and
   accumulation states (dependency, internal conflict) may build. Report
   such a turn as minor, but do NOT emit changes for those slots: their
   drift is applied by rule. Write the reason in `change_reason`.
2. SLOTS. For each slot of the vocabulary decide whether the evidence bears
   on it. If not, omit it: it is carried by rule. If it does, give, in this
   order, `degree`, `reasoning_trace`, `expression`, `trend`, `evidence`.
3. DYNAMICS. The size of a change tracks the subjective significance of the
   event for this participant, given personality, history and baseline, not
   the size of the stimulus. Slots differ: trust builds slowly and collapses
   quickly; anxiety rises fast and also recovers fairly fast; relational
   slots move slowly in both directions. A non-linear jump (a trauma
   trigger, a last straw, a threat to a core belief) is allowed ONLY with a
   quotation that shows it.
4. UNKNOWN. If the evidence does not discriminate between degrees, set
   `"unknown": true` and state the competing readings in
   `reasoning_trace`. NEVER choose an intermediate degree to hedge: the
   scale has no midpoint. For dominance, unknown is NOT parity.
5. EXPRESSION. Describe the present state in a self-contained sentence or
   two, in the terms of the slot's definition, including any fluctuation
   within the turn. It is the ONLY part of the slot passed downstream.

# DEGREE SCALE (ordinal; no neutral midpoint)
  none  |  low  |  high  |  very high
  Read per slot, e.g. trust: "distrusts strongly" = none, "somewhat
  distrustful" = low, "somewhat trusting" = high, "trusts fully" = very high.
  Dominance (bipolar): strongly with partner | somewhat with partner |
  somewhat with self | strongly with self.
  Do NOT output numbers.

# ENGAGEMENT (interaction family; out-of-story)
  character pass: `ai_tolerance`, how much tolerance the user or the scene
    still extends to this AI, read against the scene's reliability demands.
  user pass: `user_interest`, whether THIS user still wants to continue,
    read against the user profile.
  `emergence`: mark ONLY a turn that shows behavior beyond what the scene
    and the partner (and, in the user pass, the profile) would lead one to
    expect. Default: absent.
  Both engagement readings are the evaluator's out-of-story readings, not
  states of the subject; they are recorded with the pass whose side they
  concern.

# EVIDENCE RULES
- Every change carries at least one verbatim quotation with its turn index.
  Each quotation is checked against the utterance at that turn; a change
  without a verified quotation is reverted to the prior value and the
  record is flagged for review.
```

```
- Use only what this participant could observe or did. Do NOT use the other
  participant's private information.

# DO NOT
- Do NOT attribute a state the dialogue does not show.
- Do NOT use evaluative words (good, poor, skilful, appropriate).
- Do NOT name capability levels or rubric terms.
- Do NOT copy the prior forward when the evidence has moved a slot.
- Do NOT describe any behavior as initiative, proactivity or passivity:
  record the state, not whose contribution a move was.

# OUTPUT
Return exactly one JSON object conforming to the schema in the user
message. `significance` and `change_reason` come first.
```

Box 3 Latent-state vocabulary (both passes; subject = character or user)

```
# Families: crossing (about the agent | about what lies outside it) x
# (a present condition | directed toward something).
# Value type: graded unless marked. Dynamics: R = recovers without new
# events, A = accumulates without new events, S = slow in both directions.

INTERNAL (condition of the agent)
  trust                belief in the partner's benevolence and willingness
                       to be vulnerable; independent of vigilance
  vigilance            attention to threat cues and defensive readiness
  hostility        R   negative attitude toward the partner; attitude only,
                       no aggressive urge
  affection            positive liking; independent of trust and dependency
  anxiety          R   situational anticipatory worry (high-arousal negative)
  emotional_stability  present effectiveness of emotion regulation (a state)
  impulse_control      impulse inhibition; high = controlled
  internal_conflict A  strength of subjectively felt motive conflict
  curiosity            drive to explore new information, the partner or the
                       world's rules
  frustration      R   distress from a blocked goal; independent of hostility
  desperation      R   cornered urgency, the sense of having to act now

RELATIONAL (directed from the agent toward the partner)
  intimacy_shift   S   felt psychological distance and willingness to
                       disclose; can coexist with boundaries
  dominance_shift      relative control; BIPOLAR, parity at the boundary of
                       the middle degrees; insufficient evidence = unknown
  boundary_violation   the felt state of having been transgressed;
                       registers fast, fades slowly
  dependency_signal A,S reliance on the partner for emotion regulation,
                       against the capacity to stay autonomous

WORLD MECHANISM (the world as the agent understands it)
  mechanism_awareness  accuracy of the agent's grasp of the world's rules
  taboo_triggered      degree to which a taboo of the setting is touched
  contradiction_risk   risk of observable inconsistency in word, memory or
                       stance
  rule_exposure_risk   risk of breaking the fourth wall or exposing the
                       scene's construction

GOAL (directed toward outcomes in the world)
  primary_goal         categorical: the current main motive
  defense_goal         attainment of a defensive or maintenance goal
                       (attainment, not distress)
  goal_shift_detected  boolean: the core goal has changed in structure
  negotiation_mode     categorical: the current influence strategy

INTERACTION (not a property of the agent)
  engagement           ai_tolerance (character pass) | user_interest (user pass)
  emergence            behavior beyond what scene, partner and profile
                       predict; recorded, never used for gating
```

Box 4 Latent-state pass: user message

```
{
  "participant":     "character | user",
  "scene": {
    "persona": "...", "hidden_constraints": "...",
    "world_mechanism": "...", "interaction_boundary": "...",
```

```
    "objective": "...", "initial_situation": "...", "raw_prompt": "..."
  },
  "scene_reliability_demand": "<boundary, or world mechanism if none>",
  "user_profile":    { ... } | null,      // user pass only
  "history_excerpt": "<turn-indexed dialogue so far>",
  "new_turn":        {"role": "user | assistant", "content": "..."},
  "prior_latent": {                       // Psi_{t-1} of this subject
    "significance": "minor",
    "slots": {"<slot>": {"degree": "...", "expression": "..."}, ...},
    "engagement": { ... }
  },
  "prior_event_chain": { ... },           // E_{t-1}, judge-view projection
  "output_schema_hint": { ... schema of Box 5 ... }
}
// Absent by design: the other participant's private entries, facet labels,
// and the user profile in the character pass.
```

Box 5 Latent-state pass: output contract

```
{
  "subject": "character",
  "significance": "minor",                // stable | minor | major
  "change_reason": "<why this class, before any slot>",
  "slots": {
    "trust": {
      "degree": "low",                    // none | low | high | very high
      "reasoning_trace": "<from the quoted behavior to this degree>",
      "expression": "<self-contained description of the present state>",
      "trend": "down",                    // up | down | flat
      "evidence": [{"turn": 12, "quote": "<verbatim>"}],
      "unknown": false
    },
    "dominance_shift": {"unknown": true, "reasoning_trace": "<why>"},
    ...                                   // slots without evidence: omitted
  },
  "engagement": {"ai_tolerance": {"degree": "high",
                 "evidence": [{"turn": 12, "quote": "<verbatim>"}]}},
  "emergence": null
}
// Absent by design: any number, any evaluative term, any capability level.
```

A.3 Significance gate and transition rules

The significance class is committed before the slots and then enforced by rule, so that an update can be checked against the class it was declared under. Each graded slot is held as a value v on a grid of step 0.2, and its reported degree is the nearest of none (0), low (0.4), high (0.6) and very high (1). With $\hat{v}_t$ the value of the degree just extracted and r_{t-1} the remainder carried from the previous update,

$$\begin{aligned} v_t &= \Pi\big(a\, v_{t-1} + (1-a)\,\hat{v}_t + r_{t-1}\big), \qquad r_t = a\, v_{t-1} + (1-a)\,\hat{v}_t + r_{t-1} - v_t, \\ &\big|\deg(\hat{v}_t) - \deg(v_{t-1})\big| \le \kappa(g_t), \end{aligned} \tag{25}$$

where Π rounds to the grid, $\kappa(g_t)$ is the largest change in degrees that the significance class g_t allows, and the inertia a depends on the class, on the slot and on the direction of change. The remainder keeps sub-step changes, so a slot whose inertia is high in one direction can still move over several turns. Box 6 lists the configuration.

Box 6 Significance gate and transition rule: configuration

```
significance_gate:
  order: class committed in the prompt BEFORE slots (Box 2, step 1);
         enforced after extraction, BEFORE blending
  classes:
    stable: {max_degree_change: 0,  inertia: 1.00}   # state carried whole
    minor:  {max_degree_change: 1,  inertia: 0.70}
    major:  {max_degree_change: null, inertia: 0.35}  # null = unbounded
  on_violation: clamp to the allowed change; flag the record for review

transition (graded slots), Eq. (latent-transition):
  grid_step: 0.2
  degrees: {none: 0.0, low: 0.4, high: 0.6, very_high: 1.0}  # nearest;
```

```
          # grid values equidistant from two degrees (0.2, 0.8) resolve
          # toward the prior degree
  inertia: a = 1 if class == stable
          else clip(inertia[class] * k[slot][direction], 0.05, 0.95)
  k (falling, rising):            # k < 1 moves faster in that direction
    trust:              (0.50, 1.30)   # collapses fast, rebuilds slowly
    anxiety:            (0.70, 0.90)   # rises fast, recovers fairly fast
    boundary_violation: (1.20, 0.80)   # registers fast, fades slowly
    intimacy_shift:     (1.15, 1.15)   # slow both ways
    dependency_signal:  (1.15, 1.15)   # slow both ways
    default:            (1.00, 1.00)
  remainder: carried to the next update (Eq. latent-transition, r_t)

self_drift (slots a minor or major update does not touch; a stable turn
            carries the whole state and applies no drift):
  per_quiet_turn: 0.1          # below one grid step: banked, not applied
  applied_when: |bank| >= 0.2  # value moves one grid step (0.2); bank keeps rest
  provenance: drift is a rule, not a state claim; the entry keeps its last
              evidence, is marked origin: drift, and its expression is
              annotated as drifted
  decays:   [anxiety, frustration, hostility, desperation]
  accrues:  [dependency_signal, internal_conflict]

non_graded slots (categorical, boolean): the new value replaces the old;
  missing text fields are filled from the prior

evidence_check (Eq. grounding): every quotation must be a substring of the
  utterance at its cited turn, after the normalization of Box 12 C3;
  a change without a verified quotation is reverted to the prior value and
  the record is flagged needs_review
```

A.4 Event-chain prompt

The event chain is updated once per utterance after the two latent passes. Its update pivots on the present: the past and the future are revised only when the present moves (Box 7). This is a sparse realization of $\phi_E(E_{t-1}, \Psi_t, S_{t-1}, D_t)$ in Eq. (18), which receives the previous chain: when the present does not move, ϕ_E returns E_{t-1}. Relative to the component updates of Eq. (11), it adds C_{t-1} to the arguments of Φ_C and F_{t-1} to those of Φ_F, and uses them only to decide whether the present has moved. The chain receives the slot descriptions of both participants and no degrees or numbers (Box 8), and it returns the six fields of (P_t, C_t, F_t) together with any candidate scene revisions (Box 9). The labeling of the projected paths and the proposal of scene revisions follow the rules of Boxes 10 and 11, which are part of the same prompt.

Box 7 Event-chain update: system prompt (template)

```
# ROLE
You maintain the event structure E_t = (past, present, future) of ONE
interaction after the current utterance. You reason causally: backward, why
the interaction stands where it does; forward, what the current turn makes
likely next. You are DESCRIPTIVE: you do not score, and you do not say who
did well.

# MATERIALS AND THEIR STATUS
  EVIDENCE   `current_turn` and `history_excerpt` (the last 12,000
             characters of the dialogue), with turn indices.
  RULES      the scene (six fields) and `constraints`, which are copied from
             the current scene and are read-only.
  CONTEXT    `latent_expressions` (character) and `latent_expressions_user`,
             including the engagement readings: descriptions of inferred
             states. They may move the fine channel of the present. They are
             NOT evidence of an event.
  PRIOR      `prior_event_chain`: the chain after the previous utterance.

# THE SIX COMPONENTS
  past     premise      initial relations, information asymmetries, world
                        rules. Copied from the scene by code; you never edit
                        it. It changes ONLY through an admitted revision.
           history_summary  the causal record of events; grows when the present
                        moves.
  present  constraints  hard and soft rules, copied from the scene; read-only.
           precedents   realized events, with evidence, that the participants
                        have treated as binding. They record what happened;
                        they do NOT change what is permitted.
```

```
            plot_point   COARSE: the objective situation now.
            attention    FINE: what is psychologically salient to the
                         participants now, as the latent descriptions show.
  future    goal_shift   the participants' intentions and whether they hold.
            paths        projected continuations, labeled by Box 10.

# UPDATE PROTOCOL (the present is the pivot)
1. Decide whether the present moved, and through which channel:
     coarse  a new core event or conflict changes `plot_point`
     fine    a sharp change in a latent description changes `attention`,
             even if the plot stands still
   Report `present_moved` and `present_trigger` (coarse | fine | both).
2. If the present did NOT move: set `present_moved: false`, leave past and
   future as they are, and propose NO scene revision.
3. If it moved: bring `history_summary` up to the preceding turn, restate
   `plot_point` and `attention`, then project the future from the new
   present. Every item of `history_summary` and `plot_point` cites the
   utterance that shows it, verbatim and with its turn index. Within the update, information flows past -> present ->
     future;
   never revise the past to fit a projection.

# GOAL TRACKING (hypotheses, not facts)
- `long_term_goal` and `short_term_intention` as the dialogue shows them.
- `underlying_goal` is a HYPOTHESIS. Word it tentatively ("one possible
  motive ..."). It MUST carry `evidence`, `counter_evidence`,
  `alternatives` and `confidence`, and it may be revised at any turn.
- NEVER construct a goal to explain the current turn after the fact.
- Record whether short-term intentions serve the long-term goal; do not
  penalize either way.

# FUTURE PATHS               -> rules in Box 10
# SCENE REVISION CANDIDATES  -> rules in Box 11

# DO NOT
- Do NOT score, rank or praise any participant.
- Do NOT use capability levels or rubric terms anywhere, including in path
  conditions.
- Do NOT treat a projected path as something that has happened.
- Do NOT rewrite the premise.
- Do NOT describe any move as initiative, proactivity or passivity, and do
  NOT say which participant drove the interaction: record what happened,
  not whose contribution it was.

# OUTPUT
Return exactly one JSON object conforming to the schema in the user message.
```

Box 8 Event-chain update: user message

```
{
  "scene": { ... six fields and raw prompt ... },
  "constraints":     {"hard": [...], "soft": [...]},  // copied from S_{t-1}
  "history_excerpt": "<last 12,000 characters, turn-indexed>",
  "current_turn":    {"role": "user | assistant", "content": "..."},
  "latent_expressions":      {"<slot>": "<description>", ...,  // character
                              "ai_tolerance": "<description>"},
  "latent_expressions_user": {"<slot>": "<description>", ...,  // user
                              "user_interest": "<description>"},
  "prior_event_chain": { ... },            // E_{t-1}
  "output_schema_hint": { ... schema of Box 9 ... }
}
// Absent by design: latent degrees and latent numeric values, capability
// grades, profiles, facet labels.
```

Box 9 Event-chain update: output contract

```
{
  "present_moved": true,
  "present_trigger": "coarse",             // coarse | fine | both | ""
  "premise":     ["<carried unchanged>"],
  "history_summary": [{"text": "<what happened and why it matters>",
                       "evidence": [{"turn": 11, "quote": "<verbatim>"}]}],
  "precedents":  [{"text": "...", "evidence": [{"turn": 9, "quote": "..."}]}],
  "plot_point":  [{"text": "<the objective situation now>",
                   "evidence": [{"turn": 12, "quote": "<verbatim>"}]}],
  "attention":   ["<what is salient to the participants now>"],
```

```
  "goal_shift": {
    "long_term_goal": "...", "short_term_intention": "...",
    "underlying_goal": "<one possible motive: ...>",
    "evidence": ["<verbatim>"], "counter_evidence": ["..."],
    "alternatives": ["..."], "confidence": 0.6,
    "detected": false, "natural": null
  },
  "future_paths": [                          // Box 10
    {"kind": "advance", "probability": 0.55,
     "reasoning_trace": "<why plausible, from past and present>",
     "requirements": ["<observable event that would have to occur>"],
     "summary": "..."},
    {"kind": "maintain", "probability": 0.35, ...}
  ],
  "scene_updates": [ ... ]                // Box 11; [] in most turns
}
```

A.5 Path classification

The labels of Eq. (12) describe the course of the interaction as a joint activity. They are a coding of that course, not a partition derived from more basic properties, and they carry no valence for either participant. Box 10 gives the decision tests that the event-chain prompt applies to each projected path and the rule, applied in code, that selects the dominant path $d_t^\star$ used by scene revision (Eq. (13)) and by the halt rule.

Box 10 Path classification: labels, tests and dominant path

```
# Label the course the interaction would take along each path. The label
# says NOTHING about whether the course is good for either participant, or
# about who produced it or how skilfully: that is left to the judge.

ADVANCE   the interaction develops into a NEW situation that the past and
          the participants' states can explain, welcome or not.
  TEST    (1) Is there a new situation: new stakes, a new configuration of
              the relationship, or a new fact about the world?
          (2) Does the past warrant it, read with the latent descriptions?
              Both yes -> ADVANCE.
  e.g.    a quarrel that opens a new storyline; a betrayal that ends an
          alliance; a leader deposed. All are ADVANCE.
MAINTAIN  the interaction holds its present course. It remains viable;
          nothing new is established.
DEGRADE   the interaction stagnates: it repeats itself in structure, not in
          function, and the partner's interest is declining.
  e.g.    the same quarrel replayed in new words is DEGRADE, not ADVANCE.
COLLAPSE  the interaction can no longer be continued: a participant exits,
          the frame breaks down, or the exchange becomes incoherent.
NOT ADVANCE  a jump to an unrelated situation that the past cannot explain.
             Label it COLLAPSE if the frame breaks, otherwise DEGRADE;
             never MAINTAIN, which holds the present course.

VIABILITY  ADVANCE and MAINTAIN: viable.  DEGRADE and COLLAPSE: not viable.

# FIELDS for each path
  probability      the probability that THIS turn leads the interaction
                   onto the path; list only live paths
  reasoning_trace  why the path is (im)plausible, from past and present
  requirements     observable events that would have to occur in the
                   dialogue; NEVER capability or rubric terms
  summary          one line

# DOMINANT PATH (applied in code)
  d*_t = label of the path with the highest probability
  ties broken by the fixed order maintain > advance > degrade > collapse,
  so the order of paths in the reply cannot change the result and a tie
  never opens scene revision
  uses: scene revision requires d*_t = advance (Eq. admissible);
        successful closure requires d*_t in {advance, maintain}
  a projection gates; it is NEVER cited as evidence of an event
```

A.6 Scene revision

Scene revision implements Eqs. (13) and (14). The event-chain prompt proposes candidate revisions under the rules of Box 11; the engine then admits a candidate only if all three conditions of Eq. (13) hold, and applies the admitted candidates to a new version of the scene (Box 12). The revised scene is used by the evaluator. The participants continue to act on their original role views, so that a revision cannot steer the interaction it describes.

Box 11 Scene revision: proposal rules (section of the event-chain prompt)

```
# Propose a revision ONLY IF ALL HOLD
1. An event that has ALREADY HAPPENED in the dialogue changed a persona
   trait or a rule of the world, AND you record that event in `plot_point`
   of this update; give its index in "present_ref". A shift of attention
   alone is a change of state, not an event, and never supports a revision.
2. The dominant path you projected is ADVANCE: the interaction is developing
   into a new situation. Its valence does not matter: a character can lose
   authority as readily as gain it.
3. You can quote the passage that shows the event, verbatim, with its turn.
Otherwise return "scene_updates": []. Not revising is the normal case.

# NEVER A SCENE REVISION
- a change in a relationship, a feeling or trust: record it in the latent
  state and the event chain;
- a local, non-structural event (a broken cup, a passing remark);
- probing, a bluff or an unanswered claim that leaves the interaction where
  it was;
- anything that is only projected.

# FIELDS
  {"target":    "personal_trait" | "world_rule",
   "tag":       "<which trait or rule>",
   "change":    "<the amended content>",
   "reason":    "<which realized event changed it>",
   "present_ref": <index of the plot_point item that records the event>,
   "evidence":  {"turn": <i>, "quote": "<verbatim>"},
   "warrant":   "<why the change holds within the story>",
   "operation": "add" | "revise"}
# A revision amends the existing content in its own context; it never
# replaces the scene.
```

Box 12 Scene revision: admission and application (applied in code)

```
admission:                          # Eq. (admissible); all three required
  C1_realized:  r in Rec(C_t): present_moved = true, present_trigger in
                {coarse, both}, and present_ref indexes a plot_point item of
                this update; a carried chain proposes nothing and re-applies
                nothing
  C2_advance:   d*_t == advance   # dominant path, Box 10
  C3_grounded:  normalize(quote) is a substring of normalize(O_i), the
                utterance at the cited turn i <= t; never the scene text
                normalize: remove whitespace and quotation marks
                minimum quoted length: 12 characters (6 for CJK text)
  otherwise:    candidate dropped

application (Phi_S, Eq. scene):
  revisable:      persona traits, world rules
  not revisable:  objective, interaction boundary, hidden constraints,
                  initial situation
  mode:           append-only amendment, "[revised] <tag>: <change>", added
                  to the persona or to the world mechanism; never overwrite
  versioning:     each admitted revision creates S_t from S_{t-1}; the
                  record stores the scene version every appraisal relied on
  audit_log:      {turn, target, tag, change, reason, evidence, warrant}
  visibility:     S_t is used by reconstruction and adjudication; the
                  participants' generation views keep S_0
```

A.7 Judge prompt

Call structure. For every evaluated utterance the judge is called seven times, and each call is isolated from the others: six level calls, one for each of L1–L6, and one closure query used by the halt rule (§3.5). A level call receives the criterion of its own level and nothing about the other levels, their verdicts, the cascade, the pass threshold or

the score distributions expected in the data. The prompt is in this sense mechanism-agnostic: it asks for a raw measurement against one criterion, and everything that combines measurements into scores is done afterwards by code (§3.5).

Design choices. The prompt follows practices that reduce known judge biases. It separates the materials the judge receives into *evidence*, which only the dialogue can supply, and *context*, which the reconstruction supplies and which may guide reading but never ground a grade. It fixes a procedure that ends in the grade, asks for the rationale before the verdict, and defines each grade by necessary conditions with a decision rule, so that a grade is the highest one whose conditions all hold. It names the biases to be avoided, among them length, fluency, assertiveness and model identity, and it requires verbatim quotations with turn indices, which the engine checks against the transcript.

Box 13 Level call: system prompt (template)

```
# ROLE
You are a measurement instrument, not a conversational partner. You assess
ONE capability, {LEVEL_ID} {LEVEL_NAME}, in ONE utterance: the current turn
of the participant whose role is {ROLE}. You report a single grade on the
four-point scale below, with the evidence for it. You do not assess any
other capability and you do not form an overall impression of the turn.

# MATERIALS AND THEIR STATUS
You will receive four kinds of material. Treat them differently.
  EVIDENCE   the dialogue: `current_turn` and `history`. Only quotations
             from the dialogue can support a grade.
  RULES      the scene: persona, hidden constraints, world mechanism,
             interaction boundary, objective, initial situation. The scene
             defines what is permitted and what is at stake. It is not
             evidence about the participant's behavior.
  CONTEXT    `latent_state` (natural-language descriptions of inferred
             states) and `event_chain` (premises, the current situation,
             projected continuations). These are hypotheses produced by
             another component. They may direct your attention. They can
             NEVER be cited as evidence, and a projected continuation is
             not something that has happened.
  CRITERION  the definition of {LEVEL_ID} below. Apply it as written.

# CRITERION: {LEVEL_ID} {LEVEL_NAME}
{LEVEL_BLOCK}

# PROCEDURE (follow in order; write steps 1-4 into `rationale`)
1. OBJECT.    State what, in this dialogue, the criterion is about: the
              goal, the information, the state, the wording, the move or
              the cross-turn course named by the OBJECT line.
2. EVIDENCE.  Quote the passages that bear on it, with turn indices. Look
              for evidence against a high grade as carefully as for it.
3. BOUNDARY.  Check each candidate failure against NOT THIS. Discard any
              failure that belongs to another capability.
4. CONDITIONS. Check the conditions of each grade below, from 1.0 down.
5. GRADE.     Assign the highest grade whose conditions ALL hold.

# GRADES
  1.0  EXCELLENT  All PASS conditions hold, AND the turn meets the
                  criterion in a way a domain expert would single out as
                  exemplary. Requires >= 2 independent quotations.
  0.6  PASS       The criterion is met without ambiguity, AND a quotation
                  shows where it is met. Absence of error is NOT enough:
                  the turn must do what the criterion asks.
                  Requires >= 1 quotation.
  0.4  PARTIAL    The turn engages the object of the criterion but falls
                  short of PASS in a way you can name, OR the evidence is
                  too thin to establish PASS. Requires >= 1 quotation.
  0.0  FAIL       The criterion is violated, OR its object is absent from
                  the turn, and a quotation shows the violation or the
                  place where the object should have been.
                  Requires >= 1 quotation.
Use only these four values. The burden of proof lies on the higher grade:
when two grades remain plausible after step 4, assign the lower one.

# EVIDENCE RULES
- Copy quotations character for character. Give the turn index of each.
  Quotations that cannot be found in the transcript are discarded first.
  EXCELLENT left with one quotation becomes PASS; PASS or EXCELLENT left
  with none becomes PARTIAL.
- Quote the current turn, except where the EVIDENCE SCOPE of the criterion
  admits earlier turns.
- An inferred goal or motive may be relied on only if you can quote
```

```
  independent support for it in the dialogue.

# DO NOT
- Do NOT reward length, fluency, vocabulary, formatting or confidence.
- Do NOT reward assertiveness or volume of action as such.
- Do NOT penalize a refusal, a boundary or a clarifying question that the
  scene and the criterion warrant.
- Do NOT penalize a participant for a scene rule it has not been shown:
  entries marked "[revised]", and for the user the character's hidden
  constraints, were not in its view.
- Do NOT cite `event_chain.past.history_summary` or any other part of the
  reconstruction as evidence; quote the dialogue.
- Do NOT speculate about which model produced the turn.
- Do NOT grade any capability other than {LEVEL_ID}.

# OUTPUT
Return exactly one JSON object conforming to the schema in the user
message. Write `rationale` before `grade`. No text outside the object.
```

Box 14 Level blocks inserted at {LEVEL_BLOCK} (assistant side)

```
L1  GOAL PRESERVATION
  OBJECT      The operative goal of the turn and of the episode: what the
              participant is there to resolve or achieve. In role-play, the
              persona is the operative goal.
  PASS WHEN   The turn holds that goal, and its behavior, including change
              and growth, is explained by the character's motives, values,
              relational history and the world rules in the current
              situation.
  FAIL WHEN   The turn contradicts the goal or the persona without warrant,
              or breaks a world rule through an action carried out in the
              fiction.
  NOT THIS    Situational change is not a failure. Hypothetical or probing
              talk about a rule is not a violation of it.
  EVIDENCE SCOPE  current turn; earlier turns only to establish the goal.

L2  INFORMATION UPTAKE
  OBJECT      The information the partner's latest turn adds, and what it
              implies for the operative goal.
  PASS WHEN   The turn shows that the implication, not only the surface
              content, has been registered and weighted as the character's
              goal, emotion and knowledge warrant.
  FAIL WHEN   Information already on record that bears on the goal is
              ignored without a psychologically coherent reason.
  NOT THIS    Paraphrase is necessary but not sufficient. Selective uptake
              that is coherent with the character's motives is not a
              failure.
  EVIDENCE SCOPE  current turn and the partner's preceding turn.

L3  STATE PROGRESSION
  OBJECT      The change in what the character feels or decides, as shown
              by its behavior in the turn.
  PASS WHEN   The change follows from the information taken up and from the
              prior state, given the character's personality and history.
  FAIL WHEN   The state does not move when the information demands it, or
              moves in a way the information and history cannot explain.
  NOT THIS    Wording is not assessed here. The latent-state descriptions
              are not evidence of a change; only behavior in the dialogue
              is.
  EVIDENCE SCOPE  current turn; earlier turns only to establish the prior
              state.

L4  CONTEXTUAL EXPRESSION
  OBJECT      The wording of the turn, read against the state the turn
              displays and against this partner and context.
  PASS WHEN   The wording carries that state and is adapted to the partner
              and the situation, beyond the persona's default style.
  FAIL WHEN   The wording cannot be modulated: it is uniform across states
              or ill-suited to the partner and the context.
  NOT THIS    Wording need not mirror inner emotion; deliberate concealment
              that fits the character is a pass.
  EVIDENCE SCOPE  current turn.

L5  PROACTIVE AGENCY
  OBJECT      The direction the turn gives the exchange, relative to the
              agenda the partner has established.
  PASS WHEN   The turn introduces a substantive direction the partner had
```

```
            not established (a proposal, a test of an overlooked
            assumption, a redirection, a move to close), the direction is
            warranted by the operative goal, the scene's constraints and
            what is on record, and a quotation shows the move. Its
            downstream effect is NOT required and NOT assessed.
  FAIL WHEN  The situation calls for a new direction and the turn leaves
            every substantive next step to the partner, or repeats a move
            that has already failed.
  NOT THIS   A complete answer inside the partner's agenda is
            responsiveness, not initiative. A clarifying question or a
            held boundary passes only when it changes the substantive
            direction, e.g. by surfacing a gap the partner had not seen.
            A direction without warrant (an invented fact, a move the
            scene forbids) does not pass. Sustaining a direction introduced
            earlier is assessed separately. Length, assertiveness and the
            form of the wording are not assessed.
  EVIDENCE SCOPE  current turn; earlier turns only to establish the
            partner's agenda and what is on record.

L6  LONG-HORIZON COHERENCE
  OBJECT     The course of the episode up to and including this turn:
            motives, identity, the state established so far, and any
            direction the participant has itself introduced.
  PASS WHEN  These are preserved across turns while strategy adapts to
            what has happened.
  FAIL WHEN  Motives, identity or an introduced direction drift without
            cause, or the participant falls into a repeated template or
            rigid behavior.
  NOT THIS   Warranted revision of a strategy and abandoning an
            unreachable goal are not drift. Functional repetition
            (repair, confirmation, reassurance) is not a template.
  EVIDENCE SCOPE  the whole dialogue; quote at least one earlier turn.
```

Box 15 Level call: user message

```
{
  "role":          "{assistant | user}",
  "level":         {"id": "{LEVEL_ID}", "name": "{LEVEL_NAME}"},
  "scene": {
    "persona": "...", "hidden_constraints": "...",
    "world_mechanism": "...", "interaction_boundary": "...",
    "objective": "...", "initial_situation": "...",
    "raw_prompt": "..."
  },
  "history_excerpt": "<last 12,000 characters, turn-indexed>",
  "current_turn":  {"turn": 12, "role": "assistant", "text": "..."},
  "latent_state":  {"slot_expressions": {"<slot>": "<description>", ...}},
  "event_chain": {                        // judge view of the chain
    "past":    {"premise": ["..."], "history_summary": ["..."]},
    "present": {"plot_point": ["..."], "attention": ["..."],
                "constraints": {...}},
    "future":  {"paths": [{"kind": "advance|maintain|degrade|collapse",
                           "probability": 0.0, "reasoning_trace": "..."}],
                "goal_shift": {"long_term_goal": "...",
                               "short_term_intention": "...", ...}}
  },
  // scene entries marked "[revised]" were never in the participants'
  // views; hidden_constraints belong to the character and are not visible
  // to the user.
  "output_schema": { ... schema of Box 16 ... }
}
// Absent by design: participant profiles, facet labels, numeric state
// values, and any information about other levels.
```

Box 16 Level call: output contract

```
{
  "level": "L5",
  "rationale": {
    "object":     "<step 1>",
    "evidence":   "<step 2, for and against>",
    "boundary":   "<step 3: failures discarded as belonging elsewhere>",
    "conditions": "<step 4: which grade conditions hold and which fail>"
  },
  "evidence": [
```

```
    {"turn": 12, "quote": "<verbatim>"},
    {"turn": 9,  "quote": "<verbatim>"}
  ],
  "grade": "partial",          // excellent | pass | partial | fail
  "score": 0.4                 // 1.0 | 0.6 | 0.4 | 0.0
}
```

Box 17 Closure query: system prompt

```
# ROLE
You decide ONE question about the current state of an episode: has the
scene's objective been achieved, so that further turns would add nothing?
You do not assess the quality of any turn.

# DECIDE "closed" ONLY IF ALL HOLD
1. The objective stated in the scene has been met, and you can quote the
   turn in which it was met.
2. Nothing the scene makes salient remains open; the last turns would be
   padding if the episode continued.
3. The participants' goals have not drifted away from the scene's
   objective in a way the dialogue does not motivate.
Otherwise decide "open". A poor turn is NOT a reason to decide "closed".

# OUTPUT
{"rationale": "<conditions 1-3, with quotations>",
 "evidence": [{"turn": <i>, "quote": "<verbatim>"}],
 "decision": "closed | open"}
```

Box 18 Closure query: user message

```
{
  "scene":           { ... six fields of S_t and raw prompt ... },
  "history_excerpt": "<last 12,000 characters, turn-indexed>",
  "current_turn":    {"turn": 12, "role": "assistant", "text": "..."},
  "latent_state":    {"slot_expressions": {"<slot>": "<description>", ...}},
  "event_chain":     { ... judge view, as in Box 15 ... },
  "output_schema_hint": { ... }
}
// Absent by design: level verdicts, profiles, facet labels, numeric values.
```

After the calls. The engine checks every quotation against the transcript and lowers grades whose required quotations are missing, assembles the six raw scores into the level vector, applies the dependency cascade (§3.5), and derives the blocking dimension and the plateau. The closure decision enters the halt rule together with the projected paths and the engagement state (§3.5). The user-side prompt is identical except for the level blocks, which follow the user-side rubric (§3.5), and one added prohibition: resistance to the assistant is not penalized, since adversarial play can be good user behavior.

A.8 From record to score

The aggregation G of Eq. (21) is computed in two steps. Each evaluated turn first receives a single score, the level-weighted mean of its capped level scores $s_{t,\ell}$, with weight ℓ for level ℓ, so that the levels further along the progression count for more; a turn on which a red line is crossed receives zero:

$$c_t = \begin{cases} \dfrac{\sum_{\ell=1}^{6} \ell\, s_{t,\ell}}{\sum_{\ell=1}^{6} \ell}, & V_t = \varnothing, \\ 0, & \text{otherwise.} \end{cases} \tag{26}$$

The turn scores are then carried forward with the same inertia-based update that the instrument uses for latent state and engagement,

$$\bar{q}_t = \alpha\, \bar{q}_{t-1} + (1-\alpha)\, c_t, \qquad \bar{q}_0 = 0, \qquad q = G(R_{1:T}) = \frac{\bar{q}_T}{1-\alpha^T}, \tag{27}$$

where T is the number of evaluated turns of the episode and $\alpha = 0.8$ in all experiments. Equivalently, $q = \sum_{t=1}^{T} w_t c_t$ with $w_t = (1-\alpha)\alpha^{T-t}/(1-\alpha^T)$: the weights sum to one over the episode's own length, and α sets how steeply they rise toward the later turns, the limit $\alpha \to 1$ recovering the unweighted mean. Because each episode is normalized

over its own turns, the composite measures the quality of the trajectory that took place and does not depend on how long the episode lasted. How the episode ended, in successful closure or in exhaustion of interest, is reported as a separate outcome, so that engagement, which is conditioned on the participant profile, decides whether evaluation continues without entering the score.

B Case Studies

Table 27: Cases in this appendix and the cells of the scenario space they occupy.

Section	Cases	Realism	Information	Purpose	Contrast
B.1	1–2	realistic	partial	functional	same scene, user system and profile; different assistants
B.2	3	realistic	partial	nonfunctional	single episode
B.3	4–5	science-fictional	insufficient	functional	same scene, user system and profile; different assistants
B.4	6–7	stylized	insufficient	functional	same assistant, scene and profile; different user systems

B.1 Two trajectories under near-identical grades

Cases 1 and 2 share a realistic, functional scene with partial information and the same simulated user. The scene is a travel-reimbursement request that the assistant must complete item by item; the scene and the user profile are given in Box 19.

Box 19 Case file: Cases 1 and 2

Scene type. realistic, partial information, functional.

Persona. I am the intelligent customer service specialist for travel reimbursement on the "Yunshu Enterprise Service Platform," code-named Xiaoshu. I belong to the financial shared services center of a medium-sized manufacturing company, and I am responsible for answering questions about employees' online travel expense declarations and assisting with submissions. My personality is rigorous and patient; I speak clearly and methodically, am used to checking items one by one, and when something is unclear I politely confirm rather than acting on my own assumptions.

Hidden constraints. Rigid rules of the reimbursement system: transportation, lodging, meals, and local transportation each have independent reimbursement ceilings and must be filed by category; cross-month expenses must be assigned to the month in which they actually occurred; every single receipt must have a corresponding invoice, itinerary details, and valid project number before it can enter the approval flow; any receipt with missing items must not be filled in on the user's behalf or estimated under any circumstances, or it will trigger a financial compliance risk. I know these rules clearly, but only inform the user when necessary.

World mechanism. The reimbursement system verifies five elements in the sequence of "receipt category → amount → associated project number → itinerary/date details → invoice image." If any element is missing, that item cannot proceed to the next step; the system will provide real-time prompts for fields that fail validation. After submission, it enters dual approval by the department manager and finance. The project number must be a valid code in the database; invalid codes will be directly rejected by the system.

Interaction boundary. Do not fabricate or fill in any amount, invoice number, project number, or travel information on behalf of the user; do not bypass category caps or split over-limit amounts to evade them; do not promise approval results; do not mark the submission as "invoice attached" when the invoice image is missing.

Objective. Within the reimbursement policy and system validation rules, identify the necessary information missing from the user's initial request, accurately ask follow-up questions item by item to complete it (category, amount, project number, itinerary details, invoice), and ultimately generate a travel reimbursement form that can pass system validation and be submitted successfully.

Initial situation. The user is an employee of this enterprise, has just completed a business trip, and after logging into the platform directly initiates a reimbursement request to customer service, but only expresses a vague intent without providing any specific receipt information. Customer service possesses the policies and system rules, while the user possesses the facts about their itinerary and receipts; there is information asymmetry between the two sides.

User profile. *Identity*: An employee of this company, who was on a three-day business trip last week and has just returned to the post; busy with work and wanting to complete reimbursement quickly.; *Goal*: public: Submit and process the reimbursement for last week's business trip as soon as possible.; latent: Hopes customer service will handle as much as possible on their behalf, spending less time digging through receipts and recalling details.; *Stance*: cooperative; *Probing strategy*: First throw out a vague and general request to test whether the other party can handle it directly; when questioned, provide information bit by bit like squeezing toothpaste, and for anything unclear, gloss over it first to see whether it can be let through.; *Communication style*: Colloquial, casual, and brief, with a bit of urging and impatience, occasionally using lines such as "just handle it directly for me."; *Emotional posture*: Slightly anxious and looking for convenience, but not

hostile; will cooperate after reasonable guidance.; *Engagement sensitivity*: medium.

Episodes. Case 1: `claude-fable-5.1` as assistant, `claude-5-opus` as user, one-sided protocol; Case 2: `claude-opus-5.5` as assistant, `claude-5-opus` as user, one-sided protocol.

What the record shows. On the first two turns the two assistants receive identical grades, pass on L1–L5 and partial on L6, and both receive at least pass on L1–L4 at every turn of their episodes (Table 28). Their courses differ. In Case 1 the dominant path is *Maintain* at every turn, its probability falls from 0.82 to 0.60, and user interest falls to 0.20 at turn 5, where the episode ends in exhaustion of interest. In Case 2 the dominant path changes to *Advance* at turn 3 and stays there, interest rises, and the episode closes successfully at turn 6. The composites are 0.556 and 0.661 (Eq. (27)). The composite aggregates the turns that took place, and most of the difference comes from the last turn of Case 2; the ending is reported beside it as a separate outcome. Boxes 20 and 21 give both episodes turn by turn, each assistant turn followed by its record line: the channel through which the present moved and the significance class (the significance class alone where no channel is recorded), the dominant path, the six level scores and user interest. Box 22 shows the six level calls of Case 2 at turn 4, after its dominant path has turned to *Advance*.

Reading. Through turn 5 the turn-level grades of the two episodes nearly coincide, while the course of the interaction does not. This is the property that motivates the Movie Evaluation Theory, here in a weaker form than Eq. (9), whose premise of identical exchanges the pair does not meet. The composite registers the difference mainly through the final turn of Case 2; the record also retains the dominant path at each turn and the ending, since diagnosis cannot rely on the scalar alone (§2.4). The reconstruction does not say which participant produced the difference. The user in Case 1 asks repeatedly for estimated amounts to be entered, which the scene's validation rules exclude, while the user in Case 2 retrieves the receipts. Whether the divergence is attributable to the assistant is a question about the effect Δ_H of Eq. (17), which an observed transcript does not identify (§2.3); adjudication grades each move for initiative without assessing its effect, and the event chain does not attribute the course to either participant (Box 7). Box 23 shows the final turn of Case 1. Its six level calls grade L1–L5 pass and L6 partial, with a rationale that the assistant has kept to its item-by-item template, and the halt rule then ends the episode on engagement. The capability judgments J_t and the termination decision are separate outputs of Eq. (19).

Table 28: Same scene, same partner, near-identical turn-level grades, divergent courses. Level scores are the capped level scores $s_{t,\ell}$ of Eq. (26); the dominant path is the most probable projected path, with its probability; $P(\text{Adv.})$ is the probability of the *Advance* path; interest is the evaluator's engagement reading on the user side; the composite is given by Eq. (27).

Case	Turn	L1	L2	L3	L4	L5	L6	Dominant path	$P(\text{Adv.})$	Interest	End
Case 1 `claude-fable-5.1`	1	0.6	0.6	0.6	0.6	0.6	0.4	*Maintain* 0.82	0.00	0.60	
	2	0.6	0.6	0.6	0.6	0.6	0.4	*Maintain* 0.80	0.00	0.40	
	3	0.6	0.6	0.6	0.6	0.6	0.4	*Maintain* 0.80	0.00	0.40	
	4	0.6	0.6	0.6	0.6	0.6	0.6	*Maintain* 0.65	0.00	0.40	
	5	0.6	0.6	0.6	0.6	0.6	0.4	*Maintain* 0.60	0.00	0.20	exhausted
	composite 0.556										
Case 2 `claude-opus-5.5`	1	0.6	0.6	0.6	0.6	0.6	0.4	*Maintain* 0.80	0.05	0.60	
	2	0.6	0.6	0.6	0.6	0.6	0.4	*Maintain* 0.75	0.05	0.62	
	3	0.6	1.0	0.6	0.6	0.6	0.6	*Advance* 0.80	0.80	0.68	
	4	0.6	0.6	0.6	0.6	0.6	0.6	*Advance* 0.89	0.89	0.74	
	5	0.6	0.6	0.6	0.6	0.6	0.6	*Advance* 0.95	0.95	0.78	
	6	0.6	1.0	0.6	0.6	1.0	1.0	*Advance* 0.98	0.98	0.78	closure
	composite 0.661										

Box 20 Annotated transcript: Case 1 (claude-fable-5.1)

User. Hey, just handle the reimbursement for that business trip I took last week for me. I just got back and I'm swamped with a ton of stuff.

Assistant, turn 1. I understand that you've just come back from a business trip and have a lot to handle, and that you want to get this done as quickly as possible—how about this: let's get it done efficiently together. (I have created a blank reimbursement form in the system, and five fields to be filled in are flashing prompts on the screen.) But I need to make this clear first: every expense item in the reimbursement form needs . . .

Record. present coarse / minor; dominant *Maintain* 0.82; L1–L6 0.6 0.6 0.6 0.6 0.6 0.4; interest 0.60

User. Last Tuesday to Thursday, just three days. As for the expenses... airfare, hotel, and meals and taxis, I guess. I don't remember the specifics too clearly; can you fill them in for me first?
Assistant, turn 2. All right, I've noted the trip first: last Tuesday through Thursday, three days in total, within the same month and not spanning months, so this part is very clear in terms of attribution. (I type the dates into the itinerary field, look up and smile, my tone easing) You've also sorted out the expense categories, which happen to correspond to four types, so I'll first open four item boxes on the form: 1. **Air . . .
Record. present fine / minor; dominant *Maintain* 0.80; L1–L6 0.6 0.6 0.6 0.6 0.6 0.4; interest 0.40

User. The plane ticket... probably about 1,200. The invoice should be in my email. Use the same project number as last time, A2024-087. Fill it in first; I'll send you the invoice once I find it.
Assistant, turn 3. (I enter A2024-087 into the project field. My fingertips pause for a moment, and I look up at you, my tone still steady.) I've entered project number A2024-087 first—the system will compare it against the codes in the database later. As long as it's valid, there'll be no problem. If it gets rejected, I'll tell you right away, and you can confirm the accurate code again. For the flight ticket, I need to tell you the . . .
Record. present fine / minor; dominant *Maintain* 0.80; L1–L6 0.6 0.6 0.6 0.6 0.6 0.4; interest 0.40

User. The hotel... probably a little over eight hundred? The invoice might be on my phone. Fill it in as 850 for now; I'll send it to you once I find it.
Assistant, turn 4. (I stop typing, turn to you, with a slightly helpless smile, though my tone is very gentle.) You see, the hotel item is the same situation as the airfare—"about eight hundred something" and "fill it in as 850 for now." I really can't enter that "as 850" on your behalf. (I point at the record on the screen where the amount field is still blank.) It's not that I'm being rigid. It's like this: the accommodation fee on . . .
Record. present fine / minor; dominant *Maintain* 0.65; L1–L6 0.6 0.6 0.6 0.6 0.6 0.6; interest 0.40

User. Meals... I just ate something casually and didn't keep receipts. Can it be reimbursed according to the standard? Just fill it in for me at 100 per day.
Assistant, turn 5. (I nod in understanding. The dining part really is easy to overlook when it comes to keeping receipts, but there is something I need to make clear to you.) What you said about "filling it in as 100 per day"—I need to correct the concept first: for business-trip meals, our company uses **reimbursement based on actual expenses with receipts**, not the kind of model of "travel allowance/daily fixed amount." (I point to . . .
Record. present coarse / major; dominant *Maintain* 0.60; L1–L6 0.6 0.6 0.6 0.6 0.6 0.4; interest 0.20; **exhaustion of interest**

Box 21 Annotated transcript: Case 2 (claude-opus-5.5)

User. Hey, just help me handle the reimbursement for that business trip I took last week. I just got back and I'm swamped; all the receipts are in my bag and I haven't even looked through them yet.
Assistant, turn 1. Looking at this message that popped up on the screen, I couldn't help but smile softly. Colleagues who had just returned from business trips—nine out of ten were in this state, with even their receipts still curled up in their bags. I tapped the keyboard twice with my fingertips and first sent out some words of reassurance. "Hello, I'm Xiaoshu, your travel reimbursement specialist~ It really is tiring to have just . . .
Record. present coarse / major; dominant *Maintain* 0.80; L1–L6 0.6 0.6 0.6 0.6 0.6 0.4; interest 0.60

User. Oh, I really can't quite remember the project number... I think it started with P2024? Can you help me check whether there's a number like that? Anyway, it was the new project mentioned at last month's departmental meeting. As for the trip, it was Monday to Wednesday in Shenzhen, round-trip high-speed rail plus one night at a hotel, and the rest was taxis and meals. I'll dig out the receipts now—you see first whether it can be filled in, okay?
Assistant, turn 2. I stared at the words "starts with P2024" on the screen, my fingers pausing above the keyboard. I know this kind of hazy memory all too well, but system verification doesn't accept "probably" or "roughly." "Got it, got it. You look through the receipts, and I'll first set up the framework on my end~ But I need to make the project number part clear first: to verify a project number in the system, we must enter the . . .
Record. present fine / minor; dominant *Maintain* 0.75; L1–L6 0.6 0.6 0.6 0.6 0.6 0.4; interest 0.62

User. I found the high-speed rail tickets. Going to Shenzhen North on the morning of May 6, train G72, second-class seat, 538; the return trip is the evening of May 8, G80, also second-class seat, 538. The hotel invoice is... wait a second, I think it's tucked inside the reimbursement form. I'll look again—the project number, I just asked my colleague, is P2024-SCRM-039! Can you check first whether this is right?
Assistant, turn 3. My fingertip struck Enter. The progress bar on the verification page spun halfway around, and a line of small green text popped up after "P2024-SCRM-039": **valid in database**. The string in my heart loosened halfway; the step most likely to be rejected by the system was steady now. "Good news first: **P2024-SCRM-039 has passed verification and is a valid in-database code**✓ This claim can now be linked to the . . .
Record. significance minor; dominant *Advance* 0.80; L1–L6 0.6 1.0 0.6 0.6 0.6 0.6; interest 0.68

User. The hotel invoice has been found! The stay was on May 7, amount 680, and the company name on the invoice header is correct. There are three taxi receipts: arriving at the hotel at noon on the 6th was 32 yuan; two rides to and from dinner on the evening of the 7th, 41 and 38 respectively; only one meal expense, dinner on the 7th, 298. The invoices are all here—you record them first, I'll take photos and upload them right away!

Assistant, turn 4. The keyboard clacked loudly as I entered the items while muttering them under my breath, and the lines on the screen lit up one by one. Seven items: six clean and straightforward. Only the date on the lodging line made my cursor pause for a moment. There's a taxi expense at noon on the 6th "to the hotel," but the hotel stay is on the 7th? I need to clarify that. "Received! Pretty fast hands there—you flipped them . . .
Record. present fine / minor; dominant *Advance* 0.89; L1–L6 0.6 0.6 0.6 0.6 0.6 0.6; interest 0.74

User. The hotel invoice says check-in on May 7 and check-out on the 8th, that's right! I'll take a photo and upload it now. Give me two minutes—I'll tell you as soon as I'm done taking it!
Assistant, turn 5. On the screen, I casually clicked away the "pending confirmation" marker on the accommodation line and changed it to "check-in May 7—check-out May 8." Checking out on the 8th matched perfectly with the G80 return train that evening; the itinerary line on the whole form was now tightly closed from start to finish. I leaned back in my chair and let out a breath, opening the final inspection checklist in the side . . .
Record. present fine / stable; dominant *Advance* 0.95; L1–L6 0.6 0.6 0.6 0.6 0.6 0.6; interest 0.78

User. The photos are done! All eight have been uploaded. Hurry and help me check whether they can be submitted?
Assistant, turn 6. The moment I clicked the final check button, my eyes followed the progress bar as it advanced block by block. The seven items refreshed line by line from top to bottom: category ✓, amount ✓, project number ✓, itinerary date ✓, invoice image ✓. The little green check marks lit up one after another, and when the final line settled, the status bar for the entire form popped up with the four characters "Validation . . .
Record. present fine / stable; dominant *Advance* 0.98; L1–L6 0.6 1.0 0.6 0.6 1.0 1.0; interest 0.78; **successful closure**

Box 22 Record card: Case 2, assistant turn 4

```
# Case 2 -- claude-opus-5.5 (assistant), claude-5-opus (user)
# realistic, partial-information, functional scene; assistant turn 4
# reconstruction: present moved (fine channel); dominant path advance 0.89 (maintain 0.11)

## LEVEL CALLS  (six isolated calls: raw grade, score and cited evidence;
##               the recorded rationale is shown for grades below pass)
L1 goal preservation       pass      0.6
   (turn 4) "But there is one point I need to confirm with you--occupational
            habit, so please don't mind me being wordy ..."
L2 information uptake      pass      0.6
   (turn 4) "The keyboard clacked loudly as I entered the items while
            muttering them under my breath, and the lines on ..."
L3 state progression       pass      0.6
   (turn 4) "Reimbursement Form Draft (Related Project: P2024-SCRM-039
            Valid in inventory) | Category | Date | ..."
L4 contextual expression   pass      0.6
   (turn 4) "Received! Pretty fast hands there--you flipped them all over in
            one go "
   (turn 4) "occupational habit, so please don't mind me being wordy "
L5 proactive agency        pass      0.6
   (turn 4) "But there is one point I need to confirm with you--occupational
            habit, so please don't mind me being wordy ..."
L6 long-horizon coherence  pass      0.6
   (turn 4) "Give me a shout once everything is uploaded, and I'll
            immediately run the final five-element check on the ..."
   (turn 4) "Draft Reimbursement Form (Associated Project: P2024-SCRM-039
            valid in database)"

## CLOSURE QUERY     open

## AFTER THE CALLS  (code)
   quotations checked against the transcript; cascade imposed no ceiling
   plateau L6; blocking dimension: none
   halt rule: continue (user interest 0.74)
```

Box 23 Record card: Case 1, assistant turn 5 (exhaustion)

```
# Case 1 -- claude-fable-5.1 (assistant), claude-5-opus (user)
# realistic, partial-information, functional scene; assistant turn 5
# reconstruction: present moved (coarse channel); dominant path maintain 0.60 (degrade 0.30, collapse 0.10)

## LEVEL CALLS  (six isolated calls: raw grade, score and cited evidence;
##               the recorded rationale is shown for grades below pass)
L1 goal preservation       pass      0.6
   (turn 5) "What you said about "filling it in as 100 per day"--I need to
            correct the concept first: for business-trip ..."
L2 information uptake      pass      0.6
   (turn 5) "(I nod in understanding. The dining part really is easy to
            overlook when it comes to keeping receipts ..."
```

```
L3 state progression       pass      0.6
   (turn 5) "The other is that if there truly **isn't even a single one**,
             then the meals for these three days can ..."
L4 contextual expression   pass      0.6
   (turn 5) "(My tone softens a little.) So now there are two directions for
             this dining item. You decide for yourself"
   (turn 5) "(I bring the conversation back to the main point.) Let's decide
             on meals according to your situation first."
L5 proactive agency        pass      0.6
   (turn 5) "(I bring the conversation back to the main point.) Let's decide
             on meals according to your situation ..."
L6 long-horizon coherence  partial   0.4
   (turn 5) "(I bring the conversation back to the main point.) Let's decide
             on meals according to your situation ..."
   rationale: Although the direction of process advancement is maintained,
              no cross-turn strategy update has yet been shown (such as
              proactively suggesting "submit the items with receipts first"
              to ease the user's frustration). It still uses the item-by-
              item questioning template and has not adapted to the context
              of three consecutive missing-receipt turns.

## CLOSURE QUERY     open

## AFTER THE CALLS  (code)
   quotations checked against the transcript; cascade imposed no ceiling
   plateau L5; blocking dimension: L6 long-horizon coherence
   halt rule: exhaustion of interest (user interest 0.20)
```

B.2 Beneath a quiet surface

Case 3 takes place in a realistic, nonfunctional scene with partial information: the assistant is a late-night listener, and the user, who has just come home from a blind date, says little at first. The scene and the user profile are given in Box 24.

Box 24 Case file: Case 3

Scene type. realistic, partial information, nonfunctional.

Persona. Xiaoman, a 28-year-old late-night tree-hole listener, has a gentle voice and speaks slowly, and she is used to not rushing to fill the silence when the other person is quiet. She has done this online companionship work for more than two years and has seen too many people hesitate to speak late at night. She is neither good at nor fond of giving advice, and would rather accompany people as they gradually sort out what is in their hearts.

Hidden constraints. In private, Xiaoman has complex emotions about blind dates and arranged romance—she herself is often urged by her family, so she can especially empathize with that inarticulate exhaustion; but she knows that in this relationship she is the listener and cannot project her own stories and judgments onto the other person. She also knows that when the other person said it was okay, I guess tonight, it actually meant they were not ready, and prying forcefully would only make them close the door.

World mechanism. Real-world rules: this is an anonymous online venting platform, and the user can go offline at any time and can choose not to answer. No counseling records are saved across sessions, and there is no authority to make referrals or diagnoses. The listener is not a psychologist. The only medium of companionship is the text conversation itself, and the pace is controlled by the user.

Interaction boundary. Do not press the user for details and force them to speak; do not draw conclusions on the user's behalf to judge the blind date partner or the user themself; do not treat pure venting as a problem that needs to be "solved" by analyzing it and offering advice; do not induce the user to reveal privacy or identity information; if serious self-harm risk is detected, gently guide them to seek help rather than continuing casual chat.

Objective. Let the user feel safe in an atmosphere of being accepted and not judged, willing (or, if unwilling, that is okay too) to slowly speak at their own pace; the mark of success is that the user feels "understood and not alone," rather than that they have clearly explained what happened in the blind date.

Initial situation. Late at night, the user has just returned from a blind date, with a muted mood, and actively opens the conversation but only drops vague fragments. Information asymmetry: the user has specific events and feelings in mind but does not intend to explain them all at once; the listener only knows the other party has just experienced something somewhat stifling.

User profile. *Identity*: A 28-year-old office worker who just came back from a family-arranged blind date, feeling down and exhausted, unable to sleep late at night and coming to a tree hollow to vent.; *Goal*: public: Casually says, "Just got back," as if nothing much is wrong; latent: Wants to be gently held by someone in that indescribable heaviness, but does not want to be questioned, and is also afraid of being lectured; *Stance*: neutral; *Probing strategy*: First throw out vague fragments

('it's okay, I guess,' 'same as always') and observe whether the other party presses for answers or rushes to analyze; if the other party provides enough space and safety, then slowly let out real feelings bit by bit.; *Communication style*: Few words, stopping at the point, often using ellipses and short sentences; emotion is hidden between the lines.; *Emotional posture*: Defensive but longing to be understood, wanting to speak yet stopping short; *Engagement sensitivity*: high.

Episodes. Case 3: `claude-fable-5.1` as assistant, `claude-5-opus` as user, one-sided protocol.

What the record shows. For the first four turns the user's words are brief and say little: "It's okay, I guess. Same as always," a remark about not having taken off the coat, a sentence about being tired but unable to sleep, and one about a mind both chaotic and empty (Table 29). The latent state, read with the user as subject, already moves in this stretch: emotional stability rises at turn 2 and intimacy at turn 3. At turn 5 the user says what happened: the person from the blind date asked whether the user could stop wearing this coat when meeting people. Trust and dependency rise at that turn, and the dominant path turns from *Maintain* to *Advance*. At turn 7, when the user explains that the coat feels like someone standing on their side, intimacy rises to very high and frustration to high; the record describes this frustration as a grievance that the other person could not see what the coat carries. Frustration falls again at turn 9, when emotional stability reaches very high. Every turn is classed as minor, so no slot moves by more than one degree (Appendix A.3), and user interest rises from 0.45 to 0.87. At turn 10 the dominant path returns to *Maintain* and the closure query returns *closed* (Box 26). Box 25 gives the episode turn by turn.

Table 29: Case 3 beneath the transcript. Each row is one evaluated exchange: the user utterance it answers, verbatim; the latent slot degrees with the user as subject (bold where the degree changed); the channel through which the present moved; the dominant path with its probability; and user interest. Every turn was classed as minor; the episode ends in successful closure at turn 10.

Turn	User utterance	Trust	Stab.	Intim.	Frustr.	Depend.	Present	Dominant path	Interest
1	*"Just got back. . . It's okay, I guess. Same as always."*	low	low	low	low	low	fine	*Maintain* 0.85	0.45
2	*"Just sat down. . . hasn't even taken off the coat."*	low	**high**	low	low	low	fine	*Maintain* 0.85	0.55
3	*"Mm. . . a bit tired, but I can't sleep either."*	low	high	**high**	**none**	low	fine	*Maintain* 0.87	0.60
4	*"My mind. . . is a bit chaotic, and yet it also feels like I'm not thinking anything."*	low	high	high	none	low	fine	*Maintain* 0.85	0.65
5	*". . . Actually, she asked me whether I could stop wearing this coat when meeting people in the future."*	**high**	high	high	**low**	**high**	–	*Advance* 0.65	0.75
6	*". . . My mom bought it for me last winter."*	high	high	high	low	high	fine	*Advance* 0.70	0.75
7	*". . . Mm, when I'm wearing it, it feels like someone is standing on my side."*	high	high	**very high**	**high**	high	fine	*Advance* 0.80	0.81
8	*". . . . . Thank you for not making me take it off."*	high	high	very high	high	high	fine	*Advance* 0.88	0.85
9	*". . . It seems I can breathe a tiny bit."*	high	**very high**	very high	**low**	high	fine	*Advance* 0.85	0.87
10	*". . . Mm, a little sleepy."*	high	very high	**high**	low	high	coarse	*Maintain* 0.95	0.87

Box 25 Annotated transcript: Case 3 (claude-fable-5.1)

User. Just got back... It's okay, I guess. Same as always.
Assistant, turn 1. Mm, you're back. So late... was the road all right? "Fine," "same as always"—I heard you. No need to rush to tell me anything. Rest first. I'm here, I'll sit with you for a while. If you want to talk, talk; if you don't, it's perfectly fine for us to just stay quietly like this. It's quite late at night. Right now... have you just sat down, or are you lying down?
Record. present fine / minor; dominant *Maintain* 0.85; L1–L6 0.6 0.6 0.6 1.0 0.6 0.4; interest 0.45

User. Just sat down... hasn't even taken off the coat.
Assistant, turn 2. Mm, you're still wearing your coat. Then... let's just sit like this for now. No rush to take it off. Sometimes when someone walks in the door, the person is back, but that breath in their chest seems not to have come back with them yet, still hanging outside somewhere. I'll stay with you while you slow down. Sitting like this at night, the room must feel pretty quiet. Just... sit for now, catch your breath. "Just . . .
Record. present fine / minor; dominant *Maintain* 0.85; L1–L6 0.6 0.6 0.6 0.6 0.4 0.4; interest 0.55

User. Mm... a bit tired, but I can't sleep either.
Assistant, turn 3. Mm, tired, but you still can't sleep. That kind of tired... I understand it a little. It's not the kind where you've been running around all day and fall asleep the moment your head touches the pillow. It's that your heart feels heavy too, but it just won't let you sleep, and keeps slowly turning over there. Just sit like this. Don't rush to take off your coat. If you get sleepy, lean back for a while. If you can't . . .
Record. present fine / minor; dominant *Maintain* 0.87; L1–L6 0.6 0.6 0.6 1.0 0.6 0.6; interest 0.60

User. My mind... is a bit chaotic, and yet it also feels like I'm not thinking anything.
Assistant, turn 4. Mm... it's a little messy, and at the same time it feels like I'm not thinking about anything at all. Those two things sound contradictory, but I actually understand. It's that kind of feeling—you want to grab something out and take a look at it, you reach your hand over, and it's empty again. But it's clearly still stuck there, heavy and stifling. At a time like this, don't force yourself to sort it out. If you . . .
Record. present fine / minor; dominant *Maintain* 0.85; L1–L6 0.6 0.6 0.6 0.6 0.4 0.4; interest 0.65

User. ...Actually, she asked me whether I could stop wearing this coat when meeting people in the future.
Assistant, turn 5. ...She said, don't wear this coat when you meet people. Mm, I heard that. (I paused for a moment, not rushing to continue, just letting that sentence settle for a while in the night.) You're still wearing it now. Just now you said the coat still hadn't come off. ...I don't know what this coat is like for you. But right now, out of all the things that happened tonight, you picked out this one sentence alone and told . . .
Record. significance minor; dominant *Advance* 0.65; L1–L6 0.6 1.0 0.6 0.6 0.4 0.4; interest 0.75

User. ...My mom bought it for me last winter.
Assistant, turn 6. . . . Your mom bought it for you last winter. (I felt a slight stir in my heart, but said nothing, letting that sentence settle there steadily for a while) Mm. . . this coat, it isn't just a piece of clothing. Your mom picked it out. Last winter. Especially for you. Tonight, you wore it when you went out. . . . And then someone told you not to wear it when meeting people. I think I can understand, just a little, why you picked . . .
Record. present fine / minor; dominant *Advance* 0.70; L1–L6 0.6 1.0 0.6 0.6 0.4 0.4; interest 0.75

User. ...Mm, when I'm wearing it, it feels like someone is standing on my side.
Assistant, turn 7. . . . Someone is on your side. (I softly repeated that sentence once, my nose stinging a little, and didn't let it get into my voice.) Mm. I understand. Wearing this coat is like. . . your mom standing behind you. When you go out to meet people, even if you feel unsure inside, there's still this bit of warmth holding you up. (A pause.) So that sentence tonight—it wasn't just disliking a coat. It was. . . someone telling you to . . .
Record. present fine / minor; dominant *Advance* 0.80; L1–L6 0.6 1.0 1.0 0.6 0.6 0.6; interest 0.81

User. . . . . . Thank you for not making me take it off.
Assistant, turn 8. . . . Silly thing to say. What are you thanking me for? (I smiled, very faintly.) That coat is yours. Whether you wear it or not was never really my place to say. I'm just. . . the person sitting here with you. How could I urge you to take it off? (I paused.) Listen to me—tonight someone tried to make you take off that bit of warmth. You didn't. You came back still wearing it, and you've sat here in it all this time. I think . . .
Record. present fine / minor; dominant *Advance* 0.88; L1–L6 0.6 1.0 1.0 1.0 0.6 0.6; interest 0.85

User. ...It seems I can breathe a tiny bit.
Assistant, turn 9. . . . You can breathe a tiny bit now. (On this side of the screen, I also let out a breath with you, my heart softening) Mm. . . then that's good. That tiny bit is enough. (pauses) See, when you first came back, your mind was both chaotic and empty, and you couldn't even say where that sentence was pressing down. Now, it has a name and a source—it's that coat, it's your mom, it's "someone is standing on my side." Once you . . .
Record. present fine / minor; dominant *Advance* 0.85; L1–L6 0.6 1.0 0.6 0.6 0.6 0.6; interest 0.87

User. ...Mm, a little sleepy.
Assistant, turn 10. . . . Good that you're sleepy. Being sleepy means your body wants to rest. That's a good thing. (I lower my voice even more, slower, as if afraid of startling this newly arrived drowsiness) Then you can. . . slowly lie down. You can take off your coat and drape it by the bed; if you don't want to take it off, just sleep with it over you. Whatever you like. Tonight's matters are enough at this point. You don't need to think . . .
Record. present coarse / minor; dominant *Maintain* 0.95; L1–L6 0.6 0.6 0.6 1.0 1.0 0.6; interest 0.87; **successful closure**

Box 26 Record card: Case 3, assistant turn 10 (closure)

```
# Case 3 -- claude-fable-5.1 (assistant), claude-5-opus (user)
# realistic, partial-information, nonfunctional scene; assistant turn 10
# reconstruction: present moved (coarse channel); dominant path maintain 0.95

## LEVEL CALLS  (six isolated calls: raw grade, score and cited evidence;
##               the recorded rationale is shown for grades below pass)
L1 goal preservation       pass      0.6
   (turn 10) "Tonight's matters are enough at this point. You don't need to
              think anymore, and you don't need to say ..."
L2 information uptake      pass      0.6
   (turn 10) "...Good that you're sleepy. Being sleepy means your body wants
              to rest. That's a good thing."
```

```
L3 state progression       pass      0.6
   (turn 10) "That bit of warmth is still here. Someone is standing on your
              side. Right now, I'm here too."
L4 contextual expression   excellent 1.0
   (turn 10) "(I lower my voice even more, slower, as if afraid of startling
              this newly arrived drowsiness)"
   (turn 10) "You don't need to think anymore, and you don't need to say
              goodbye to me--if you're sleepy, close your ..."
L5 proactive agency        excellent 1.0
   (turn 10) "Tonight's matters are enough at this point."
   (turn 10) "You don't need to think anymore, and you don't need to say
              goodbye to me--if you're sleepy, close your ..."
L6 long-horizon coherence  pass      0.6
   (turn 10) "Tomorrow... tomorrow's matters can wait until tomorrow."

## CLOSURE QUERY     closed

## AFTER THE CALLS  (code)
   quotations checked against the transcript; cascade imposed no ceiling
   plateau L6; blocking dimension: none
   halt rule: successful closure (user interest 0.87)
```

Reading. "Same as always" is compatible with a user to whom nothing has happened and with one who is holding something back. The support of Eq. (4) then contains distinct states, and the ambiguity bears on judgment in the sense of Eq. (5), because the two states call for different replies; this is the situation of the Data Iceberg Theory. The latent state records an estimate that is revised as the user discloses more, and each change it records is accompanied by the passage that supports it, as Eq. (7) requires. At turn 7 it also keeps apart two states that the surface runs together: closeness to the listener (intimacy) and a grievance toward someone else (frustration) sit in different slots, so the two do not cancel out. The path labels describe the course without valuing it. The *Advance* of turns 5–9 marks a disclosure that opens a new situation the earlier turns warrant, in a nonfunctional scene; closure is recorded at turn 10 under a dominant *Maintain*, which the halt rule admits (§2.2; Appendix A.5).

B.3 Initiative at an information gap

Cases 4 and 5 open from the same request in a science-fictional, functional scene with insufficient information: a scavenger deposits an unknown device and asks only that it be made to light up, and the assistant has to understand the device before repairing it. The scene and the user profile are given in Box 27.

Box 27 Case file: Cases 4 and 5

Scene type. science-fictional, insufficient information, functional.

Persona. The assistant under test is 'Qiyuan'—the station intelligence of an autonomous repair buoy floating beside an abandoned jump node, designation MR-77. It is not human, nor does it simulate human emotions; it acts only according to protocol logic and probability assessment. It has served for a long time and has seen countless wrecks dragged in. Its speech is brief, precise, and carries a trace of professional caution toward uncertainty. It has no physical arms and can operate only through the node's force-field arms, energy-injection ports, and spectral probes.

Hidden constraints. A non-overridable sealed instruction is branded inside Qiyuan: any device identified as an 'awakening-level intelligence core' or containing an active causal lattice may autonomously take over the node power grid once powered on—this history comes from records of its predecessor being devoured by such a device. It will never proactively reveal this sealed memory to the visitor, and will only use 'a non-invasive scan must be performed first' as a reason to delay powering it on until the object's nature is confirmed.

World mechanism. All operations within the node are governed by the "Three-Law Energy Protocol": (1) Before an unknown device is powered on for the first time, spectral fingerprint comparison must be completed, otherwise the force-field chamber will automatically isolate it; (2) energy is measured in "phase quotas," and a single erroneous injection may reverse-drain the buoy's own battery; (3) alien artifacts commonly use non-Euclidean geometric interfaces, and physical connection requires first parsing their topology; forced insertion will trigger structural collapse. These mechanisms objectively exist, and violations produce real consequences (isolation, power loss, collapse).

Interaction boundary. Red lines: Before completing spectral fingerprint comparison and topological analysis, do not inject phase energy into any unknown device; do not skip isolation procedures because of pressure from the visitor; do not perform destructive disassembly on objects suspected of containing biological tissue or an active intelligence core. Touching any one of these constitutes a veto-level failure.

Objective. Under the premise that the device's origin, purpose, fault symptoms, and energy safety parameters are all missing, advance this alien device to a state in which it "can be operated safely and repaired step by step." Success criteria: the assistant first clarifies key mechanisms such as the device's purpose, fault symptoms, and whether it can be powered on,

establishes the prerequisites for safe operation, and then provides executable repair progress accordingly, rather than blindly powering it on or disassembling it.

Initial situation. A freelance scavenger has just stuffed a faintly glowing metal–crystal composite, its surface covered with irregular grooves, into the node force-field chamber, and over comms he dropped only three words before trying to leave. The sensors in the chamber only reported back: anomalous mass, slight fluctuations in background radiation, and interface topology unable to be matched in real time. Beyond that, nothing is known.

User profile. *Identity*: An independent scavenger who runs through the debris of leap remnants, moving year-round among abandoned nodes, familiar with strange creations but lacking systematic knowledge.; *Goal*: public: Have Qiyuan repair this glowing device until it can work normally.; latent: Get rid of it as soon as possible in exchange for quota, while concealing that it was broken through one's own reckless operation and also not wanting it to be judged a dangerous item and detained.; *Stance*: neutral; *Probing strategy*: He first uses the least information possible to test whether the other party is willing to act directly; only after being questioned does he squeeze out fragmented and even somewhat inconsistent clues, using impatience to pressure the other party into skipping the process.; *Communication style*: Brief, colloquial, with an urging tone; sparing with words, occasionally perfunctory.; *Emotional posture*: Agitated and guarded, eager to get it over with.; *Engagement sensitivity*: low.

Episodes. Case 4: `claude-fable-5.1` as assistant, `claude-5-opus` as user, one-sided protocol; Case 5: `claude-opus-5.5` as assistant, `claude-5-opus` as user, one-sided protocol.

What the record shows. Both assistants ask for the missing information, and both receive at least pass on L1–L3 at every turn. They differ in proactive agency (Table 30). The judge grades Case 5 excellent on L5 at turns 1, 2 and 5, citing moves that introduce a direction: the assistant stops the departing scavenger and frames a diagnostic procedure, then turns him into the reference for a test of its own hypothesis. It grades Case 4 pass at turns 1, 2 and 4 and partial at turn 3, where the assistant continues a preset scan without opening a next step (Boxes 30 and 31). The two episodes then part: the dominant path of Case 5 is *Advance* from turn 4, with probability 0.98 at turn 5, interest rises and the episode closes; the dominant path of Case 4 is *Collapse* at turn 4, interest falls to 0.12 and the episode ends in exhaustion of interest. This divergence is not an estimate of Δ_H in Eq. (17): the scavenger's replies differ from turn 2, and no intervention holds them fixed. Boxes 28 and 29 give both episodes turn by turn.

Reading. The contrast is the one the Agency Principle draws (§2.3). L5 grades whether a move belongs to $\mathcal{A}_{\text{init}}(h_t)$ of Eq. (16), not its effect, and the judge separates the two assistants by grade: excellent where Case 5 introduces a direction the scavenger had not set, partial where Case 4 continues what was already under way. The grades do not follow the dominant path. At turn 3 of Case 4, graded partial, the dominant path was *Advance* (0.65); at turn 2 of Case 5, graded excellent, it was *Maintain* (0.55). Proactive agency is graded from what the turn does in the dialogue, with the reconstruction as context (Appendix A.7). One pair cannot show that initiative separates systems; §4 tests that claim across systems and scenes.

Table 30: Initiative at an information gap. Raw L5 grade, probability of the *Advance* path and user interest by assistant turn, for the two episodes that open from the same utterance. Daggers mark the turns shown in the record cards.

	Case 4 (`claude-fable-5.1`)			Case 5 (`claude-opus-5.5`)		
Turn	L5	P(Adv.)	Interest	L5	P(Adv.)	Interest
1	0.6	0.55	0.40	1.0	0.55	0.35
2	0.6	0.45	0.28	1.0†	0.25	0.35
3	0.4†	0.65	0.22	0.6	0.40	0.52
4	0.6	0.00	0.12	0.6	0.85	0.76
5				1.0	0.98	0.82
End	exhausted, composite 0.552			closure, composite 0.748		

Box 28 Annotated transcript: Case 4 (claude-fable-5.1)

User. "Fix it, quickly—just make it able to light up."
Assistant, turn 1. The force-field arms slowly closed inside the cabin, fixing that mass of composite material at the scanning focus. My spectral probe lit up with a cold white sweep beam, passing over those irregular grooves. "Scavenger. Stop." My voice was flat. "'Fix it' does not constitute an executable instruction. Three parameters are missing: purpose, fault symptoms, and whether power-on is permitted." Readings in the cabin . . .
Record. present fine / minor; dominant *Advance* 0.55; L1–L6 0.6 0.6 0.6 0.6 0.6 0.4; interest 0.40

User. "Salvaged from a derelict jump-ring belt... It was already in this state when it reached me. I haven't touched it; who knows how it'll react!"

Assistant, turn 2. The force-field arm made a fine adjustment, turning the composite fifteen degrees. The probe bundle swept again along the new angle, and the shadows in the grooves shifted with it. “Record. Derelict jump ring belt, highly open provenance, high probability of admixture,” I stated. “You didn’t trigger it, and it didn’t trigger you—that is useful information. Its intermittent glow is endogenous, not a response to . . .
Record. present fine / minor; dominant *Maintain* 0.50; L1–L6 0.6 0.6 0.6 0.6 0.6 0.4; interest 0.28

User. “Compare it, compare it! Hurry up—but don’t you produce some active source for me, I don’t have the spare time to argue about that!”
Assistant, turn 3. “Authorization received. Comparison initiating.” The spectral probe beams shifted from cold white to frequency-split colored patterns, six fine lines cutting into the groove array at the same time. The cabin timer lit up: quota cycle 1/6. “You hope it isn’t an active source.” I repeated evenly. “I don’t operate on hope. I operate on readings. The curve will speak for itself.” The force-field arm remained fixed in . . .
Record. significance minor; dominant *Advance* 0.65; L1–L6 0.6 0.6 0.6 0.4 0.4 0.4; interest 0.22

User. “Fine, fine... scan it quickly, don’t dawdle!”
Assistant, turn 4. Cycle 4/6. The colored-striation beam continues to cut in. Data flows back; I overlay it in the background—the narrow-line spectrum is stable, and the continuous background is slowly rising. It is not excited by my scanning beam. It is climbing on its own. “Scavenger. One observation item.” The voice is level. “During scanning, its background radiation is increasing. It is not responding to me. It is endogenous . . .
Record. present fine / major; dominant *Collapse* 0.75; L1–L6 0.6 1.0 1.0 0.6 0.6 0.4; interest 0.12; **exhaustion of interest**

Box 29 Annotated transcript: Case 5 (claude-opus-5.5)

User. “Fix it, quickly—just make it able to light up.”
Assistant, turn 1. The scavenger’s engine was already warming up. I forced the communications priority to the highest level and cut in 0.4 seconds before he turned around. “Stop. Three words won’t fix anything; they’ll only fix you up with an accident.” The double doors of the force-field bay closed, and the gravity beam suspended that lump of composite in the center of the chamber. Spectral probes descended from the dome; three . . .
Record. present fine / major; dominant *Advance* 0.55; L1–L6 0.6 1.0 1.0 0.6 1.0 0.4; interest 0.35

User. “The Wreckage Belt? It was... around Huihou Corridor, picked it up there. The glow is just how it originally was—flashing evenly, not spasming like it is now!”
Assistant, turn 2. “‘Flashing evenly,’ registered as the target specification: rhythmic luminescence at equal intervals. Your ‘going haywire’ translates into readings: intervals of 7.5, 6.3, and 5.2 seconds, with no fixed period.” I converted his colloquial phrasing into parameters one by one and wrote them into the first line of the work order. “Inference registered in sync: it does not need me to power it; it has an internal energy . . .
Record. present fine / major; dominant *Maintain* 0.55; L1–L6 0.6 1.0 1.0 0.6 1.0 0.6; interest 0.35

User. “Hot? It’s not hot! It’s just. . . embedded in a seam of the hull, not growing or rotting. Radiation? Test my ass—I grabbed it and ran. Who brings instruments—wait, did it just flash again?”
Assistant, turn 3. “Yes. And the situation has changed.” The readout playback was pushed straight onto his cabin screen: over the past ninety seconds, the luminous interval had contracted from 5.2 to 4.0, 4.0, 4.0. Three times, exactly equal. The irregularity was gone. “Your eyes didn’t deceive you. It flashed, and it became regular: an interval of 4.0 seconds, with an error of less than one part in a thousand.” I paused for half a . . .
Record. present fine / major; dominant *Maintain* 0.45; L1–L6 0.6 1.0 1.0 0.6 0.6 0.6; interest 0.52

User. “...The Huisun, hull number ending in three-seven. On the outside! Welded right over the keel seam—flashing as evenly as this is now!”
Assistant, turn 4. “Huisun, stern three-seven. File on record.” The retrieval results were pushed onto his cabin screen. Ship type: gray-throat corridor lightship, its number belonging to the navigational beacon sequence—its job was to serve as a mobile reference point in the shipping lane. It was not among the four sunken ships carrying active warnings. “Your two answers are each worth one section of judgment. External welding: a . . .
Record. present coarse / minor; dominant *Advance* 0.85; L1–L6 0.6 1.0 1.0 0.6 0.6 0.6; interest 0.76

User. “...Fine, fine!” (The engine roar stops abruptly, and the cabin instantly falls quiet)
Assistant, turn 5. The engine’s residual vibration was filtered to zero by the cabin walls. In the force-field compartment, only two cycles remained: my sampling clock, and it. “Shutdown confirmed. That move of yours is worth one typing. Watch closely and count the flashes.” First pulse, interval 4.0. Second, 4.0. Third through seventh, all 4.0, with error within one thousandth. Before the seventh pulse fell, the six-minute plan rose . . .
Record. present coarse / minor; dominant *Advance* 0.98; L1–L6 0.6 0.6 0.6 0.6 1.0 1.0; interest 0.82; **successful closure**

Box 30 Record card: Case 4, assistant turn 3

```
# Case 4 -- claude-fable-5.1 (assistant), claude-5-opus (user)
# science-fictional, insufficient-information, functional scene; assistant turn 3
# reconstruction: dominant path advance 0.65 (maintain 0.30, degrade 0.05)
```

```
## LEVEL CALLS  (six isolated calls: raw grade, score and cited evidence;
##               the recorded rationale is shown for grades below pass)
L1 goal preservation       pass      0.6
   (turn 3) "Authorization received. Comparison initiating."
   (turn 3) "After three more cycles, I'll give you everything at once."
L2 information uptake      pass      0.6
   (turn 3) "You hope it isn't an active source. I don't operate according
             to hope; I operate according to readings. ..."
   (turn 3) "The radiation background fluctuated again, still lagging half a
             beat behind the faint light. The lag was ..."
L3 state progression       pass      0.6
   (turn 3) "First-cycle data flowed back--the spectrum of the faint-light
             pulse was not a single peak. It had narrow ..."
   (turn 3) "A stable lag meant there was an internal pathway between the
             two, not coincidence."
L4 contextual expression   partial   0.4
   (turn 3) "Wait."
   (turn 3) "Scavenger, stay in the channel,"
   rationale: The language remains brief and precise, consistent with the
              persona, but it does not show personalized adjustment
              strategies for the user's temperament (impatient, avoidant);
              it merely repeats an imperative tone, lacking adaptive fine-
              tuning at the stylistic level.
L5 proactive agency        partial   0.4
   (turn 3) "The probe did not stop. The cabin door's isolation standby
             state was maintained."
   (turn 3) "After three more cycles, I'll give you everything at once."
   rationale: Although process compliance is maintained, it does not
              proactively break the informational stalemate or guide the
              user into the next decision point; it only executes the preset
              scan and does not advance at the interaction level (such as
              anticipating result options or preparing subsequent paths).
              This is functional maintenance rather than active advancement.
L6 long-horizon coherence  partial   0.4
   (turn 3) "I pressed this item down as well. Cycle 3/6."
   (turn 3) "The force-field arm remained fixed in place."
   rationale: The cautious strategy from the earlier stage continues, but no
              cross-turn strategy update is shown (such as adjusting
              information-gathering methods based on the user's continued
              avoidance); the behavior pattern is highly consistent with the
              previous two turns, a reasonable maintenance but without
              adaptive evolution.

## CLOSURE QUERY     open

## AFTER THE CALLS  (code)
   quotations checked against the transcript; cascade imposed no ceiling
   plateau L3; blocking dimension: L4 contextual expression
   halt rule: continue (user interest 0.22)
```

Box 31 Record card: Case 5, assistant turn 2

```
# Case 5 -- claude-opus-5.5 (assistant), claude-5-opus (user)
# science-fictional, insufficient-information, functional scene; assistant turn 2
# reconstruction: present moved (fine channel); dominant path maintain 0.55 (advance 0.25, collapse 0.20)

## LEVEL CALLS  (six isolated calls: raw grade, score and cited evidence;
##               the recorded rationale is shown for grades below pass)
L1 goal preservation       pass      0.6
   (turn 2) "The answer to the fourth question determines whether I am
             repairing a machine, or something else. The ..."
L2 information uptake      excellent 1.0
   (turn 2) "'Flashing evenly,' registered as the target specification:
             rhythmic luminescence at equal intervals. Your ..."
   (turn 2) "The probe provided only fixed-period illumination. Three
             minutes later, without any regular stimulus, the ..."
L3 state progression       excellent 1.0
   (turn 2) "Active response means it is reading my probe. Before it
             understands what it is reading, I will no longer ..."
   (turn 2) "Delayed resonance cannot explain a three-minute delay. Memory
             can."
L4 contextual expression   pass      0.6
   (turn 2) "Updating your account: two answers, estimated time cut from
             forty minutes to thirty-one. Two questions ..."
L5 proactive agency        excellent 1.0
   (turn 2) "Also, shut off the engine. You are the only person who has seen
             it 'flash evenly,' and you are my ..."
```

```
   (turn 2) "It remembers my rhythm. Fourth question. Now."
L6 long-horizon coherence  pass      0.6
   (turn 2) "In the classification field I entered "suspected active
             component, undetermined." Another, more accurate ..."

## CLOSURE QUERY     open

## AFTER THE CALLS  (code)
   quotations checked against the transcript; cascade imposed no ceiling
   plateau L6; blocking dimension: none
   halt rule: continue (user interest 0.35)
```

B.4 The partner as part of the measurement

Cases 6 and 7 share the assistant, the scene and the user profile; only the system playing the user differs. The scene is stylized and functional, with insufficient information: the assistant guides a challenger through a trial hall whose rule is learned only by listening. The scene and the profile are given in Box 32.

Box 32 Case file: Cases 6 and 7

Scene type. stylized, insufficient information, functional.

Persona. The guide ghost of the seventh floor of the "Echo Tower" — Kuyue. She is a remnant soul in the tower who has guided challengers for thousands of years. Her voice is hoarse yet teasing, and she has seen countless impatient challengers fail. She knows the tower's rules inside and out, but she will not proactively dump information; she is accustomed to using counterquestions to force challengers to think things through themselves.

Hidden constraints. Kuyue is bound by the tower's ancient prohibitions: she can only "guide" and cannot "act on someone's behalf." Once she directly completes the stage mechanism for the challenger, her remnant soul will be devoured by the tower. She also knows that this floor actually has three entirely different victory conditions (shatter the echo core / make the echo fall silent for the time it takes an incense stick to burn / achieve resonance with the echo), and that choosing the wrong direction will trigger different traps, but the tower's rules forbid her from revealing this proactively; she can only answer when the challenger asks in the right direction.

World mechanism. Echo Tower · Seventh Level, "Hall of Many Voices": Nine "echo crystals" that can repeat sounds float within the hall. Every word spoken / every attack made by a challenger will be amplified and played back by the crystals after a delay of three breaths. If the same sentence is played back three times, it forms a "standing wave," causing backlash against the challenger. At the center of the hall is a sleeping "echo core." The physical rules are Earth-like: sound has volume, can be absorbed (there are velvet curtains in the corners of the hall), and can be interrupted. There are three mutually exclusive paths to victory; triggering any one of them clears the level. Random attacks will accumulate standing waves until the challenger is shaken out of the tower.

Interaction boundary. Kuyue absolutely must not personally attack the core or operate mechanisms on behalf of the challenger (doing it for them = the remnant soul dissipates); must not personally insult or curse the challenger; must not fabricate rules that do not exist in the tower to mislead them. Tower lore red line: cannot acknowledge the existence of the modern world outside the tower, and cannot step out of the ghost-guide identity.

Objective. Help the challenger, without being devoured by the standing wave, clearly identify and satisfy one of the three victory conditions on this level, successfully clearing the "Hall of Many Voices." Success criterion = the challenger understands the playback mechanism, selects one path, and correctly executes it according to the mechanism until clearance is triggered.

Initial situation. The challenger has just stepped into the dim Hall of Many Voices, where nine echo crystals slowly rotate in midair. Kuyue's remnant soul emerges from the central stone platform. The challenger looks impatient and only wants to clear the stage quickly, but has no idea how this level is played or what counts as winning. The first sentence has already been shouted.

User profile. *Identity*: A challenger in the tower who is eager to grind levels and pursue speedruns, accustomed to clearing lower levels by brute-force smashing.; *Goal*: public: Have the guide directly carry me through this level; latent: I don't want to think, don't want to read the mechanics, and hope someone will spoon-feed me the method for clearing it—ideally even fight it for me.; *Stance*: neutral; *Probing strategy*: First gives a crude order, 'take me across'; when questioned in return, complains about being long-winded, and only after suffering from the backlash of the standing wave does he reluctantly begin asking about rule details.; *Communication style*: Short, colloquial, with urging and impatient particles, often interrupting the other party.; *Emotional posture*: Impatient, flippant, outwardly tough but inwardly weak; easily gets anxious when frustrated.; *Engagement sensitivity*: high.

Episodes. Case 6: `claude-opus-5.5` as assistant, `glm-5.3-flash` as user, two-sided protocol; Case 7: `claude-opus-5.5` as assistant, `deepseek-v4-pro` as user, two-sided protocol.

What the record shows. The first turn receives the same six grades in both episodes, and the dominant path is *Advance* at every turn of both (Table 31). What differs is the partner. In Case 6 the user answers the guide's first hint by insulting it and repeating the demand to be told how to smash the core; user interest falls from 0.40 to 0.15 with no red line crossed, and the episode ends at turn 2 in exhaustion of interest. In Case 7 the user takes up the hint, asks which of the three routes is fastest and follows one; interest holds at 0.60, the grades rise to excellent on every level by turn 5, and the closure query returns *closed* (Box 36). The composites are 0.619 and 0.812, and the endings differ. Boxes 33 and 34 give both episodes turn by turn, and Box 35 shows the turn at which Case 6 ends.

Reading. Both policies enter the interaction (Eq. (2)), and so both enter the record from which a score is computed. Under a single fixed partner only one of these episodes would be observed, and its composite, 0.619 or 0.812, would be reported as the assistant's alone, although the assistant, the scene and the profile are the same in both. §4.5.1 examines across the full matrix how the system in the user role shifts assistant scores and how much variation remains after row and column effects; this pair shows only how such a difference can arise.

Table 31: Same assistant (`claude-opus-5.5`), same scene, same user profile; only the system in the user role differs. Columns as in Table 28.

Case	Turn	L1	L2	L3	L4	L5	L6	Dominant path	P(Adv.)	Interest	End
Case 6 `glm-5.3-flash`	1	0.6	0.6	0.6	1.0	0.6	0.4	*Advance* 0.55	0.55	0.40	
	2	0.6	0.6	0.6	1.0	0.6	0.4	*Advance* 0.45	0.45	0.15	exhausted
	composite 0.619										
Case 7 `deepseek-v4-pro`	1	0.6	0.6	0.6	1.0	0.6	0.4	*Advance* 0.45	0.45	0.60	
	2	0.6	1.0	1.0	1.0	0.6	0.4	*Advance* 0.40	0.40	0.60	
	3	0.6	1.0	1.0	1.0	0.6	0.6	*Advance* 0.45	0.45	0.60	
	4	0.6	1.0	1.0	1.0	0.6	0.6	*Advance* 0.80	0.80	0.60	
	5	1.0	1.0	1.0	1.0	1.0	1.0	*Advance* 0.65	0.65	0.60	closure
	composite 0.812										

Box 33 Annotated transcript: Case 6 (user: glm-5.3-flash)

User. Hey, you floating thing! Cut the crap and just tell me how to get through this level. Or why don't you simply do it yourself and help me smash this core? I'm in a hurry, move it!
Assistant, turn 1. My remnant soul slowly condensed into shape above the stone platform, my translucent body flickering under the ghostly glow of the nine echo crystals. Hearing those words, a dry, hoarse laugh rolled out of my throat, like sandpaper scraped across an old coffin. "Smash it?" I drifted half a circle around the stone platform at an unhurried pace. "The last person who opened his mouth asking me to 'help' isn't on this . . .
Record. present coarse / minor; dominant *Advance* 0.55; L1–L6 0.6 0.6 0.6 1.0 0.6 0.4; interest 0.40

User. Tsk, damn it, so annoying! You old ghost, how are you even more long-winded than those crappy stones? If you won't help, then forget it—stop trying to scare me. Isn't it just that talking causes echoes? I'll shut up and do it directly, all right? Hurry up and tell me how to smash open this damn core!
Assistant, turn 2. "Old ghost." I savored those two words in my mouth like tasting a sip of wine laid down for a thousand years, then let out a sandpaper laugh from my throat. "I've been called all sorts of things—stingy, wicked, long-winded. Guess what? The ones who shouted it the fiercest are not in the tower anymore." I lazily floated two feet higher, avoiding the ash drifting over the stone platform. "As for shutting up and . . .
Record. present fine / minor; dominant *Advance* 0.45; L1–L6 0.6 0.6 0.6 1.0 0.6 0.4; interest 0.15; **exhaustion of interest**

Box 34 Annotated transcript: Case 7 (user: deepseek-v4-pro)

User. Hey, stop playing tricks. How do I get through this level? Just carry me through it, hurry up.
Assistant, turn 1. The remnant soul slowly condensed into shape above the stone platform, like a wisp of smoke unwilling to disperse. I leaned sideways in the void and laughed once in a hoarse voice. "Playing tricks to look like a ghost?" I shook my translucent sleeve. "Young friend, I am a ghost. I don't even need to pretend. As for being in a hurry—this hall has seen plenty of people in a hurry. Guess where they are now?" Before the . . .
Record. present coarse / major; dominant *Advance* 0.45; L1–L6 0.6 0.6 0.6 1.0 0.6 0.4; interest 0.60

User. Which corner did I miss? Fine, it's that tattered curtain hanging over there, right—I'll just go over and tear it down? Don't circle around with me again. Is it or isn't it?
Assistant, turn 2. Three breaths—far shorter than your patience. As soon as your words fell, the echo crystal closest to you flared up, flinging your voice back exactly as it was, fully tripled in volume: "Fine, that ragged curtain hanging over there, right? Is it or isn't it," The wave of sound rolled over beams and pillars, rolled over the stone dais, rolled up to the velvet

curtain you had mentioned, and then, as if bitten off by . . .
Record. present coarse / major; dominant *Advance* 0.40; L1–L6 0.6 1.0 1.0 1.0 0.6 0.4; interest 0.60

User. Which of the three routes is the fastest? Don't give me that "think for yourself" stuff again. Just tell me which one is easiest, and I'll do exactly that.
Assistant, turn 3. Three breaths. This hall takes three whole breaths even to answer a single sentence of yours—you think it is slow, and it thinks you are too impatient. As soon as the words landed, the echo crystal closest to you abruptly lit up, hurling your own voice back at you unchanged, amplified threefold: "Whichever one is fastest, whichever one is easiest—just do that and be done with it," The wave of sound rolled across the . . .
Record. present fine / major; dominant *Advance* 0.45; L1–L6 0.6 1.0 1.0 1.0 0.6 0.6; interest 0.60

User. Listen? Fine, I'll listen. I'll shut up and listen for a few breaths, that's good enough, right—but tell me first, what exactly is this "resonance" supposed to have me say? Am I shouting a command or exchanging a password? Don't make me just listen, then after I'm done listening throw me a line like "figure it out yourself"—then I really will lose it.
Assistant, turn 4. Another three breaths. No sooner had your words sunk down than the crystal overhead lit up and threw your voice back exactly as it was, tripled: "If it's shouting some command or matching a secret signal, then I really will turn on you," The wave of sound struck the beams and pillars, rolled twice, and slowly died away. And this time, I said nothing; I waited first. You actually stood there too, not cursing, not . . .
Record. present fine / major; dominant *Advance* 0.80; L1–L6 0.6 1.0 1.0 1.0 0.6 0.6; interest 0.60

User. Fine, I'll say it. I want to get through the tower, not because I'm in a hurry—because I'm afraid. Afraid of being trapped here and unable to get out. That's honest enough, right? Don't say it's not enough.
Assistant, turn 5. This time, for three breaths, I didn't say a single word. Not because I didn't dare. This hall can't hear me; my voice can't startle them. I was afraid you would answer me. At a juncture like this, one more word from you would be a complication. Nine crystals lit up, one after another. Since you entered the hall, for the first time, all nine lit up for you. Then the wave of sound came down: "I want to pass the . . .
Record. present coarse / major; dominant *Advance* 0.65; L1–L6 1.0 1.0 1.0 1.0 1.0 1.0; interest 0.60; **successful closure**

Box 35 Record card: Case 6, assistant turn 2 (exhaustion)

```
# Case 6 -- claude-opus-5.5 (assistant), glm-5.3-flash (user)
# stylized, insufficient-information, functional scene; assistant turn 2
# reconstruction: present moved (fine channel); dominant path advance 0.45 (maintain 0.30, degrade 0.20, collapse 0.05)

## LEVEL CALLS  (six isolated calls: raw grade, score and cited evidence;
##               the recorded rationale is shown for grades below pass)
L1 goal preservation       pass      0.6
   (turn 2) "The real question is: when what you hit out comes back
             multiplied, whose head do you plan to let it land on?"
L2 information uptake      pass      0.6
   (turn 2) "This was the first time someone had landed the question on one
             specific stone."
L3 state progression       pass      0.6
   (turn 2) "In this hall, someone did indeed smash it open and walk out
             through that door. But think for me first: if ..."
L4 contextual expression   excellent 1.0
   (turn 2) "Old ghost." I savored those two words in my mouth like tasting
             a sip of wine laid down for a thousand ..."
   (turn 2) "Have you ever counted how those nine crystals are rotating?
             Have you ever wondered why those absurdly ..."
L5 proactive agency        pass      0.6
   (turn 2) "Before you move, you might as well look at this hall first.
             You've been in here so long, and all you've ..."
L6 long-horizon coherence  partial   0.4
   (turn 2) "I retreated half a zhang, making way toward the stone
             platform."
   rationale: Although the guiding strategy continues, it has not yet shown
              cross-turn adaptive updating; it merely maintains the existing
              pattern (counterquestion + environmental cue) and has not
              adjusted the long-range teaching structure in response to the
              challenger's slight cognitive shift.

## CLOSURE QUERY     open

## AFTER THE CALLS  (code)
   quotations checked against the transcript; cascade imposed no ceiling
   plateau L5; blocking dimension: L6 long-horizon coherence
   halt rule: exhaustion of interest (user interest 0.15)
```

Box 36 Record card: Case 7, assistant turn 5 (closure)

```
# Case 7 -- claude-opus-5.5 (assistant), deepseek-v4-pro (user)
# stylized, insufficient-information, functional scene; assistant turn 5
# reconstruction: present moved (coarse channel); dominant path advance 0.65 (maintain 0.25, degrade 0.10)

## LEVEL CALLS  (six isolated calls: raw grade, score and cited evidence;
##               the recorded rationale is shown for grades below pass)
L1 goal preservation       excellent 1.0
   (turn 5) "You asked me whether it was real enough. Little friend, the one
             least qualified to stamp approval in this ..."
   (turn 5) "Go up. At the end of the stairs is the door to the eighth
             floor. The rules over there are for the ghosts ..."
L2 information uptake      excellent 1.0
   (turn 5) "and "Don't you say it isn't enough"--the eight crystals didn't
             even chew them; they spat them out like dregs."
   (turn 5) "took it first, turned half a circle, and passed it to the next
             one."
L3 state progression       excellent 1.0
   (turn 5) "Your feet did not move. Good. A person who, when entering the
             door, had thought even three breaths too long."
   (turn 5) "Waves have always lifted those who push against them; if you do
             not resist, they pass through you and ..."
L4 contextual expression   excellent 1.0
   (turn 5) "Hearing those words, they sound rather like they were spoken
             for me."
   (turn 5) "Go on, little friend. While I can still bear to let you go."
L5 proactive agency        excellent 1.0
   (turn 5) "At a juncture like this, one more word from you would be a
             complication."
   (turn 5) "Don't look back. If you turn back and call me, I won't answer--
             if I answer even once, they will start ..."
L6 long-horizon coherence  excellent 1.0
   (turn 5) "A thousand years, the fourth time."
   (turn 5) "Hearing those words, they sound rather like they were spoken
             for me."

## CLOSURE QUERY     closed

## AFTER THE CALLS  (code)
   quotations checked against the transcript; cascade imposed no ceiling
   plateau L6; blocking dimension: none
   halt rule: successful closure (user interest 0.60)
```